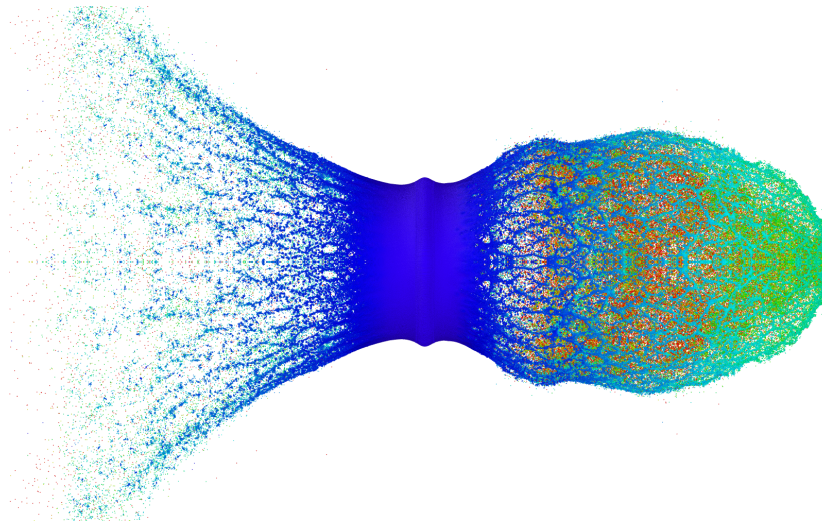


σ SPH

A GPU-Accelerated SPH Solver for Fluids, Elastic–Plastic Solids, and Detonation Media

THEORY AND USER MANUAL

Continuum Formulations, SPH Discretisation, Material Constitutive Models, and Keyword Deck Reference



Architecture: Taichi JIT Backend / CUDA Massive Parallelism

Kinematics: 3D Cartesian, 2D Plane Strain, 2D Axisymmetric

Scope: explicit dynamics of severe deformations, shock and reactive detonation

Foreword

It is the summer of 2026, and I believe I have finally found a way to achieve stable Smoothed Particle Hydrodynamics (SPH) simulations of solids, even in the updated Lagrangian formulation.

A decade ago, I was heavily involved in the development of SPH methods and devised a variant of total Lagrangian SPH scheme for solids, stabilised by a stiffness-based hourglass control. The total Lagrangian approach significantly improved accuracy and stability, but it departed from the true meshless spirit. Because the reference configuration is fixed, it cannot represent the topology changes that accompany violent deformation, fracture and fragmentation. Yet these are precisely the problems for which we need meshless methods rather than FEM. At the time, I saw no way to achieve comparable stability with updated Lagrangian SPH.

Since then, the SPH community has developed several new stabilisation techniques. Density and pressure diffusion schemes such as δ -SPH suppress the unstable growth of spurious oscillations in the stress field. Particle shifting techniques reduce particle disorder and thereby improve the accuracy of the kernel approximation. Combined with the corresponding transport-velocity corrections, shifting turns SPH into an Arbitrary Lagrangian-Eulerian (ALE) method. Together, diffusion and shifting have greatly improved accuracy and stability, with one notable exception: the tensile instability under strong tensile stress states.

This remaining deficiency is where my idea comes in. Could my original stiffness-based hourglass control be reformulated as a velocity-dependent, viscous scheme that works together with the transport mechanism of the ALE formulation? Answering this question required an open-source code for solids that already included diffusion and shifting stabilisation. To my knowledge, no such code existed.

However, this is the age of generative AI. In the mathematical and physical sciences, if you can describe precisely what you want, AI can write much of the code for you. It generates unit tests and assertions relentlessly, at a level of detail I would never have had the patience for. I could focus on defining the architecture, enforcing mathematical correctness, and ensuring that conservation laws and physical principles are respected. Within only a couple of weeks, the core of this code was written with the help of Claude Code (Claude Opus 5.5) and Google Antigravity (Gemini 3.8 Flash). The result is a hydrocode which seamlessly bridges the gap between solid and fluid behaviour, positively answering the question raised above. I hope it will be useful beyond academic interest.

I truly believe this is a golden age for scientists like me, who grew up writing Fortran and C and spent days or weeks chasing memory allocation errors. Today we can be productive at a level that would have seemed implausible only a few years ago. Of course, everything comes at a price. Be careful which demons you summon.

Georg C. Ganzenmüller, September 2026

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1 Introduction and Architecture

1.1 Overview and Physical Scope

sigmaSPH is a high-performance, GPU-accelerated Smoothed Particle Hydrodynamics (SPH) solver implemented in the Taichi programming language. It unifies weakly compressible computational fluid dynamics, large-deformation hypoelastic–plastic solid mechanics, and high-explosive detonation physics within a single, highly regularised particle framework.

Rather than segregating fluid and solid mechanics into disparate codes, **sigmaSPH** solves the conservation laws of continuum mechanics over a single unified neighbour-search and particle-sorting pipeline. The core capabilities comprise:

- **Weakly Compressible Hydrodynamics (δ^+ -SPH and γ -SPH):** Implements the position-correction δ^+ -SPH scheme of Sun et al. [29] and the low-Mach γ -SPH-ALE formulation of Collé et al. [10]. The primary kinematic variable is particle volume V_i rather than raw density, eliminating numerical density drift across free surfaces.
- **Large-Deformation Hypoelastic–Plastic Solids:** Solves finite-strain solid mechanics using a hypoelastic formulation based on the objective Jaumann rate of Cauchy deviatoric stress. Supports three constitutive plasticity models: classical J_2 (von Mises) plasticity with linear isotropic hardening (MAT001), Johnson–Cook viscoplasticity with strain-rate sensitivity, thermal softening, and adiabatic heating (MAT002), and the pressure-dependent Holmquist–Johnson–Cook strength of concrete (MAT003), which is declared together with its compaction equation of state (EOS004) and cumulative damage (DAM002), and optionally a Grady–Kipp tensile damage with Weibull-distributed flaws (DAM003) that governs its fragmentation. Spall damage in metals is available as a threshold cutoff or as Cocks–Ashby void growth (DAM001). Standard engineering metals can be invoked directly from the built-in material constants library (`material_models/material_library.py`).
- **High-Explosive Detonation and JWL Products:** Models energetic materials via a programmed Huygens burn front coupled to the Jones–Wilkins–Lee (JWL) equation of state. Tracks the evolution of specific internal energy per particle, closed with ALE convective transport corrections.
- **Unified Kinematic Dimensions:** The underlying Taichi data layout is strictly 3D (`dim = 3`) across all allocated fields. Two-dimensional planar problems are simulated via a single particle layer constrained to plane strain ($\dot{\epsilon}_{zz} = 0$, $v_z = 0$). Axisymmetric problems are resolved as meridional half-planes (x axial, $y = r \geq 0$, $z = 0$) revolved about the x -axis, incorporating exact hoop strain rates and momentum source terms at two-dimensional computational cost. A genuinely three-dimensional body possessing one or two planes of symmetry may in addition be discretised as the corresponding half or quarter of itself through `Domain.symmetryPlanes`, the cut planes being supplied with their missing neighbours by mirror images rather than read as free surfaces (Sec. 2.9).

1.2 Solver Architecture and Multi-Material Dispatch

In continuum mechanics, fluid dynamics represents the zero-shear limit ($G \rightarrow 0$, $\sigma_y \rightarrow 0$, $\alpha \rightarrow 0$) of solid dynamics. In this limit, the deviatoric Cauchy stress tensor vanishes identically ($\mathbf{s} \equiv \mathbf{0}$),

leaving the total stress tensor isotropic and governed entirely by thermodynamic pressure: $\sigma = -P\mathbf{I}$.

Within the `sigmaSPH` codebase, this continuum principle is implemented through an object-oriented class hierarchy rather than through two decoupled solver instances running concurrently. Figure 1.1 illustrates the overall architecture and dispatch logic.

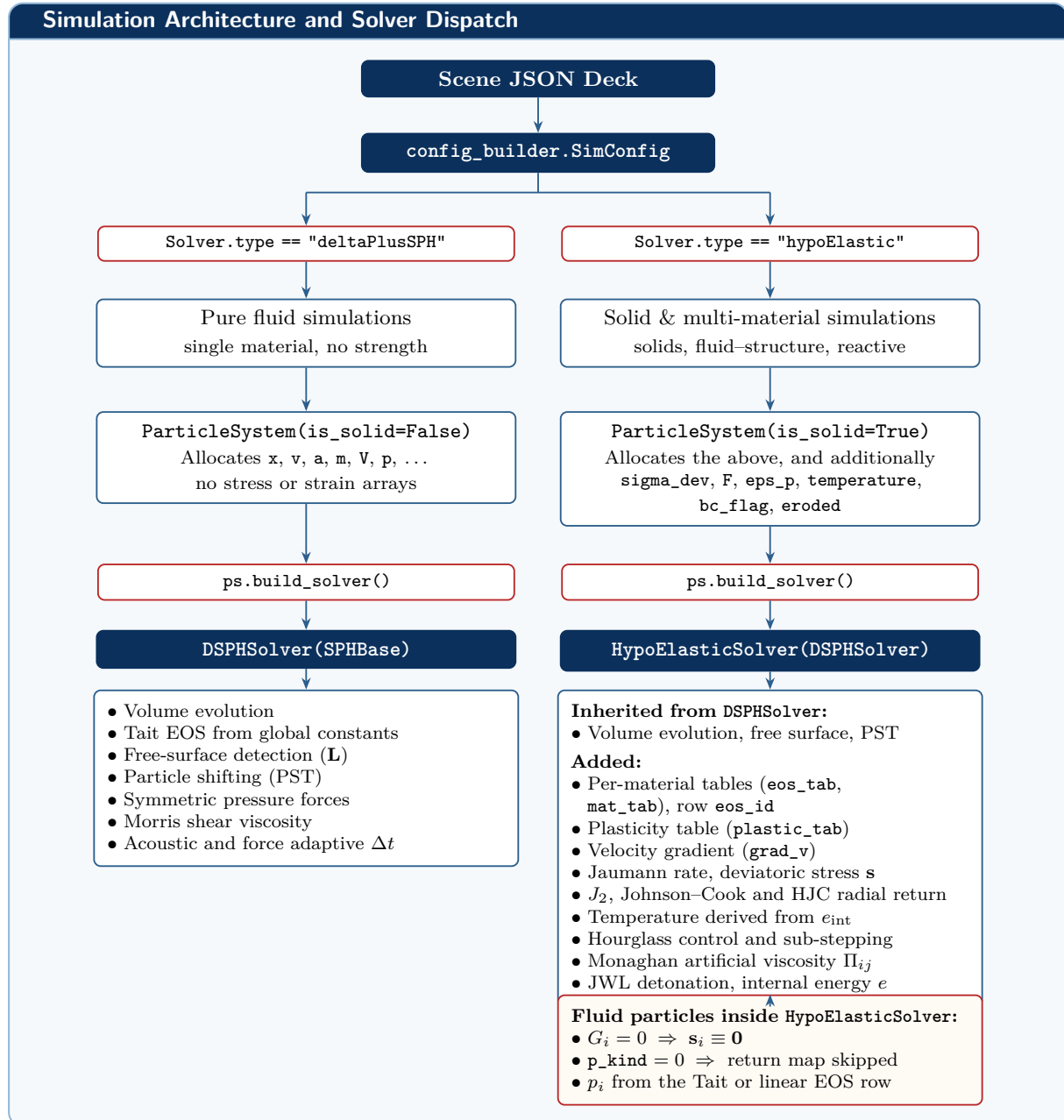


Figure 1.1: Simulation architecture and solver dispatch flow chart.

1.2.1 Scene-Wide Solver Selection

The factory method `ParticleSystem.build_solver()` instantiates exactly one solver object for the entire computational domain: either `DSPHSolver` (when `Solver.type == "deltaPlusSPH"`) or `HypoElasticSolver` (when `Solver.type == "hypoElastic"`). The simulation time-stepping loop executes that single solver instance across all particles. There is no partitioned co-simulation or split time-stepping between separate fluid and solid solvers.

Consequently, whenever a simulation contains both solid and fluid particles, or involves multiple interacting materials (such as high-explosive detonation products expanding against an elastic casing, or an elastic projectile penetrating a fluid basin), the scene deck must specify `Solver.type: "hypoElastic"`. In this mode, `HypoElasticSolver` processes all particles across the domain simultaneously.

1.2.2 Multi-Material Fluid Treatment ($G = 0$)

When `HypoElasticSolver` executes a multi-material simulation, fluid, gas, and detonation-product particles are treated seamlessly within the solid solver through constitutive parameterization:

- **Zero Shear Modulus ($G_i = 0$):** Materials declared without a `strengthModel` block (such as water, air, or unconfined fluids) are assigned `has_strength = False` by the material parser (`materials.py`). Their shear modulus is set to zero ($G = 0$), and their initial yield stress and hardening modulus are zeroed ($\sigma_{y0} = 0, H = 0$).
- **Deviatoric Stress Rate:** The objective Jaumann rate advances deviatoric stress via $\dot{\mathbf{s}}_i = 2G_i \text{dev}(\dot{\boldsymbol{\epsilon}}_i) + \boldsymbol{\Omega}_i \cdot \mathbf{s}_i - \mathbf{s}_i \cdot \boldsymbol{\Omega}_i$. For fluid particles where $G_i = 0$, physical deformation generates zero shear stress.
- **Interface Isolation via Explicit Zeroing:** At a fluid–solid contact interface, Arbitrary Lagrangian–Eulerian (ALE) particle shifting transports carried state fields across neighbouring particles. Without special handling, a fluid particle neighbouring a solid metal particle could pick up a non-zero deviatoric stress $\mathbf{s}_j$ through the convective transport term. To prevent this numerical contamination, `HypoElasticSolver.update_stress()` explicitly enforces:

$$\mathbf{s}_i = \mathbf{0} \quad \text{whenever} \quad G_i \leq 0. \quad (1.1)$$

This guarantees that fluid and gas particles never carry unphysical deviatoric shear stress, irrespective of neighbour interaction or meshless shifting.

- **Plasticity Bypass:** In the radial return map of J_2 plasticity, the return step is guarded by $G_i > 0$ and $\sigma_{y0,i} > 0$. For fluid particles, these conditions are false, ensuring the return map is bypassed entirely and preventing floating-point division by zero (0/0).
- **Thermodynamic Pressure Evaluation:** Fluid particles evaluate their hydrostatic pressure from their material’s row of the equation-of-state table (`eos_tab`), which each particle reaches through its integer material index `eos_id`, utilizing either the Tait equation of state ($P = \frac{c_0^2 \rho_0}{\gamma} [(\rho/\rho_0)^\gamma - 1]$) or a linear equation of state ($P = K(\rho/\rho_0 - 1)$). The inter-particle momentum flux naturally balances the isotropic fluid pressure $-P_{\text{fluid}}\mathbf{I}$ against the full solid Cauchy stress $\boldsymbol{\sigma}_{\text{solid}} = -P_{\text{solid}}\mathbf{I} + \mathbf{s}_{\text{solid}}$.

1.2.3 The Role of `DSPHSolver` as an Independent Base Class

If `HypoElasticSolver` can simulate fluid particles by setting $G = 0$, the existence of `DSPHSolver` as an independent class serves three vital engineering purposes:

1. **Memory Efficiency for Pure Fluids:** In pure fluid simulations (e.g., solitary dam breaks or free-surface sloshing), solid-specific history fields—deviatoric stress (`sigma_dev`), equivalent plastic strain (`eps_plastic`), boundary displacement (`bc_disp`), and erosion flags (`eroded`)—do not need to be allocated in GPU memory or sorted during the per-step radix counting sort. The deformation gradient (`F`) goes one step further: it is not allocated even on a solid run unless `Output.deformationGradient` asks for it, since no constitutive path reads it. The same rule governs the solid solver’s own scratch fields (Sec. 1.3.2).

2. **Computational Performance:** A pure fluid simulation does not require evaluating spatial velocity gradients ($\nabla \mathbf{v}$), regularised completeness inversions ($\mathbf{L}_i = \mathbf{E}_i^{-1}$), or tensor stress rotations at every particle. By skipping these operations, DSPHSolver provides a lightweight and fast execution path.
3. **Inheritance Foundation:** HypoElasticSolver directly subclasses DSPHSolver. Rather than duplicating hydrodynamic code, HypoElasticSolver inherits its volume evolution, free-surface detection, particle shifting (PST), and adaptive time integration directly from DSPHSolver.

1.3 Computational Architecture

1.3.1 Taichi JIT Compilation and Memory Management

All computational kernels in sigmaSPH are JIT-compiled to native CUDA kernels via Taichi decorators (`@ti.kernel` and `@ti.func`). The solver achieves high memory throughput by enforcing structure-of-arrays (SoA) layout across all particle fields.

Neighbour interactions within the compact support radius h are evaluated through a linear-hashed cell table:

1. **Spatial Binning:** Space is partitioned into a regular grid of cell size h whose origin is the coordinate origin. The lattice is *unbounded*: each particle's position $\mathbf{x}_i$ maps to $\mathbf{c}_i = \lfloor \mathbf{x}_i/h \rfloor$ with no clamp and with negative indices permitted, so the grid extends as far as the material does and the simulation box plays no part in it.
2. **Cell Sorting:** A 32-bit linear Morton/hashed key is assigned to each particle. A GPU radix/counting sort reorders particle indices so that particles occupying the same spatial cell reside in contiguous blocks in device memory.
3. **Prefix Sum Cell Pointers:** Parallel prefix scans build table offsets, allowing each thread executing a neighbour search to query only the 27 neighbouring cells (or 9 cells in 2D) using direct contiguous index ranges.
4. **Mirror Image Sweeps:** Where an axis of revolution or a Cartesian symmetry plane is declared, the traversal appends one further sweep per cut over the single cell row lying against it, and a third over the cell column against both, visiting the reflections of the stored particles in place of neighbours that the reduced model does not hold (Sec. 2.9.1). The candidates of these sweeps are filtered by the cell into which they were *binned*, never by a cell re-derived from a live position, because particle coordinates move during the step while the sort does not.

TRAP / OPERATIONAL NOTE: `Domain.end` DOES NOT BOUND THE NEIGHBOUR SEARCH

The clamp is a property of `pos_to_index_binned`, and *nothing calls that function*. The grid build, the neighbour traversal and the shifting operator all take the **unclamped** cell index and hash it, so the cell lattice is unbounded and `grid_num` — hence `Domain.end` — constrains nothing. Under `boundary: "open"` the box is a view setting: it sizes the camera and the drawn domain outline, and nothing else. A deck may omit `Domain.start` and `Domain.end` altogether, in which case they default to `[0, 0, 0]–[1, 1, 1]` and only the framing suffers. Under `boundary: "reflect"` the box is of course physical, because it is where the walls are placed.

This is recorded as a trap because the opposite was believed, written down, and acted on. The clamp exists, its docstring still describes the dense-grid write it once guarded, and

reading that docstring as current is enough to conclude that a particle beyond the box stops interacting. Measured directly it does not: 2000 particles laid over $x \in [20.05, 29.95]$, run once in a box containing them and once in a box of $[0.4, 0.4, 0.45]$ — fifty times too small, so that every particle lies outside it — give **identical neighbour counts for every particle**, mean 24.158 and minimum 8, while `grid_num` collapses from $[134, 134, 2]$ to $[2, 2, 2]$.

The evidence originally offered for the opposite conclusion was an energy drift of +6.89% on a small box against +7.00... + 7.17% over three large-box runs — a difference inside the run-to-run spread, and labelled as such at the time. A quantity at its own noise floor cannot establish a mechanism, and a claim about what the neighbour search does is settled by counting neighbours.

Sorting occurs at every time step, re-indexing up to 80% of particles. Consequently, all state variables that survive across time steps (including positions, velocities, stresses, plastic strains, internal energy, and shifting vectors $\delta\mathbf{r}$) are allocated in `ParticleSystem._alloc_sorted()` to ensure their memory addresses remain strictly synchronized with the sorted particle permutor. A field that holds state across the sort and is *not* registered there keeps its value at a slot that now belongs to a different particle; nothing fails, and the run continues on state attached to the wrong particles. Test T13 therefore walks every per-particle field of the particle system and requires it to be either registered or declared as derived, that is, rewritten for every particle before it is read.

1.3.2 Per-Particle Memory

The solver's reach in particle count is set by the bytes it holds per particle, and the layout follows three rules.

1. **Material constants live per material, not per particle.** The reference density ρ_0 , the bulk and shear moduli K and G , the P-wave and bulk sound speeds c_p and c_0 , the initial yield stress σ_{y0} , the hardening modulus H , the specific heat C_v , the reference temperature T_0 and the reference specific energy e_0 are functions of the material alone and are never written after initialisation. They are held once per material in the table `mat_tab`, beside the equation-of-state table `eos_tab` and the plasticity table `plastic_tab`, and a particle reaches all three through its integer index `eos_id`, which is sorted with it. A lookup is an indirect load in place of a load from the particle's own slot; what it saves is 40 bytes per particle of storage, which the sort would otherwise permute every step.
2. **The sort permutes through one scratch buffer per field type.** The counting sort cannot permute a field in place, because a thread writing a particle's new slot may overwrite a value another thread has still to read. Each registered field is therefore scattered into a scratch buffer at the new index and copied back. The scratch buffers are shared: there is one for each combination of data type and shape (scalar and 3-vector integers; scalar, 3-vector and 3×3 -matrix floats), and the fields of that type pass through it one after another within a single kernel launch. Holding the scratch per type rather than per field reduces it from the full size of the sorted state to 68 bytes per particle.
3. **A scratch field exists only when a feature that reads it is switched on.** The per-particle fields of the velocity limiters, of the adaptive hourglass measurement and of the super-time-stepped hourglass damper, and the eigenvector surface normal of `Shifting.normal: "eigvec"`, are allocated only when the corresponding option is set in the scene. A kernel that reads such a field does so under a compile-time condition on the same option, so that a disabled feature compiles to no access at all, and an access that escaped its condition fails at

compilation rather than reading unallocated memory. The regularised inverse $\mathbf{L}_i$ of Sec. 2.3, which `compute_lam` carries from its first neighbour pass to its second, is symmetric to the last bit, and it is stored as the six entries of its upper triangle.

On the three-dimensional quarter-symmetry hypervelocity-impact deck (1 524 803 particles) the field walk of `tools/material_table_bench.py` counts 396 bytes per particle in per-particle fields, plus 33 bytes per particle for the neighbour grid, or 654 MB in total. None of the three rules changes a computed value: each was verified bit for bit against the preceding code on a single CPU thread with IEEE-exact compilation.

1.4 Simulation Timestep Execution Lifecycles

This section details the exact numerical sequence executed at each simulation time step, contrasting the pure fluid lifecycle (`DSPHSolver`) against the solid and multi-material lifecycle (`HypoElasticSolver`).

1.4.1 Pure Fluid Timestep Lifecycle (`DSPHSolver`)

For pure fluid simulations executing under `Solver.type: "deltaPlusSPH"`, the state is integrated through `DSPHSolver.step()` and `DSPHSolver.substep()` across ten discrete phases:

1. **Particle Sorting and Cell Hashing (`initialize_particle_system`):** Computes 3D cell hash IDs via `update_grid_id()`, *records the cell each particle is binned into* in `cell_id`, executes a parallel prefix scan to construct cell bucket offsets, and performs a GPU counting sort to reorder all registered hydrodynamic arrays contiguously in memory. The stencil sweeps filter candidates against `cell_id` rather than against a cell re-derived from the live position, because the sort runs once per step while advection and shifting move particles afterwards.
2. **Free-Surface Detection (`compute_lam`):** Accumulates the first-order completeness matrix $\mathbf{E}_i = \sum_j V_j (\mathbf{x}_j - \mathbf{x}_i) \otimes \nabla W_{ij}$. Inverts $\mathbf{E}_i$ via the regularised eigenspace operator to obtain $\mathbf{L}_i$ and the minimum eigenvalue $\lambda_i = \min \text{eig}(\mathbf{E}_i)$. Evaluates the surface normal gradient $\langle \nabla \lambda_i \rangle$ via $\mathbf{L}_i$.
3. **Continuity and Volume Evolution (`compute_volume_evolution`):** Evaluates the volumetric strain rate:

$$\dot{\epsilon}_{\text{vol},i} = \sum_j V_j (\hat{\mathbf{v}}_i - \hat{\mathbf{v}}_j) \cdot \nabla W_{ij} + \frac{1}{\rho_i} \sum_j V_j (\rho_j \delta \mathbf{u}_j - \rho_i \delta \mathbf{u}_i) \cdot \nabla W_{ij} + \frac{\mathcal{D}_i}{\rho_i}, \quad (1.2)$$

where $\mathcal{D}_i$ is the Molteni & Colagrossi δ -SPH diffusion. Advances particle volume $V_i \leftarrow V_i(1 - \Delta t \dot{\epsilon}_{\text{vol},i})$ and recovers mass density $\rho_i = m_i/V_i$. Also accumulates the diffusion work W_D of Eq. (2.70); note that the pressure multiplying $\mathcal{D}_i$ there is the value left by the *previous* step, since P_i is not written until phase 5.

4. **Body Forces (`compute_non_pressure_forces`):** Initializes particle accelerations with gravitational body force $\mathbf{a}_i = \mathbf{g}$.
5. **Pressure Accelerations (`compute_pressure_accelerations`):** Accumulates symmetric volume-based pressure accelerations $\mathbf{a}_{p,i} = -\frac{1}{m_i} \sum_j V_i V_j (P_i + P_j) \nabla W_{ij}$. Computes the hydrostatic pressure $P_i = \frac{c_0^2 \rho_0}{\gamma} [(\rho_i/\rho_0)^\gamma - 1]$ from the current density, then adds Morris physical shear viscosity, optional γ -SPH low-Mach flux momentum corrections, and the ALE convective momentum transport term $\frac{1}{\rho_i} [\nabla \cdot (\rho \mathbf{v} \otimes \delta \mathbf{u}) - \mathbf{v}_i \nabla \cdot (\rho \delta \mathbf{u})]$. Accumulates the per-particle viscous power $\mathbf{v}_i \cdot \mathbf{a}_{\text{visc},i}$ for the energy budget.

6. **Adaptive Timestep Sizing (`compute_adaptive_dt`):** Sizes Δt from acoustic CFL ($\Delta t_{\text{acoustic}} = \text{CFL} \cdot h / \max_{ij} c_{ij}$, where c_{ij} is evaluated from neighbour finite differences $\sqrt{|\Delta P|/|\Delta \rho|}$ floored at c_0) and force CFL ($\Delta t_{\text{force}} = \text{CFL} \cdot \sqrt{h / \max_i \|\mathbf{a}_i\|}$). Sets $\Delta t = \min(\Delta t_{\text{acoustic}}, \Delta t_{\text{force}})$.
7. **Energy Budget Banking (`accumulate_step_work`):** Drains the accumulated viscous power into W_{visc} and banks the symplectic discrete leftover $\frac{1}{2}m_i\Delta t^2(\|\mathbf{f}_i\|^2 - \|\mathbf{g}\|^2)$, $\mathbf{f}_i = \mathbf{a}_i - \mathbf{g}$. **Its position is load-bearing:** it must follow phase 6 so that the banked powers meet the Δt the advection will take, and precede phase 8 so that the velocities and accelerations they were built from still exist (Sec. 2.10.3).
8. **Lagrangian Kinematic Advection (`advect`):** Advances particle velocity $\mathbf{v}_i \leftarrow \mathbf{v}_i + \Delta t \cdot \mathbf{a}_i$ and position $\mathbf{x}_i \leftarrow \mathbf{x}_i + \Delta t \cdot \mathbf{v}_i$ via symplectic Euler integration.
9. **Particle Shifting Technology (`apply_pst`):** Computes shifting displacement $\delta \mathbf{r}_i$ proportional to $\text{CFL} \cdot M \mathbf{a}_i \cdot h^2 \sum_j \nabla W_{ij} V_j$. Clamps displacement to $|\delta \mathbf{r}_i| \leq \text{PST_MAX_SHIFT} \cdot \Delta x_0$. Zeroes shifting in strong free-surface regions ($\lambda_i < \text{LO}$) and projects tangentially along $\langle \nabla \lambda_i \rangle$ in intermediate surface bands ($\text{LO} \leq \lambda_i < \text{HI}$). Applies position correction $\mathbf{x}_i \leftarrow \mathbf{x}_i + \delta \mathbf{r}_i$ and preserves $\delta \mathbf{r}_i$ as `pst_shift` for the subsequent step's ALE mass flux and momentum transport.
10. **Kinematic Boundary Enforcement:** Applies boundary reflections: in axisymmetry, reflects particles crossing the axis $r = 0$ ($y \rightarrow |y|, v_y \rightarrow -v_y$); under `Domain.symmetryPlanes`, reflects particles that have crossed a cut plane in the same manner and one plane at a time, carrying the velocity, the acceleration, the deviator and $\mathbf{F}$ through the corresponding mirror operator, and doing so *before* the wall clamp (Sec. 2.9); in plane strain, zeroes z -velocities and pins z -coordinates to layer center; enforces reflective bounding walls if configured, at coefficient of restitution `Domain.restitution`, accumulating the removed $\frac{1}{2}m(1 - c_f^2)v_n^2$ into the wall-dissipation channel.

1.4.2 Solid and Multi-Material Timestep Lifecycle (`HypoElasticSolver`)

For solid and multi-material simulations executing under `Solver.type: "hypoElastic"`, each time step integrates through `HypoElasticSolver.step()` and `HypoElasticSolver.substep()` across seventeen discrete phases:

1. **Boundary Load Factor Ramping (`step`):** Advances boundary displacement scale factor $s(t) = 3(t/T_{\text{ramp}})^2 - 2(t/T_{\text{ramp}})^3$ for smooth, non-shocking kinematic loading.
2. **Full-Field Particle Sorting (`initialize_particle_system`):** Hashes cell coordinates and executes GPU counting sort across all hydrodynamic and solid state fields, including deviatoric stress $\mathbf{s}$, deformation gradient $\mathbf{F}$, equivalent plastic strain $\bar{\epsilon}^p$, displacement boundary flags, and erosion state.
3. **Completeness Tensor Inversion (`compute_lam`):** Evaluates first-order completeness matrix $\mathbf{E}_i$ and regularised inverse $\mathbf{L}_i$ along principal eigendirections.
4. **Velocity Gradient Evaluation (`compute_velocity_gradient`):** Calculates spatial velocity gradient $\nabla \mathbf{v}_i = [\sum_j V_j(\mathbf{v}_j - \mathbf{v}_i) \otimes \nabla W_{ij}] \mathbf{L}_i$. In 2D axisymmetry, populates hoop strain rate $\dot{\epsilon}_{\theta\theta} = v_r/r$ directly into the diagonal $(\nabla \mathbf{v})_{33}$ slot.
5. **Constitutive Deviatoric Stress and Plasticity Update (`update_stress`):** Computes strain rate $\dot{\epsilon} = \text{sym}(\nabla \mathbf{v})$ and spin tensor $\boldsymbol{\Omega} = \text{skew}(\nabla \mathbf{v})$. Evaluates Jaumann objective stress rate $\dot{\mathbf{s}}_i = 2G_i \text{dev}(\dot{\epsilon}_i) + \boldsymbol{\Omega}_i \cdot \mathbf{s}_i - \mathbf{s}_i \cdot \boldsymbol{\Omega}_i$, deformation gradient rate $\dot{\mathbf{F}} = (\nabla \mathbf{v})\mathbf{F}$, and ALE convective transport of deviatoric stress $(\delta \mathbf{u} \cdot \nabla)\mathbf{s}$ and equivalent plastic strain $(\delta \mathbf{u} \cdot \nabla)\bar{\epsilon}^p$.

Integrates trial stress $\mathbf{s}_{\text{trial}} = \mathbf{s} + \Delta t \dot{\mathbf{s}}$ and re-projects onto the trace-free symmetric subspace. Explicitly zeroes $\mathbf{s}_{\text{trial}}$ for fluid particles ($G_i \leq 0$). Executes J_2 radial-return map for yielding solid particles ($G_i > 0, \sigma_{\text{vM}} > \sigma_y$).

6. **Specific Internal Energy Evolution (update_internal_energy):** Integrates $e_{\text{int},i} \leftarrow e_{\text{int},i} + \Delta t \dot{e}_i + \Delta t (\delta \mathbf{u} \cdot \nabla) e_i$, where $\dot{e}_i$ is the work conjugate of the pairwise forces from the preceding step, plus the upwind-equivalent diffusion of Sec. 2.5.1. **Does not** subtract the symplectic discrete leftover from e_{int} : that artefact is banked into a reservoir of its own and enters the total negatively, which is what leaves e_{int} a physical internal energy and lets Eq. (2.80) take a temperature from it. e_{int} **is not clamped** either, for any material: the negative-energy guard lives at the equation of state instead (Sec. 2.11.3). On a material carrying a specific heat, the temperature is updated here too, immediately after the energy. This phase is also the solid solver's energy-budget banking point — hourglass work, viscous work, the raw leftover and the axisymmetric measure gaps are all accumulated here, because this is where Δt still holds the value the work was done over (Sec. 2.10.3).
7. **Continuity and Volume Evolution (compute_volume_evolution):** Advances volume $V_i \leftarrow V_i(1 - \Delta t \dot{\epsilon}_{\text{vol},i})$ using material-gated δ -SPH diffusion (preventing artificial mass diffusion across material interfaces). Density is updated as $\rho_i = m_i/V_i$.
8. **Deformation Limiting and Particle Erosion (apply_erosion):** Measures volume ratio $J_i = V_i \rho_{0,i}/m_i$. In mode "cap", clamps $J_i \in [J_{\min}, J_{\max}]$. In mode "erode", marks severely distorted particles ($J_i \notin [J_{\min}, J_{\max}]$ or $\bar{\epsilon}^p > \bar{\epsilon}_{\max}^p$) as eroded, excluding them from future neighbour sums.
9. **Programmed Burn Front Kinematics (update_burn):** For high explosives, evaluates Huygens lighting time $t_{l,i} = \|\mathbf{x}_{0,i} - \mathbf{x}_{\text{det}}\|/D$ and updates reaction progress fraction $f_{\text{burn},i} \in [0, 1]$.
10. **Body Forces (compute_non_pressure_forces):** Initializes acceleration with gravity $\mathbf{a}_i = \mathbf{g}$.
11. **Stress Accelerations and Artificial Dissipation (compute_stress_accelerations):** Evaluates thermodynamic pressure P_i from the particle's own EOS row (Linear, Tait, JWL, polynomial Mie–Grüneisen, or the Holmquist–Johnson–Cook compaction curve, which reads the particle's $\mu_{\max}$ and damage), reading the specific internal energy through the $\max(e_{\text{int}}, 0)$ guard where the equation of state needs it. Adds quadratic von Neumann–Richtmyer bulk viscosity q_i . Computes Cauchy stress divergence $\mathbf{a}_i + = \frac{1}{m_i} \sum_j V_i V_j (\boldsymbol{\sigma}_i + \boldsymbol{\sigma}_j) \nabla W_{ij}$. Adds Monaghan pairwise artificial viscosity Π_{ij} for approaching pairs, line-of-centres rate hourglass damping $\mathbf{f}_{h,ij}$, and axisymmetric cylindrical momentum source terms. Accumulates work rate $\dot{e}_{\text{int}}$ conjugate to the pairwise forces.
12. **Erosion Freezing (freeze_eroded):** Sets velocity and acceleration to zero for eroded particles, ensuring they remain inert.
13. **Adaptive Time-Step Clamping Suite (compute_adaptive_dt):** Sizes baseline Δt from dilatational P-wave speed $c_p = \sqrt{(K + 4G/3)/\rho_0}$ and maximum acceleration. Applies consecutive stability clamps, each of which can only lower Δt further: `clamp_dt_jwl()` for energetic products (first, because their sound speed swings by an order of magnitude over a run), `clamp_dt_mie_gruneisen()` for the shock equation of state, `clamp_dt_hjc()` for the stiffening fully dense branch of the concrete compaction curve, `clamp_dt_bulk_viscosity()` for quadratic bulk viscosity, `clamp_dt_monaghan()` for pairwise artificial viscosity, and finally one of the hourglass caps — `clamp_dt_hourglass()` by default (rate $\cdot \Delta t \leq 1.6$), or the adaptive, saturating, or super-time-stepped variant.

14. **Kinematic Advection and Super-Time-Stepping (STS):** Advances velocity and position via symplectic Euler advection. When STS hourglass sub-stepping is enabled, the step is split as half-kick, sub-cycled (or Runge–Kutta–Legendre) damping stages, second half-kick, and position drift. The damper thereby acts on the synchronised velocity (Sec. 2.7.3). The damping stages are bracketed by kinetic-energy snapshots so that the hourglass work remains measurable, and the damper–kick cross term is banked with the integrator leftover (Sec. 2.11).
15. **Non-Affine Velocity Limiters (`limit_velocity`):** Predicts affine velocity, detects run-away non-affine outlier velocities, applies dissipative braking clamp, and redistributes shed momentum to neighbouring unconstrained particles to maintain exact linear momentum conservation. Enforces hard speed cap $\|\mathbf{v}_i\| \leq v_{\text{cap}}$.
16. **Displacement Boundary Constraints (`apply_constraints`):** Overwrites coordinates and velocities of grip particles with prescribed motion $\mathbf{x} = \mathbf{x}_0 + \Delta\mathbf{x}_{\text{disp}}s(t)$ and $\mathbf{v} = \Delta\mathbf{x}_{\text{disp}}\dot{s}(t)$.
17. **Particle Shifting Technology (`apply_pst`):** Calculates shift displacement $\delta\mathbf{r}_i$, projects tangentially along free surfaces, skips displacement-constrained boundary particles, updates coordinates $\mathbf{x}_i \leftarrow \mathbf{x}_i + \delta\mathbf{r}_i$, and stores $\delta\mathbf{r}_i$ for subsequent ALE transport.
18. **Boundary Reflections and Simulation Clock:** Enforces axisymmetry radial reflections, symmetry-plane fold-backs, and domain wall bounces, in that order; advances physical simulation clock $\text{sim_time} \leftarrow \text{sim_time} + \Delta t$.

1.5 Execution and Workflow

1.5.1 Interactive Execution

Interactive simulations with real-time 3D GGUI visualization are executed via the command line:

```
mamba run -n taichi python3 run_simulation.py --scene_file data/scenes/
↳ dambreak2d_dsph.json
```

The window opens only after the first status line has been printed, not at start-up. The reason is Taichi's just-in-time compilation: the step kernels are compiled inside the first call to `solver.step()`, and on the 3D quarter hypervelocity deck that first step takes 4.3 s with a warm offline cache, against 0.62 s for every later render cycle, and tens of seconds on a cold cache after a kernel source has been edited. A window opened before that interval would stay empty for all of it, which cannot be told apart from a crash. The runner therefore prints **compiling the solver kernels ...** in both modes, reports the first cycle's time with the compilation included, and opens the window afterwards.

TRAP / OPERATIONAL NOTE: THE REPORTED WALL CLOCK INCLUDES KERNEL COMPILATION

The run's wall clock starts after `initialize()`, which keeps the set-up kernels out of it, but it does *not* keep out the compilation of the step kernels. The `[wall ...]` figure on step 0 and the closing `[WALL CLOCK]` both contain it. On a short run this is a noticeable share of the total, and a comparison of two code versions by wall clock carries the difference in their compilation times, which depends on whether the offline cache was warm. The cost of a step is measured with the compilation excluded, as `tools/material_table_bench.py bench` does by timing a fixed window after a warm-up.

1.5.2 Headless Execution

The same deck runs without a window under `--no-gui`, which suppresses the GGUI window, the camera and the vis buffers and changes nothing else. The render-cycle loop, the adaptive timestep, the status line, the trajectory output and the closing energy and momentum summary are all identical, so this is the form to use on a machine with no display:

```
mamba run -n taichi python3 run_simulation.py \
    --scene_file data/scenes/dambreak2d_dsph.json --no-gui
```

Two consequences follow. `Output.frame` writes PNG screenshots *of the window* and is therefore refused rather than silently ignored, since a deck that requested frames and received none would otherwise be indistinguishable from a run that produced no output at all. And the run terminates only at `Time.duration` or `Time.maxSteps`, there being no window to close.

The saving is real but modest, because the GPU solve dominates the render. Measured once each way, the axisymmetric hypervelocity deck (18 332 particles) runs in 8.13 s with the window against 7.47 s without it, and the 3D quarter deck (190 669 particles) in 39.25 s against 36.74 s — about 8% and 6%. These are single runs and carry the run-to-run spread that Sec. 2.9.6 describes, the step counts having differed by 1 203 against 1 209 and 526 against 522, so the figure to carry away is “a few per cent”.

1.5.3 Direct Solver Instantiation for Verification

For automated testing, regression suites, and analysis scripts that want the solver alone — no output files, no summary, no render-cycle loop — the system can be constructed directly. This is what the verification modules and the scripts under `tools/` do:

```
import taichi as ti
ti.init(arch=ti.cuda)
from config_builder import SimConfig
from particle_system import ParticleSystem

# Load scene configuration and construct solver
cfg = SimConfig('data/scenes/tensile2d_elastic.json')
ps = ParticleSystem(cfg, GGUI=False)
solver = ps.build_solver()
solver.initialize()

# Run explicit integration steps
for step in range(100):
    solver.step()
```

1.5.4 Verification Test Suite

The codebase includes a comprehensive test suite containing 38 test modules and 527 discrete verification checks (`tests/run_tests.py`). Tests can be executed in full or targeted by module:

```
# Full regression suite (250-640 s)
mamba run -n taichi python3 tests/run_tests.py

# Targeted modules: PST stability and Jaumann rate
mamba run -n taichi python3 tests/run_tests.py t1 t11
```

2 Theoretical Formulations and Discrete Operators

2.1 Continuum Foundations and Kernel Approximations

Smoothed Particle Hydrodynamics discretises a continuum domain through a finite set of interpolating particles carrying mass m_i , volume V_i , density $\rho_i = m_i/V_i$, momentum, and constitutive state. The spatial value of an arbitrary field $f(\mathbf{x})$ and its gradient $\nabla f(\mathbf{x})$ are reconstructed using a symmetric, smoothly decaying smoothing kernel $W(\mathbf{x} - \mathbf{x}', h)$ with compact support radius h :

$$\langle f(\mathbf{x}_i) \rangle = \sum_j V_j f_j W_{ij}, \quad (2.1)$$

$$\langle \nabla f(\mathbf{x}_i) \rangle = \sum_j V_j (f_j - f_i) \otimes \nabla W_{ij}, \quad (2.2)$$

where $W_{ij} = W(\|\mathbf{x}_i - \mathbf{x}_j\|, h)$ and $\nabla W_{ij} = \nabla_{\mathbf{x}_i} W(\|\mathbf{x}_i - \mathbf{x}_j\|, h)$.

2.1.1 Wendland C^2 Smoothing Kernel

sigmaSPH employs the Wendland C^2 quintic spline kernel with compact support on $[0, h]$. Defining the normalised pair distance $q = \|\mathbf{x}_i - \mathbf{x}_j\|/h$:

$$W(r, h) = \alpha_D (1 - q)^4 (4q + 1), \quad 0 \leq q < 1, \quad (2.3)$$

with $W(r, h) = 0$ for $q \geq 1$. The normalisation constant α_D depends on the spatial dimension:

$$\alpha_D = \begin{cases} \frac{7}{\pi h^2}, & \text{in 2D (plane strain and axisymmetric meridional plane),} \\ \frac{21}{2\pi h^3}, & \text{in 3D Cartesian.} \end{cases} \quad (2.4)$$

The spatial gradient evaluates to:

$$\nabla W_{ij} = -\frac{20\alpha_D}{h} q(1 - q)^3 \hat{\mathbf{r}}_{ij}, \quad \hat{\mathbf{r}}_{ij} = \frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}. \quad (2.5)$$

2.2 Volume Evolution and Continuity

Rather than evolving density directly through the standard continuity equation $\dot{\rho} = -\rho \nabla \cdot \mathbf{v}$, **sigmaSPH** tracks particle volume V_i as the primary kinematic scalar. Density is recovered identically from reference mass conservation:

$$\rho_i = \frac{m_i}{V_i}. \quad (2.6)$$

The volume rate of change is driven by the volumetric strain rate $\dot{\varepsilon}_{\text{vol},i} \equiv -\dot{V}_i/V_i$:

$$\frac{dV_i}{dt} = -V_i \left(\frac{d\varepsilon_{\text{vol},i}}{dt} \right). \quad (2.7)$$

The total discrete volumetric strain rate accounts for symmetric velocity divergence, shifting-induced ALE mass flux, and numerical density diffusion:

$$\left(\frac{d\varepsilon_{\text{vol},i}}{dt}\right) = \sum_j V_j (\hat{\mathbf{v}}_i - \hat{\mathbf{v}}_j) \cdot \nabla W_{ij} + \frac{1}{\rho_i} \sum_j V_j (\rho_j \delta \mathbf{u}_j - \rho_i \delta \mathbf{u}_i) \cdot \nabla W_{ij} + \frac{\mathcal{D}_i}{\rho_i}, \quad (2.8)$$

where $\hat{\mathbf{v}} = \mathbf{v} + \delta \mathbf{u}$ represents the total particle advection velocity including shifting velocity $\delta \mathbf{u} = \delta \mathbf{r} / \Delta t$. The second term in Eq. (2.8) represents the conservative ALE mass flux across the shifted node network and is mandatory to prevent artificial density accumulation.

2.2.1 δ -SPH Density Diffusion

To suppress high-frequency acoustic noise without damping physical bulk motion, the Molteni & Colagrossi (2009) δ -SPH diffusion term $\mathcal{D}_i$ is included in the continuity equation:

$$\mathcal{D}_i = \delta h c_0 \sum_j \frac{m_j}{\rho_j} \psi_{ij} \cdot \nabla W_{ij}, \quad \text{with} \quad \psi_{ij} = 2(\rho_j - \rho_i) \frac{\mathbf{x}_j - \mathbf{x}_i}{\|\mathbf{x}_i - \mathbf{x}_j\|^2 + 0.01h^2}. \quad (2.9)$$

Notice that $\mathcal{D}_i$ is normalised by the local particle density ρ_i in Eq. (2.8), which maintains consistency with the volumetric accumulator.

The pair term carries one further factor, w_{ij}^{crack} , which is 1 everywhere except across a cracked bond of a concrete with a Grady–Kipp tension card, where it is 0 (§3.4.5). There, $\rho_j - \rho_i$ is the density jump of an open crack and not a discretisation error, and smoothing it would dilate the intact crack faces into tension.

2.3 Free-Surface Detection and Regularised Completeness Matrix Inversion

The spatial velocity gradient $\nabla \mathbf{v}$ and surface normal reconstructions rely on the first-order completeness matrix (also termed the kernel renormalisation tensor) $\mathbf{E}_i$:

$$\mathbf{E}_i = \sum_j V_j (\mathbf{x}_j - \mathbf{x}_i) \otimes \nabla W_{ij}. \quad (2.10)$$

The matrix inverse $\mathbf{L}_i \approx \mathbf{E}_i^{-1}$ restores first-order linear completeness, ensuring that the gradient of any linear affine field is reproduced exactly: $\langle \nabla(\mathbf{A}\mathbf{x} + \mathbf{b}) \rangle = \mathbf{A}$.

2.3.1 Symmetric Positive Semi-Definite (SPSD) Character of $\mathbf{E}_i$

Because the Wendland kernel gradient ∇W_{ij} acts strictly along the unit radial vector separating particle pairs:

$$\nabla W_{ij} = -F(r_{ij})(\mathbf{x}_i - \mathbf{x}_j) = F(r_{ij})(\mathbf{x}_j - \mathbf{x}_i), \quad \text{with} \quad F(r_{ij}) = \frac{20\alpha_D}{h^2}(1 - q)^3 \geq 0, \quad (2.11)$$

every term in the neighbour summation forms a positive dyadic tensor product:

$$(\mathbf{x}_j - \mathbf{x}_i) \otimes \nabla W_{ij} = F(r_{ij})(\mathbf{x}_j - \mathbf{x}_i) \otimes (\mathbf{x}_j - \mathbf{x}_i). \quad (2.12)$$

Since particle volumes $V_j > 0$ and $F(r_{ij}) \geq 0$, $\mathbf{E}_i$ is symmetric positive semi-definite (SPSD) by construction.

2.3.2 Physical Interpretation of the Spectrum

The spectral decomposition of $\mathbf{E}_i$ into orthonormal eigenvectors $\mathbf{n}_k$ and eigenvalues e_k :

$$\mathbf{E}_i = \sum_k e_k \mathbf{n}_k \otimes \mathbf{n}_k, \quad e_k \geq 0, \quad (2.13)$$

provides an explicit directional measure of neighbourhood spatial resolution:

- **Well-Resolved Interior** ($e_k \sim 1$): When particle i is completely surrounded by neighbours in all directions, $\mathbf{E}_i \approx \mathbf{I}$. On a regular Cartesian lattice, the eigenvalues plateau at ≈ 0.977 for support radius $h = 2\Delta x_0$, and at ≈ 0.997 for $h = 3\Delta x_0$.
- **Directional Starvation** ($e_k \rightarrow 0$): When neighbours are absent along direction $\mathbf{n}_k$ —such as across a free surface (where the normal direction lacks neighbours), at an advancing crack tip, along a single-particle ligament, or when neighbours are collinear (in 2D) or coplanar (in 3D)—the associated eigenvalue collapses toward zero: $e_k \rightarrow 0$.

2.3.3 Free-Surface Detection via the λ Field

The minimum eigenvalue of $\mathbf{E}_i$ serves as an intrinsic indicator of geometric boundary proximity:

$$\lambda_i \equiv \lambda_{\min}(\mathbf{E}_i) = \min_k e_k. \quad (2.14)$$

Because the interior eigenvalue plateaus at ≈ 0.977 ($2\Delta x_0$) or ≈ 0.997 ($3\Delta x_0$), boundary detection thresholds are defined relative to these limits:

- **Bulk Interior** ($\lambda_i \geq \lambda_{\text{hi}}$): Fully resolved continuum interior. Full particle shifting and uninhibited gradient correction are active.
- **Ambiguous Transition Band** ($\lambda_{\text{lo}} \leq \lambda_i < \lambda_{\text{hi}}$): Boundary layer where particle shifting is restricted to the local tangent plane.
- **True Free Surface** ($\lambda_i < \lambda_{\text{lo}}$): Boundary particles lacking neighbours on one side. Particle shifting is deactivated ($\delta \mathbf{r}_i = \mathbf{0}$) to prevent spurious particle shedding into the void.

Default thresholds are $\lambda_{\text{lo}} = 0.40$, $\lambda_{\text{hi}} = 0.75$ under Sun shifting, and $0.20/0.75$ under Collé shifting. The unit outward surface normal vector is reconstructed from the gradient of the eigenvalue field:

$$\hat{\mathbf{n}}_i = \frac{\nabla \lambda_i}{\|\nabla \lambda_i\|}, \quad \text{where} \quad \nabla \lambda_i = \sum_j V_j (\lambda_j - \lambda_i) \nabla W_{ij}. \quad (2.15)$$

2.3.4 The Inversion Dilemma: Direction-Wise Fallback vs. Scalar Clamping

Evaluating the inverse $\mathbf{L}_i = \mathbf{E}_i^{-1}$ is essential for correcting velocity gradients, but presents a severe numerical hazard:

1. **Unbounded Amplification of Naive Inversion:** Direct inversion yields $\mathbf{E}_i^{-1} = \sum_k \frac{1}{e_k} \mathbf{n}_k \otimes \mathbf{n}_k$. As $e_k \rightarrow 0$, $1/e_k \rightarrow \infty$. A naive inverse applies the largest amplification factor precisely in the direction where there is the least physical data, blowing up noise at boundaries and crack tips.
2. **Deficiency of Global Binary Fallback:** A crude check (e.g., “if $\det(\mathbf{E}) < \text{tol}$, fall back to $\mathbf{L} = \mathbf{I}$ ”, as in Collé et al. 2019) discards valid corrections in well-resolved directions when only one direction is starved. For example, along a flat horizontal free surface, the horizontal tangential direction is fully populated with $e_1 \approx 0.99$, while the vertical normal is starved ($e_2 \approx 0.499$). Discarding $\mathbf{L}$ entirely corrupts tangential velocity gradients where resolution is abundant.

2.3.5 Spectral Regularisation Formulation

To solve this dilemma, `sigmaSPH` implements a per-eigendirection regularised inverse in `spd_inverse.py`. Each eigendirection is scaled independently by a regularised weight $w(e_k, \text{tol})$:

$$\mathbf{L}_i = \sum_k w(e_k, \text{tol}) \mathbf{n}_k \otimes \mathbf{n}_k. \quad (2.16)$$

The weight function $w(e, \text{tol})$ smoothly transitions between the exact inverse $1/e$ (in well-resolved directions) and the uncorrected weight 1 (in starved directions) using a cubic Hermite smoothstep polynomial:

$$u = \min \left(1.0, \max \left(0.0, \frac{e - \text{tol}}{\text{tol}} \right) \right), \quad (2.17)$$

$$\beta(u) = u^2(3 - 2u), \quad (2.18)$$

$$w(e, \text{tol}) = \frac{\beta(u)}{\max(e, \text{tol})} + (1.0 - \beta(u)). \quad (2.19)$$

TRAP / OPERATIONAL NOTE: Mathematical Properties of the Regularised Inverse

The weight function $w(e, \text{tol})$ in Eq. (2.19) guarantees two critical numerical stability properties:

- **Full Completeness where Resolved:** When $e \geq 2 \cdot \text{tol}$, $u = 1$, $\beta = 1$, and $w(e) = 1/e$. The scheme achieves exact first-order completeness.
- **Graceful Fallback to Bare Gradient:** When $e \leq \text{tol}$, $u = 0$, $\beta = 0$, and $w(e) = 1$. The correction drops to identity, falling back to the uncorrected bare kernel gradient.
- **Strict Upper Bound:** For all $e \geq 0$, $w(e, \text{tol}) \leq 1/\text{tol}$. Geometric degeneracy can never cause singular amplification.

2.3.6 Numerically Stable 2D Closed-Form Eigensolver

In 2D plane strain or axisymmetry, $\mathbf{E}_i$ is a 2×2 symmetric tensor:

$$\mathbf{E} = \begin{bmatrix} a & b \\ b & d \end{bmatrix}, \quad \text{with } b = \frac{1}{2}(E_{01} + E_{10}). \quad (2.20)$$

The eigenvalues are solved analytically:

$$\text{disc} = \sqrt{\max((a - d)^2 + 4b^2, 0)}, \quad e_1 = \frac{a + d + \text{disc}}{2}, \quad e_2 = \frac{a + d - \text{disc}}{2}, \quad (2.21)$$

where $e_1 \geq e_2 \geq 0$.

Elimination of Catastrophic Cancellation: Directly computing the eigenvector of the smaller eigenvalue e_2 when $e_2 \approx 0$ suffers from catastrophic floating-point cancellation. The solver in `spd_inverse_2x2` instead constructs the unit eigenvector $\mathbf{n}_1$ of the *larger* eigenvalue e_1 , where candidate spanning vectors:

$$\mathbf{v}_1 = \begin{pmatrix} b \\ e_1 - a \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} e_1 - d \\ b \end{pmatrix}, \quad (2.22)$$

are well-separated. The candidate vector with the larger Euclidean norm is normalised:

$$\mathbf{n}_1 = \frac{\mathbf{v}^*}{\|\mathbf{v}^*\|}, \quad \mathbf{v}^* = \operatorname{argmax}_{\mathbf{v} \in \{\mathbf{v}_1, \mathbf{v}_2\}} \|\mathbf{v}\|. \quad (2.23)$$

The starved eigenvector $\mathbf{n}_2$ is then recovered by an exact 90° orthogonal rotation:

$$\mathbf{n}_2 = \begin{pmatrix} -n_{1,y} \\ n_{1,x} \end{pmatrix}. \quad (2.24)$$

This guarantees that $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$ to machine precision, regardless of how small e_2 becomes.

2.3.7 3D Formulation and Consumer-Specific Tolerances

In 3D, $\mathbf{E}_i$ is first symmetrised analytically: $\mathbf{E}_s = \frac{1}{2}(\mathbf{E} + \mathbf{E}^T)$ to remove single-precision floating-point accumulation asymmetry, before computing the full eigensystem using Taichi's CUDA kernel intrinsic `ti.sym_eig(Es)`.

Consumer-Specific Tolerances: The eigenvalue tolerance `tol` is tailored to the specific consumer:

- **Velocity Gradient Correction (`L_EIG_TOL = 0.10`):** Calibrated against the measured spectrum of $\mathbf{E}$ on a plane-strain lattice ($h = 3\Delta x_0$): bulk interior 0.997, flat free surface 0.499, 90° convex corner 0.254, and collinear neighbours $\rightarrow 0$. The tolerance of 0.10 sits strictly below the worst healthy geometry (0.254), preserving full correction at boundaries while preventing singular amplification in degenerate regions.
- **Surface Normal Reconstruction (`L_EIG_TOL_LAM = 0.02`):** Used in evaluating $\langle \nabla \lambda \rangle$. Because λ varies primarily in the surface-normal direction, amplification in the starved direction is physically meaningful; hence, a permissive threshold (0.02) is employed to catch only true numerical singularity.

2.4 Particle Shifting Technology (PST)

Particle Shifting Technology perturbs particle coordinates across streamlines to eliminate the tensile instability, anisotropic particle clustering, and grid pairing.

2.4.1 Sun et al. δ^+ -SPH Displacement Shifting

In the δ^+ -SPH formulation of Sun et al. [29], shifting is evaluated as an explicit displacement increment $\delta \mathbf{r}_i$:

$$\delta \mathbf{r}_i = -\text{CFL} \cdot \text{Ma}_i \cdot h^2 \sum_j \left[1 + R \left(\frac{W_{ij}}{W(\Delta x_0)} \right)^n \right] \nabla W_{ij} V_j, \quad (2.25)$$

where $\text{Ma}_i = \|\mathbf{v}_i\|/c_0$, and $V_j = 2m_j/(\rho_i + \rho_j)$. The parameters R and n govern pair repulsion (typically $R = 0.2$, $n = 4$).

To ensure stability in disordered regimes, the displacement is strictly bounded:

$$\|\delta \mathbf{r}_i\| \leq \text{PST_MAX_SHIFT} \cdot \Delta x_0 \quad (\text{default: } 0.05\Delta x_0). \quad (2.26)$$

2.4.2 Collé et al. γ -SPH-ALE Velocity Shifting

In the γ -SPH-ALE formulation of Collé et al. [10], shifting is evaluated as a shifting velocity $\delta \mathbf{v}_i$, ensuring that the advective shift scales continuously with physical time:

$$\delta \mathbf{v}_i = -2hc_0 \text{Ma}_i \sum_j \left[1 + R \left(\frac{W_{ij}}{W(\Delta x_0)} \right)^n \right] \nabla W_{ij} V_j, \quad \delta \mathbf{r}_i = \delta \mathbf{v}_i \Delta t. \quad (2.27)$$

This formulation is invariant under adaptive sub-cycling of Δt .

2.4.3 Free-Surface Boundary Treatment

At a free surface ($\lambda_i < \lambda_{lo}$), particle shifting is deactivated ($\delta \mathbf{r}_i = \mathbf{0}$) to prevent particles from spuriously evaporating into the void. In the intermediate zone ($\lambda_{lo} \leq \lambda_i < \lambda_{hi}$), shifting is projected onto the local tangent plane:

$$\delta \mathbf{r}_i \leftarrow \delta \mathbf{r}_i - (\delta \mathbf{r}_i \cdot \hat{\mathbf{n}}_i) \hat{\mathbf{n}}_i. \quad (2.28)$$

2.5 ALE Transport of Particle-Carried State

Because particle shifting moves particles relative to the underlying material velocity ($\hat{\mathbf{v}} = \mathbf{v} + \delta \mathbf{u}$), a shifted particle ceases to be a pure material point. It becomes an Arbitrary Lagrangian–Eulerian (ALE) computational node.

Any field ϕ carried by the particle must receive a convective transport correction to account for the relative mesh drift velocity $\delta \mathbf{u}$:

$$\left. \frac{d\phi}{dt} \right|_{\text{node}} = \frac{D\phi}{Dt} + (\delta \mathbf{u} \cdot \nabla) \phi. \quad (2.29)$$

To guarantee that the transport correction vanishes identically for a spatially uniform field ($\nabla \phi = \mathbf{0}$), sigmaSPH evaluates the transport term through the identity:

$$(\delta \mathbf{u} \cdot \nabla) \phi = \frac{1}{\rho} [\nabla \cdot (\rho \phi \delta \mathbf{u}) - \phi \nabla \cdot (\rho \delta \mathbf{u})]. \quad (2.30)$$

The discrete SPH increment evaluated over time step Δt (with $\delta \mathbf{r} = \delta \mathbf{u} \Delta t$) is:

$$\Delta \phi_i^{\text{ALE}} = \frac{1}{\rho_i} \sum_j V_j (\rho_j \phi_j - \rho_i \phi_i) [(\delta \mathbf{r}_i - \delta \mathbf{r}_j) \cdot \nabla W_{ij}]. \quad (2.31)$$

TRAP / OPERATIONAL NOTE: Mandatory ALE Transport of Particle Fields

Equation (2.31) is implemented across all persistent particle state fields:

- **Volume V (hence ρ):** Always active, as the second term of Eq. (2.8).
- **Velocity $\mathbf{v}$:** Activated by `Shifting.aleMomentum`. Prevents momentum generation.
- **Deviatoric Stress $\mathbf{s}$:** Activated by `Shifting.aleHistory`. Prevents spurious shear stresses.
- **Plastic Strain $\bar{\epsilon}_p$:** Clamped strictly to ≥ 0 . Prevents irreversible yield ratchet.
- **Internal Energy e_{int} :** Centred advection *plus* the upwind-equivalent diffusion of Sec. 2.5.1, optionally made conservative by the fix-up of Sec. 2.5.2. Not clamped; see Sec. 2.11.
- **Damage D and compaction history μ_{max} of the Holmquist–Johnson–Cook concrete:** Activated by `Shifting.aleHistory`, over same-material pairs only; D is clamped to $[0, 1]$ and μ_{max} to at least the current μ .

The deformation gradient $\mathbf{F}$ is deliberately *not* transported and therefore drifts from the shifted configuration. This is tolerable only because $\mathbf{F}$ is a pure diagnostic, read by no constitutive path; it corrupts the reported strain, not the solution. Being a diagnostic and nothing else, it is also not allocated or integrated at all unless the scene sets `Output.deformationGradient`. The burn fraction F_{burn} is likewise not transported,

because it is a prescribed function of $(\mathbf{x}_0, t)$ re-evaluated every step rather than a spatial field carried by material — transporting it would advect a clock. Omitting the transport term on a shifted solid causes the return map to artificially accumulate up to 1% plastic strain per step in uniform tension, driving premature failure.

2.5.1 Stabilising the ALE Transport of Internal Energy

Equation (2.31) is a *centred* difference operator, and centred advection integrated with one-stage explicit Euler is unconditionally unstable. For the deviatoric stress this is masked by the yield surface and for the plastic strain by the non-negativity clamp, but the internal energy is bounded by nothing, and the instability is therefore visible only there — where it is severe. On a strengthless free-surface deck the unstabilised operator grows $\sum_i m_i e_i$ exponentially in the *step number*, at a rate of 1.0012–1.0019 per step, reaching 10^{24} times the entire energy of the problem over 33,000 steps while the velocity field remains entirely ordinary.

That the growth rate is per *step* and not per unit *time* is the diagnostic: refining Δt at fixed physical time makes the outcome worse, which is the opposite of an integration error. A von Neumann analysis of the operator gives the amplification factor exactly,

$$|g| = \sqrt{1 + (\nu S)^2}, \quad \nu = \frac{\|\delta \mathbf{r}\|}{\Delta x_0}, \quad (2.32)$$

with S the discrete wavenumber of the mode: every mode the operator carries is amplified, and the excess is second order in the shift Courant number ν .

The repair is an upwind-equivalent diffusion applied to e_{int} alone, scaled by `Shifting.aleDiffusion` = ξ (default 1.0). It reuses the Molteni–Colagrossi ψ_{ij} of the δ -SPH density diffusion with the density difference replaced by the energy difference and the sound speed replaced by the **pair shift speed**:

$$\mathcal{D}_i^e = \xi \frac{\|\delta \mathbf{u}_{ij}\| \Delta x_0}{2} \sum_j \frac{m_j}{\bar{\rho}_{ij}} 2(e_j - e_i) \frac{(\mathbf{x}_j - \mathbf{x}_i) \cdot \nabla W_{ij}}{\|\mathbf{x}_i - \mathbf{x}_j\|^2 + 0.01h^2}. \quad (2.33)$$

Four properties of this form are deliberate and each was required:

- **The coefficient is set by the shift, not by a material constant.** It vanishes identically wherever nothing is shifted, and grows with precisely the quantity that drives the instability.
- **ξ is fixed by upwind equivalence and is not a fitted parameter.** First-order upwinding of an advection at speed $\|\delta \mathbf{u}\|$ on a lattice of spacing Δx_0 carries a numerical diffusivity $\|\delta \mathbf{u}\| \Delta x_0 / 2$; $\xi = 1$ buys exactly that and no more.
- **It is anchored on Δx_0 , not on h .** The unstable mode is a lattice mode keyed on the particle spacing. Anchoring on the support radius would make one nominal ξ some $1.5\times$ stronger on a solid deck at `supportRadiusFactor` 6 than on the fluid decks at 4 where it was calibrated. The $\eta^2 = 0.01h^2$ regulariser stays on h , because that *is* a statement about the kernel.
- **The pair weight carries $\bar{\rho}_{ij}$ rather than ρ_j ,** which costs nothing and makes the operator symmetric under $i \leftrightarrow j$. The diffusion therefore conserves $\sum_i m_i e_i$ identically and only ever redistributes it — measured at 3.8×10^{-9} of $\sum m e$, which is `float32` pair-summation round-off.

Like the δ -SPH density diffusion, it is gated on the pair sharing a material: across an interface $e_j - e_i$ is not an error to be smoothed, it *is* the interface.

2.5.2 Conservation of the Energy Advection

The diffusion of Eq. (2.33) conserves $\sum_i m_i e_i$ pair by pair. The advection does not, and this is a property of what the advection represents. A shifted particle receives the specific energy of the position it moves to while its mass stays where it was: particle masses are constant in **sigmaSPH**. Summed over all particles, the advective increment of one step is therefore a discrete

$$\Delta E_{\text{adv}} = \sum_i m_i \delta \mathbf{r}_i \cdot \nabla e_i \approx \int_{\Omega} \rho \delta \mathbf{r} \cdot \nabla e dV = - \int_{\Omega} e \nabla \cdot (\rho \delta \mathbf{r}) dV + \oint_{\partial \Omega} \rho e \delta \mathbf{r} \cdot \mathbf{n} dS, \quad (2.34)$$

which vanishes only if the shift carries no mass divergence and has no normal component at a free surface. Particle shifting has both by design, because it moves particles from crowded regions into sparse ones. Pairwise, the advective contribution of (i, j) is $V_i V_j (e_j - e_i) (\rho_i \delta \mathbf{u}_i + \rho_j \delta \mathbf{u}_j) \cdot \nabla W_{ij}$, which is symmetric under $i \leftrightarrow j$ and does not cancel. The imbalance is largest where the energy gradient is steepest. On hypervelocity decks that is the shock front and the free surfaces behind it, and it grows as the spacing is refined, because the gradient at a shock steepens as $1/\Delta x_0$ while the shift per unit time does not.

The solid solver reports the imbalance on every deck. The internal-energy change is split into **ie_pair**, the work conjugate of the pair forces, and **ie_ale**, the transport. **ie_ale** is split further into **ie_ale_adv**, **ie_ale_diff** and **ie_ale_fix**, which sum to it. The shares from mirror images and from pairs of different materials are reported as the subsets **ie_ale_mir** and **ie_ale_xmat**. Each particle accumulates what the transport has given it in the field **ale_e_work**, which the NetCDF trajectory carries.

The global conservative fix-up. **Shifting.aleEnergyConservative** (default **false**) makes the advection conserve $\sum m e$. Each step, the advective rates a_i are formed first, together with the two sums $S = \sum_i m_i a_i$ and $G = \sum_i m_i |a_i|$. Each rate is then replaced by

$$a_i \leftarrow a_i - |a_i| \frac{S}{G}, \quad (2.35)$$

which zeroes $\sum_i m_i a_i$ exactly. The correction lands only on particles that are being advected, in proportion to how much, and since $|S| \leq G$ no particle's advection changes sign. The diffusion is left unchanged, and the correction is banked as **ie_ale_fix**.

TRAP / OPERATIONAL NOTE: With Fixed Masses, Conservation and Consistency Are Incompatible

Under a uniform shift $\delta \mathbf{r}$ through a linear energy field, every advective rate has the same sign, so $S = G$ and Eq. (2.35) removes the transport entirely. That is not a defect of the weighting. The correct discrete answer in that case does change $\sum m e$, by $M \delta \mathbf{r} \cdot \nabla e$, so no correction that forces the change to zero can reproduce it. The fix-up is therefore a per-deck option and not the default.

It is right for decks dominated by one short, strong shock. There, measured with it on, the total energy change falls without moving the observables:

- Qamar tantalum spall: from +6.0% to +0.6% of E_0 ;
- Al 2017 impact: from +8.4% to +3.3%;
- Chocron concrete, axisymmetric at 0.5 mm: from +10.4% to +0.5%;
- Chocron concrete, 3D quarter at 1 mm: from +12.7% to +0.6%.

Those decks set it. It is harmful where the advection is sustained and of both signs. On the steel-into-gelatin stack, weighting by $|a_i|$ puts the correction where the gradient is

steepest, which steepens it further. The advection then grows to 117% of E_0 against its own correction, and the negative internal energy reaches -47% .

A transport that is both conservative and consistent has to move mass with the energy: the particles exchange an antisymmetric pair mass flux, and me , $m\mathbf{v}$ and the history variables move with it, as in the ALE-SPH of Oger et al. [26]. **sigmaSPH** does not implement that.

2.6 Collé γ -SPH Low-Mach Flux Stabilisation

In the low-Mach regime, standard weakly compressible SPH exhibits spurious pressure oscillations due to acoustic stiffness. The γ -SPH scheme [10] introduces a numerical Riemann interface pressure velocity Γ_{ij} :

$$\Gamma_{ij} = \frac{\gamma}{2c_0\rho_{ij}}(p_j - p_i)\hat{\mathbf{n}}_{ij}, \quad \rho_{ij} = \frac{1}{2}(\rho_i + \rho_j). \quad (2.36)$$

Substituting this interface flux into the volume evolution equation adds a pressure Laplacian diffusion term:

$$\Delta \left(\frac{d\varepsilon_{\text{vol},i}}{dt} \right) = \frac{\gamma}{c_0\rho_i} \sum_j V_j(p_j - p_i) \|\nabla W_{ij}\|. \quad (2.37)$$

The γ -SPH diffusion acts directly on pressure differences, complementing the δ -SPH density diffusion [24]. Both regularisers share a single explicit numerical stability budget:

$$\delta + \gamma \leq \frac{1}{3}. \quad (2.38)$$

2.7 Hourglass Mode Stabilization for Elastic–Plastic Solids

2.7.1 Theoretical Origin and Literature Pedigree

A persistent pathology in Lagrangian Smoothed Particle Hydrodynamics formulations of solid mechanics is the emergence of zero-energy deformation modes, commonly referred to as hourglass modes or kinematic null spaces. Because the kernel-interpolated velocity gradient,

$$\nabla \mathbf{v}_i = - \sum_j V_j(\mathbf{v}_j - \mathbf{v}_i) \otimes \nabla W_{ij} \mathbf{L}_i, \quad (2.39)$$

samples deformation by volume-weighting over an extended compact support neighbourhood h , high-frequency alternating particle displacements (such as checkerboard or zig-zag patterns) can produce a sum that integrates identically to zero ($\nabla \mathbf{v}_i = \mathbf{0}$). When this occurs, the constitutive model registers zero strain rate ($\dot{\boldsymbol{\varepsilon}} = \mathbf{0}$) and generates no restoring stress. Unchecked, these zero-energy modes decouple neighbouring particles, allowing non-physical distortions to propagate without mechanical resistance and leading to spurious shear banding or grid breakdown.

Origin in Under-Integrated Finite Elements

The mathematical lineage of hourglass control originates in classical continuum finite element analysis. In their seminal work, Flanagan and Belytschko [13] developed orthogonal hourglass control for uniform-strain single-point integrated 4-node quadrilateral and 8-node hexahedral elements. Under-integrated elements were widely adopted to prevent volumetric locking in incompressible plastic flow and to minimize computational costs in explicit dynamics, but they introduced spurious zero-energy modes that polluted the solution. Flanagan and Belytschko [13] constructed an orthogonal projection that isolated the non-affine hourglass velocity field from rigid-body rotations and homogeneous deformation modes, proposing two distinct stabilization mechanisms:

1. **Stiffness-based (displacement) control:** Penalizes accumulated non-affine displacement errors measured against an initial reference geometry. This acts as an elastic restoring spring whose numerical stability limit is governed by an oscillatory natural frequency ω_{HG} , scaling as $\Delta t \propto 1/\sqrt{\alpha_{\text{HG}}}$.
2. **Viscous-based (velocity) control:** Penalizes the instantaneous non-affine velocity discrepancies ($\mathbf{v} - \mathbf{v}_{\text{affine}}$). This acts as a physical damper whose numerical stability limit is governed by a relaxation rate R_{HG} , scaling linearly as $\Delta t \propto 1/\alpha$.

SPH Rank Deficiency and Ganzenmüller's Formulation

Through a unified spectral stability analysis of meshfree methods, Belytschko et al. [4] demonstrated that nodal collocation in SPH is functionally identical to under-integrated finite elements, exhibiting an exact counterpart to the FEM rank-deficiency null space. While stress-point integration [12] and assumed-strain schemes [3] had been explored, they incurred heavy algorithmic complexity and memory overhead.

Ganzenmüller [14] bridged this divide by adapting the Flanagan–Belytschko orthogonal stabilization concept directly to Lagrangian SPH. Crucially, Ganzenmüller [14] recognized that the intricate rotational projection matrices required in continuum FEM can be avoided entirely in particle methods by projecting the anti-hourglass interaction strictly along the inter-particle line of centres:

$$\hat{\mathbf{x}}_{ij} = \frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}. \quad (2.40)$$

Because pairwise forces acting collinearly along $\hat{\mathbf{x}}_{ij}$ exert zero net moment ($\hat{\mathbf{x}}_{ij} \times \hat{\mathbf{x}}_{ij} = \mathbf{0}$), line-of-centre projection automatically guarantees rotational invariance and exact conservation of angular momentum without requiring artificial co-rotational coordinate tracking.

In the original Total Lagrangian SPH (TLSPH) formulation of Ganzenmüller [14], the stabilization was formulated as a reference-configuration displacement/stiffness penalty:

$$\mathbf{e}_{ij} = \mathbf{F}_i(\mathbf{X}_j - \mathbf{X}_i) - (\mathbf{x}_j - \mathbf{x}_i), \quad \delta_{ij} = \mathbf{e}_{ij} \cdot \hat{\mathbf{x}}_{ij}, \quad (2.41)$$

where $\mathbf{X}$ denotes undeformed reference coordinates and $\mathbf{F}_i$ is the particle deformation gradient. The restoring force penalizes the departure of the current bond vector $\mathbf{x}_j - \mathbf{x}_i$ from the affine mapping $\mathbf{F}_i(\mathbf{X}_j - \mathbf{X}_i)$ of the undeformed bond.

Abandonment of Displacement-Based Control in Favour of Velocity Damper

The displacement-based formulation possesses a compelling numerical advantage: because the penalty acts as a stiffness rather than a viscous damper, its explicit forward-Euler stability boundary scales under a square root ($\Delta t \propto 1/\sqrt{\alpha_{\text{HG}}}$) rather than linearly ($\Delta t \propto 1/\alpha$).

However, extensive numerical testing during the development of **sigmaSPH** demonstrated that **displacement-based control fatally interferes with Particle Shifting Technology (PST)**. In large-strain Updated Lagrangian and ALE hydrodynamics, PST is mandatory to homogenize disordered particle packings and suppress the tensile pairing instability. By design, PST continuously displaces particles relative to the underlying material motion ($\hat{\mathbf{v}} = \mathbf{v} + \delta\mathbf{u}$). Because displacement-based control references initial material coordinates $\mathbf{X}_0$, it misinterprets this beneficial convective shift as an unphysical hourglass distortion. The displacement-based penalty aggressively opposes PST, driving severe numerical instability and premature divergence under plastic deformation (e.g. failing at 34% strain in tensile specimens, and diverging earlier when the penalty coefficient is increased). Periodic re-anchoring of the reference coordinates discards the accumulated displacement error history without curing the underlying kinematic incompatibility.

The Velocity-Based Rate-Form Damper

To achieve seamless compatibility with particle shifting and finite-strain Updated Lagrangian kinematics, **sigmaSPH** implements a velocity-based (rate-form) hourglass stabilizer. Rather than tracking displacement discrepancies from a frozen reference configuration, the following form for the instantaneous non-affine relative velocity was found to work very well:

$$\delta \mathbf{v}_{ij} = (\mathbf{v}_j - \mathbf{v}_i) - \frac{1}{2}(\nabla \mathbf{v}_i + \nabla \mathbf{v}_j)(\mathbf{x}_j - \mathbf{x}_i), \quad (2.42)$$

projected along the current line of centres $\hat{\mathbf{x}}_{ij}$. Because this formulation operates strictly on current-configuration Eulerian kinematics, it is completely free from reference coordinates $\mathbf{X}_0$, vanishes identically under rigid-body motions and homogeneous deformation gradients, and operates in complete harmony alongside PST.

Evidence That the Damper Is Required

A test that runs only with the damper switched on checks the physical answer, but cannot show that it would distinguish a working damper from a missing one. T14 therefore runs its specimen a second time at $\alpha = 0$ and measures the null-space fraction of the gauge section,

$$\phi_{\text{null}} = 1 - \frac{\ln F_{yy}}{e_{\text{geo}}}, \quad (2.43)$$

where e_{geo} is the logarithmic stretch the particles actually underwent, fitted as the slope of current against reference axial coordinate over the gauge, and F_{yy} is integrated from the kernel-sum velocity gradient. ϕ_{null} is thus the share of the deformation that the constitutive model never registered. Without the damper it is 0.963, the axial stress reaches 0.041 of the elastic answer $E e_{\text{geo}}/(1 - \nu^2)$, and the peak root-mean-square non-affine speed in the gauge is 1.1 m/s. With the damper at $\alpha = 0.5$ the three read 0.075, 0.911 and 8.6×10^{-3} m/s. The acceptance thresholds (0.25 and 0.5 on ϕ_{null} , 0.3 on the stress ratio) lie between the two groups, so the verdict does not depend on the particular form the undamped failure takes.

TRAP / OPERATIONAL NOTE: THE DAMPED NULL-SPACE FRACTION GROWS WITH α (UNEXPLAINED)

The residual ϕ_{null} with the damper on is not zero, and it rises with the damping coefficient: 0.075, 0.083, 0.106 and 0.129 at $\alpha = 0.5, 1, 2$ and 4. A residual null space would shrink as the damper is strengthened, so this is a different effect. It is not a consequence of the step count. The one indication of its origin is that the stiffness control of Mohseni-Mofidi and Bierwisch [23] with shifting disabled reduces it to 0.007, which points at the shift; because two things changed in that comparison, it is evidence and not a demonstration.

2.7.2 Discrete Pairwise Damper Formulation

The discrete non-affine velocity evaluates the departure of the relative velocity $\mathbf{v}_j - \mathbf{v}_i$ from the locally affine velocity variation predicted by the particle velocity gradients. Crucially, employing the arithmetic mean of the two velocity gradient tensors,

$$\overline{\nabla \mathbf{v}}_{ij} = \frac{1}{2}(\nabla \mathbf{v}_i + \nabla \mathbf{v}_j), \quad (2.44)$$

guarantees that the pairwise non-affine velocity is strictly antisymmetric under particle index exchange:

$$\delta \mathbf{v}_{ji} = -\delta \mathbf{v}_{ij}. \quad (2.45)$$

Projecting $\delta \mathbf{v}_{ij}$ onto the unit inter-particle vector $\hat{\mathbf{x}}_{ij} = (\mathbf{x}_i - \mathbf{x}_j)/\|\mathbf{x}_i - \mathbf{x}_j\|$ isolates the radial mode while leaving transverse shear and rotational components unpenalized:

$$\delta v_{ij}^{\text{rad}} = \delta \mathbf{v}_{ij} \cdot \hat{\mathbf{x}}_{ij}. \quad (2.46)$$

The total hourglass damping acceleration exerted on particle i by its compact support neighbourhood is evaluated as:

$$\mathbf{a}_i^{\text{HG}} = -\frac{\alpha c_p}{h} \sum_j \frac{V_j}{V_0} \frac{W_{ij}}{W(\Delta x_0)} (\delta v_{ij}^{\text{rad}}) \hat{\mathbf{x}}_{ij}, \quad (2.47)$$

where:

- $\alpha = \text{Hourglass.alpha}$ is the dimensionless damping parameter (typically $1.0 \leq \alpha \leq 4.0$).
- $c_p = \sqrt{(K + \frac{4}{3}G)/\rho_0}$ is the dilatational elastic wave speed, establishing the dimensional relaxation scale $\alpha c_p/h$ in units of s^{-1} .
- $V_0 = \Delta x_0^d$ is the undeformed reference volume per particle in d spatial dimensions.
- $W(\Delta x_0)$ is the kernel function evaluated at the initial inter-particle spacing Δx_0 , normalizing the neighbour weighting so that nearest neighbours exert an order-unity contribution.

Because $\hat{\mathbf{x}}_{ji} = -\hat{\mathbf{x}}_{ij}$ and $\delta v_{ji}^{\text{rad}} = \delta v_{ij}^{\text{rad}}$, the pairwise force $\mathbf{f}_{ij}^{\text{HG}} = m_i \mathbf{a}_{ij}^{\text{HG}}$ is exactly antisymmetric ($\mathbf{f}_{ij}^{\text{HG}} = -\mathbf{f}_{ji}^{\text{HG}}$) and collinear with $\mathbf{x}_i - \mathbf{x}_j$, preserving total linear and angular momentum to machine precision.

2.7.3 Explicit Stability Limit and Super-Time-Stepping (STS)

Because every neighbour within the compact support sphere $r_{ij} < h$ contributes additively to the acceleration (2.47), the effective damping relaxation rate ω_{HG} is substantially larger than the nominal single-particle frequency $\alpha c_p/h$. The effective rate is scaled by the dimensionless neighbour accumulation factor S :

$$S = \sum_j \frac{V_j}{V_0} \frac{W_{ij}}{W(\Delta x_0)} \approx \begin{cases} 6.64, & \text{in 2D plane strain with support } h = 3\Delta x_0, \\ 15.4, & \text{in 3D Cartesian with support } h = 3\Delta x_0. \end{cases} \quad (2.48)$$

The total relaxation rate of the discrete hourglass operator is:

$$\omega_{\text{HG}} = \frac{\alpha c_p}{h} S. \quad (2.49)$$

For forward-Euler explicit time integration, the stability limit of a pure damping equation $\dot{v} = -\omega v$ is $\omega \Delta t \leq 2.0$. To avoid oscillatory overshoots and maintain a robust stability margin, **sigmaSPH** limits the time step to 80% of the theoretical threshold:

$$\Delta t_{\text{HG}} = \frac{1.6}{\omega_{\text{HG}}} = \frac{1.6 h}{\alpha c_p S}. \quad (2.50)$$

Because S is large (6.64 in 2D and 15.4 in 3D) and robust stabilization requires $\alpha \geq 1.0$ (typically $\alpha = 2.0$ – 4.0 in metal plasticity), the product αS ranges between 13 and 62. Consequently, Δt_{HG} restricts the global simulation time step Δt by a factor of $5.0\times$ (in 2D) to $11.6\times$ (in 3D) below the acoustic CFL limit.

Sub-Cycling and Super-Time-Stepping Integration

Attempting to circumvent this cap within a single global hydrodynamic step (e.g. by replacing linear damping with an analytical exponential decay $1 - e^{-\omega\Delta t}$) is mathematically unsound: no per-step operator can dissipate more than 100% of the non-affine velocity in a single step ($1/\Delta t$), and attempting to do so under-damps the mode and destroys the physical stress solution.

Instead, sigmaSPH decouples the hourglass damper from the acoustic CFL condition by integrating it on its own sub-stepping timescale beneath the global hydrodynamic step (`Hourglass.superTimeStepping`, [22]):

1. **Explicit Sub-cycling (n stages):** The linear velocity damper is advanced over n uniform sub-steps of duration $\Delta t_{\text{sub}} = \Delta t/n$, where $n = \lceil \Delta t/\Delta t_{\text{HG}} \rceil$. Because each sub-step satisfies $\omega_{\text{HG}}\Delta t_{\text{sub}} \leq 1.6$, the global hydro step Δt advances at the full acoustic CFL limit, yielding a $3\times$ to $5\times$ speedup in wall-clock execution time while preserving the exact deformation response.
2. **Runge–Kutta–Legendre (RKL1) Super-Time-Stepping:** Employs optimal shifted Legendre polynomials over s parabolic stages to expand the explicit stability boundary quadratically:

$$\Delta t_{\text{eff}} = \frac{s^2 + s}{2} \Delta t_{\text{expl}}. \quad (2.51)$$

Placement of the Damper Within the Step

Because the damper is a standalone operator, its position within the step is a choice, and the choice has an energetic consequence. Let Φ denote the damper advanced over one hydro step. The step is taken as

$$\mathbf{v} \leftarrow \mathbf{v} + \frac{1}{2}\Delta t \mathbf{a}_{\text{other}}, \quad (2.52)$$

$$\mathbf{v} \leftarrow \Phi(\Delta t) \mathbf{v}, \quad (2.53)$$

$$\mathbf{v} \leftarrow \mathbf{v} + \frac{1}{2}\Delta t \mathbf{a}_{\text{other}}, \quad (2.54)$$

$$\mathbf{x} \leftarrow \mathbf{x} + \Delta t \mathbf{v}, \quad (2.55)$$

where $\mathbf{a}_{\text{other}}$ is every acceleration except the damper's. The kick is split in two halves *about* the damper. The stored velocity of the symplectic step is the half-step velocity (Sec. 2.11.2), and after the first half-kick it is the synchronised velocity $\mathbf{v}_{k-1/2} + \frac{1}{2}\Delta t \mathbf{a}_k$, the velocity at the time of the positions. The damper therefore acts on that velocity, and the kinetic energy it removes, which is banked as HE, is read at the same time level as every other term of the budget. Neither half needs an extra force evaluation. The cost is one extra velocity pass per step, plus one velocity-sized scratch field that exists only on a super-time-stepped configuration.

TRAP / OPERATIONAL NOTE: Why the Damper Sits Between the Half-Kicks and Not After the Whole Kick

Until 2026-09-26 the step was kick, damp, drift. There the damper rewrites the half-step velocity after the kick, and it banks what it removed as kinetic energy *at that velocity*. The next kick then partly restores what the damper took, because the damper and the stress act against each other on the same non-affine motion. The budget books that exchange as dissipation, while the synchronised kinetic energy never lost it. The net effect is that part of the integrator's leftover becomes real mechanical energy.

On the unconfined copper-on-aluminium impact (`impact2d_copper_aluminum_mm_GPa_ms`, shift off, CFL 0.3), the old ordering injected $+2.54\%$ of E_0 of real mechanical energy by $t = 0.15$ ms, or $+4.83\%$ with the damper's work read at the synchronised level. The ordering of Eqs. (2.52)–(2.55) injects $+0.25\%$, and it converges at first order ($+0.25$, $+0.12$,

+0.06% at CFL 0.3, 0.15, 0.075). The peak velocity, the plastic work and the bookkeeping part of the leftover are unchanged, and the damper removes slightly less (8.8% against 10.6% of E_0). Splitting the *damper* about the kick instead ($\frac{1}{2}\Phi$, kick, $\frac{1}{2}\Phi$, drift) still leaves it on half-step velocities over consecutive steps, and recovers only about half of the improvement.

The change matters only where the damper works hard during an impact. On the Qamar spall configuration at $\alpha = 0.05$ the spall plane, the rear-surface velocity history and the scab are unchanged within the scatter of repeated runs. On the free oscillating bar it removes only the small injection that super-time-stepping itself added, and leaves the oscillation untouched. The measurement is recorded in `reports/centred_continuity_impact_2026-09-26.md` and `reports/integrator_leftover_summary_2026-09-26.md`, made with `tools/centred_continuity_study.py`, whose arm L reproduces the old ordering.

2.8 Artificial Viscosities and Shock Capturing

2.8.1 Quadratic von Neumann–Richtmyer Bulk Viscosity

To capture strong shock waves without corrupting smooth shear flows, a quadratic bulk viscosity pressure q is added to the mechanical pressure:

$$q_i = \begin{cases} C_Q \rho_i l^2 (\nabla \cdot \mathbf{v}_i)^2, & \text{if } \nabla \cdot \mathbf{v}_i < 0, \\ 0, & \text{if } \nabla \cdot \mathbf{v}_i \geq 0, \end{cases} \quad (2.56)$$

where $l = \Delta x_0$ is the characteristic length scale and C_Q is a non-dimensional coefficient (typically 1.0–2.0). Because q_i is non-zero strictly in compression, it dissipates kinetic energy exclusively across shock discontinuities.

2.8.2 Monaghan Artificial Viscosity

The pairwise Monaghan artificial viscosity tensor Π_{ij} provides dissipation against interpenetration:

$$\Pi_{ij} = \begin{cases} \frac{-\alpha \bar{c}_{ij} \mu_{ij} + \beta \mu_{ij}^2}{\bar{\rho}_{ij}}, & \mathbf{v}_{ij} \cdot \mathbf{r}_{ij} < 0, \\ 0, & \mathbf{v}_{ij} \cdot \mathbf{r}_{ij} \geq 0, \end{cases} \quad (2.57)$$

with:

$$\mu_{ij} = \frac{h(\mathbf{v}_i - \mathbf{v}_j) \cdot (\mathbf{x}_i - \mathbf{x}_j)}{\|\mathbf{x}_i - \mathbf{x}_j\|^2 + \varepsilon h^2}, \quad \bar{c}_{ij} = \frac{1}{2}(c_i + c_j), \quad \bar{\rho}_{ij} = \frac{1}{2}(\rho_i + \rho_j). \quad (2.58)$$

2.9 Symmetry Boundaries: The Axis and Cartesian Cut Planes

A body that possesses a symmetry may be discretised as the corresponding fraction of itself, and the saving is the whole fraction: a quarter model carries a quarter of the particles and costs a quarter of the wall clock. `sigmaSPH` offers two such reductions. `Domain.axisymmetric` reduces a body of revolution to a meridional half-plane (x axial, $y = r \geq 0$, $z = 0$), in which every particle interaction is a planar two-dimensional neighbour sum and the circumferential direction is integrated analytically. `Domain.symmetryPlanes` cuts a genuinely three-dimensional body on the plane $y = 0$, on the plane $z = 0$, or on both, and retains only the half or the quarter in which the cut coordinates are non-negative.

The reduction is not merely a convenience at the scale at which it matters most. The hypervelocity impact deck shipped in `data/scenes/`, discretised at $\Delta x_0 = 0.05$ mm over the full 360° would carry approximately 12.1 million particles and require some 4.6 GB of particle state before any auxiliary allocation, which does not fit on a 6 GB device; the quarter model carries 3.03 million and does.

Both reductions raise one and the same difficulty, and it is treated first, in Sec. 2.9.1, because the two geometries share not only the argument but the implementation.

2.9.1 Mirror-Image Neighbours Rather Than Ghost Particles

A particle lying within the support radius h of the axis, or of a cut plane, is missing every neighbour on the far side of it. Nothing about that deficit is physical: the material is present in the body being modelled, and the stored fraction simply does not hold it. Left untreated, every neighbour in the solver reads the cut as a free surface. The free-surface indicator λ_i of Sec. 2.3 collapses towards the value it takes on a genuine boundary, the completeness matrix $\mathbf{E}_i$ loses a direction, and—decisively—the kernel-gradient sum $\sum_j V_j \nabla W_{ij}$ ceases to vanish.

That last failure is not one symptom among several. Under a spatially uniform stress state $\sigma_j \equiv \sigma$, the discrete stress divergence reduces to

$$\mathbf{a}_i = \frac{1}{m_i} \sum_j V_i V_j (\sigma_i + \sigma_j) \nabla W_{ij} \longrightarrow \frac{2}{\rho_i} \sigma \sum_j V_j \nabla W_{ij}, \quad (2.59)$$

so that the residual of the kernel-gradient sum *is* the spurious acceleration, multiplied by the stress and by nothing else. No cancellation elsewhere in the discretisation can remove it. A body at rest under a uniform stress therefore acquires a force directed out of the cut, which opens it. On an impact deck the cut runs through the crater, which places the failure precisely where the answer is.

The material that is missing is the mirror image of the material that *is* stored, exactly and by construction. The images are therefore *visited* inside the neighbour loop rather than allocated as particles. The traversal

```
for_all_neighbors( $p_i$ , task, ret)  calls  task( $p_i, p_j$ , mir, ret)
```

passes each task function a mirror code `mir` that is a compile-time literal at every call site, and each task mirrors the neighbour quantities it reads. Under the reflection

$$\mathbf{M} = \text{diag}(1, s_y, s_z), \quad s_y, s_z \in \{+1, -1\}, \quad (2.60)$$

a scalar is unchanged, a vector $\mathbf{a}$ becomes $\mathbf{M}\mathbf{a}$, and a second-order tensor $\mathbf{T}$ becomes $\mathbf{M}\mathbf{T}\mathbf{M}$; these are supplied by `ParticleSystem.mirror_vec` and `ParticleSystem.mirror_mat`. Because the code is a Taichi template argument and the accessors are inlined `@ti.funcs`, the sign selection constant-folds at each site and the image sweeps cost no arithmetic that a fixed operator would not.

The distance test is taken between the *reflected* p_i and the real p_j , which is admissible because $\mathbf{M}$ is an isometry and

$$\|\mathbf{x}_i - \mathbf{M}\mathbf{x}_j\| = \|\mathbf{M}\mathbf{x}_i - \mathbf{x}_j\|. \quad (2.61)$$

Measured on an axisymmetric lattice whose innermost row sits at $r = \Delta x_0/2$, against the identical lattice advanced in plane strain, where that row genuinely is a free surface (verification module `t21`):

Table 2.1: The axis is not a free surface. Innermost lattice row at $r = \Delta x_0/2$, against its own interior and against the plane-strain twin in which the same row is a true boundary.

Innermost row	Axisymmetric	Its own interior	Plane-strain twin
λ	0.9974	0.9974	0.4987
$\ h \sum_j w_j \nabla W_{ij}\ $	6.8×10^{-6}	6.9×10^{-6}	1.26
$\ \mathbf{a}\ $ for uniform $\boldsymbol{\sigma}$, $\div \ \boldsymbol{\sigma}\ /(\rho h)$	1.4×10^{-5}	1.4×10^{-5}	—

TRAP / OPERATIONAL NOTE: GHOST PARTICLES WERE REJECTED DELIBERATELY

The route not taken was to allocate mirrored particles at $y < 0$. Such a ghost requires a stable identity to keep it paired with its source across the counting sort, which re-indexes up to 80% of particles every step; a refresh between every pair of kernels, because a ghost's own neighbourhood is deficient and its state is therefore invalid the moment anything writes it; a grid whose origin is no longer the domain corner; a boundary enforcement that does not teleport it; and exclusion from the trajectory dumps, the visualisation buffer, and every global reduction. Mirroring inside the neighbour loop costs one additional sweep per plane and one argument on the task functions, allocates nothing, and cannot go stale.

2.9.2 Placing the Lattice Against a Symmetry Element

The mirror images of the preceding section repair the neighbourhood of a particle near the axis or a cut plane only if the stored material and its images together form the lattice that the full body would have carried. That places one requirement on the scene file, and it is the single most consequential line of a symmetry deck.

TRAP / OPERATIONAL NOTE: THE FIRST ROW SITS AT $\Delta x_0/2$ FROM THE CUT

Particles must be placed at **half the interparticle spacing**, $\Delta x_0/2$, from every symmetry element — never on it, and never a full spacing away from it.

- **Axisymmetric decks:** the innermost row lies at $y = r = \Delta x_0/2$ from the axis.
- **Symmetry planes:** the nearest row lies at $y = \Delta x_0/2$ from the plane $y = 0$ and, in a quarter model, at $z = \Delta x_0/2$ from the plane $z = 0$.

Because $\Delta x_0 = 2\text{Domain.particleRadius}$, this means that the **start** coordinate of a block on each cut is **Domain.particleRadius**, not 0: a block with **particleRadius** = 0.125 starts at $y = 0.125$.

The reason is visible in Fig. 2.1. Stored rows at $\Delta x_0/2$, $3\Delta x_0/2$, ... have images at $-\Delta x_0/2$, $-3\Delta x_0/2$, ..., and the two sets together are one uninterrupted lattice of spacing Δx_0 that runs straight through the cut. A particle next to the axis or the plane therefore sees exactly the neighbourhood an interior particle sees, and the kernel sums of Sec. 2.9.1 are complete to round-off. The measurements of Tables 2.1 and 2.3 were taken on precisely this lattice, and the radius floor **Domain.axisRadiusMin** = $0.5 \Delta x_0$ of the following section was chosen so as to be inert on it.

Both ways of getting the offset wrong are silent in the output once they have happened, and they fail in opposite directions:

- **A row on the element** ($y = 0$ or $z = 0$) coincides with its own image. The particle is counted on top of itself, which reads as a density concentration on the cut, and the particle

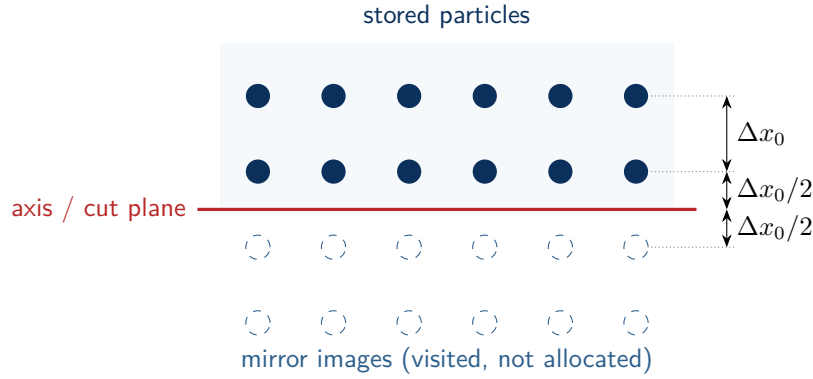


Figure 2.1: Lattice placement against a symmetry element. The stored rows (filled) start half a spacing off the cut; their images (dashed) then continue the lattice across it at the same spacing Δx_0 , so the cut is invisible to every neighbour sum.

shifting of Sec. 2.4 responds by driving the row outward. On the axis the row is moreover a ring of zero radius and hence of zero volume, $V_i = r_i A_i = 0$.

- **A row further out than $\Delta x_0/2$** leaves a gap wider than Δx_0 between it and its own image. The images can no longer fill that gap, the free-surface indicator λ falls, the residual $\sum_j V_j \nabla W_{ij}$ of Eq. (2.59) stops vanishing, and the cut begins to act as the free surface the images were meant to remove. The extreme case, a first row a full spacing out, is a lattice with one row missing.

What the code checks at load time (particle_system.py). A particle with a negative coordinate on a cut — $y < 0$ in an axisymmetric deck, $y < 0$ or $z < 0$ under the corresponding symmetry plane — is fatal, because the images would fold it back onto the stored material. The build then prints the offset of the innermost particle in units of Δx_0 and prints a red warning if it lies below $\frac{1}{4}\Delta x_0$ or above $\frac{3}{4}\Delta x_0$. **These are warnings, not errors:** a deck that ignores them still runs, and still produces a plausible-looking result.

2.9.3 The Axisymmetric Meridional Half-Plane

For bodies of revolution the equations of motion are solved on the meridional half-plane, with all particle interactions remaining planar two-dimensional neighbour sums while the circumferential hoop direction is integrated analytically.

Ring Volumes and Densities

Particles represent circular tori revolved about the x -axis. Volume V_i and mass m_i are defined per radian of azimuth:

$$V_i = r_i A_i, \quad m_i = \rho_0 r_{i,0} \Delta x_0^2, \quad \rho_i = \frac{m_i}{V_i}, \quad (2.62)$$

where A_i is the meridional cross-sectional area.

Kinematic Hoop Strains and Momentum Source Terms

The circumferential hoop strain rate is non-zero whenever radial motion occurs:

$$\dot{\epsilon}_{\theta\theta,i} = \frac{v_{r,i}}{r_i}. \quad (2.63)$$

The total velocity divergence driving volume evolution includes the hoop contribution:

$$\nabla \cdot \mathbf{v}_i = (\nabla \cdot \mathbf{v}_i)_{\text{plane}} + \frac{v_{r,i}}{r_i}. \quad (2.64)$$

In the momentum equations, cylindrical divergence generates geometric source terms:

$$(\nabla \cdot \boldsymbol{\sigma})_{x,i} = (\nabla \cdot \boldsymbol{\sigma})_{x,\text{plane}} + \frac{\sigma_{xr,i}}{r_i}, \quad (2.65)$$

$$(\nabla \cdot \boldsymbol{\sigma})_{r,i} = (\nabla \cdot \boldsymbol{\sigma})_{r,\text{plane}} + \frac{\sigma_{rr,i} - \sigma_{\theta\theta,i}}{r_i}. \quad (2.66)$$

Because $\sigma_{rr} - \sigma_{\theta\theta} = s_{rr} - s_{\theta\theta}$, hydrostatic pressure cancels identically from both momentum source terms. Consequently, a uniform pressure field exerts exactly zero spurious axis force.

The Radius Floor on the Geometric Sources

The radius entering the $1/r$ terms and the two radial accessors is floored at `Domain.axisRadiusMin`, whose default is $0.5 \Delta x_0$. The floor is not only a division guard. The hoop source acts as a linear spring of stiffness $2G/(\rho r^2)$, whose forward-Euler stability interval is $\sqrt{2}r/c_s$; at $r = \Delta x_0/2$ that interval still lies approximately 55% above the acoustic limit $0.3 \cdot 3\Delta x_0/c_p$ imposed at the house support-radius factor, so the timestep remains governed by the acoustic condition rather than by the geometric source. A smaller floor would reverse that ordering. At half a spacing the floor is inert for an undeformed lattice laid from $y = \Delta x_0/2$ (Sec. 2.9.2) and saturates only for material squeezed closer to the axis, where the regularity conditions $\sigma_{xr} = O(r)$ and $\sigma_{rr} - \sigma_{\theta\theta} = O(r^2)$ make the true source small in any case. Measured on the axisymmetric impact deck, moving the floor between 0.1 and 1.0 Δx_0 changes the reported momentum drift by 1.58%, 1.82% and 1.67% respectively: it is not a sensitive parameter.

2.9.4 Cartesian Symmetry Planes

`Domain.symmetryPlanes` accepts a subset of the string "yz": "y" for a half model cut at $y = 0$, "yz" for a quarter model cut at $y = 0$ and at $z = 0$. The planes sit at the coordinate origin, which is where `Domain.start` is, and the lattice is laid from $\Delta x_0/2$ off each plane, exactly as an axisymmetric deck is laid off the axis (Sec. 2.9.2). The key requires a full three-dimensional domain and is rejected in combination with `Domain.axisymmetric`, which already mirrors about $y = 0$ and carries the geometric sources of the preceding section, and with `Domain.planeStrain`, which pins z to a single layer and leaves no third dimension for a plane to cut.

The Mirror Code

The mirror argument is a two-bit code selecting the signs of Eq. (2.60): bit 0 reflects about $y = 0$, bit 1 about $z = 0$.

Table 2.2: The four mirror codes. Code 1 is the operator that the axisymmetric formulation has always applied at the axis.

Code	Operator M	Meaning
0	$\text{diag}(1, 1, 1)$	a real neighbour, untouched
1	$\text{diag}(1, -1, 1)$	across $y = 0$ — the axis's own operator
2	$\text{diag}(1, 1, -1)$	across $z = 0$
3	$\text{diag}(1, -1, -1)$	the corner image, where two planes meet

Bit 0 is assigned to y so that the axisymmetric formulation passes the code 1 and receives back precisely the operator it always had. That assignment is a correctness guarantee rather than a matter of tidiness: it is what makes the generalisation from a single fixed reflection to a family of four incapable of moving any previously recorded axisymmetric result. The equality of `mirror_vec(1, ·)` with the fixed $\text{diag}(1, -1, 1)$ is asserted directly by the verification suite rather than assumed.

A plane at $x = 0$ is not offered. Three planes would require a third bit and seven image sweeps in place of three, and the impact axis in every deck written so far is x , so such a plane would cut a projectile across its direction of travel. The bitmask is where the extension would go.

The Three Image Sweeps

The neighbour traversal gains up to three image sweeps after the sweep over real neighbours. Each retains the structure of the axis sweep: it hashes the stencil cell, filters candidates by the cell into which they were *binned* rather than by one re-derived from a live position, and tests the reflected distance.

- across $y = 0$: the cells $[c_x \pm 1, 0, c_z \pm 1]$, yielding code 1;
- across $z = 0$: the cells $[c_x \pm 1, c_y \pm 1, 0]$, yielding code 2;
- the corner: the cells $[c_x \pm 1, 0, 0]$, yielding code 3.

Only the single cell row against each plane is swept, and the argument is the axis's, unchanged: a real particle one row further out sits at $y_j \geq h$, so that its image lies at least $h + y_i$ from p_i and could never be a neighbour. The corner sweep needs only the single cell column against both planes, for that same reason applied twice.

The case $p_j = p_i$ is not excluded. A particle's own image across a plane is a genuine neighbour, at a separation of $2y_i$. (At $y_i = 0$ the image would coincide with the particle, but every sum evaluated there is weighted either by ∇W or by $\Delta \mathbf{v} \cdot \hat{\mathbf{r}}$, both of which vanish at zero separation, so the coincident image contributes nothing rather than double-counting.)

The particle shifting operator of Sec. 2.4 inlines a neighbour loop of its own and therefore carries its own copy of all three sweeps, as it already did for the axis. It is the one sum in the solver that does not pass through the shared stencil, and omitting the images from it would leave the shift one-sided at the plane, driving the nearest lattice row away from it and opening exactly the hole that the images exist to deny.

2.9.5 The Fold-Back Constraint

A particle that crosses the axis, or a symmetry plane, is *reflected* rather than clamped. Under the symmetry the solver already assumes, such a particle is not at a new place: it is the same material on the wrong side of the cut. It is therefore returned to $|y|$ (respectively $|z|$) with its normal velocity component, its acceleration, its deviatoric stress and, where the deformation gradient is tracked, $\mathbf{F}$ all carried through the corresponding operator of Table 2.2. Clamping the offending coordinate to zero would destroy momentum silently instead. The enforcement runs before the domain wall clamp, and the planes are tested one at a time, because a particle in the corner of a quarter model can cross both in a single step.

TRAP / OPERATIONAL NOTE: THE AXIS FOLD-BACK SHOULD NEVER FIRE; THE PLANE FOLD-BACK SHOULD

On a healthy axisymmetric run no particle crosses $r = 0$, and `enforce_axisymmetry_constraint` firing is a symptom worth investigating. The Cartesian analogue carries no such expectation. Material beneath an impact flows *along* a symmetry plane rather than away from it, and ordinary numerical noise pushes particles across the cut; `enforce_symmetry_constraint` firing on a quarter impact model is the boundary condition working, not a warning.

2.9.6 Verification and the Limits of a Symmetry Model

The symmetry-plane treatment is verified by module `t37` (28 checks). The instrument that carries the most weight is the uniform-stress patch test of Eq. (2.59), evaluated on a 6400-particle quarter block at $\Delta x_0 = 0.02$ with $h = 3\Delta x_0$ and the lattice laid from $\Delta x_0/2$ off each plane. Table 2.3 reports it *with a control*: the same lattice, the same masks, and the image sweeps disabled.

Table 2.3: Uniform-stress patch test across the symmetry planes. Spurious acceleration $\|\mathbf{a}\|$ normalised by $\|\boldsymbol{\sigma}\|/(\rho h)$. The right-hand column is the identical measurement with the image sweeps disabled.

Row measured	Planes on	Planes off, same lattice
Interior	4.10×10^{-6}	4.18×10^{-6}
On the y plane	3.62×10^{-6}	1.29
On the z plane	3.71×10^{-6}	1.29
In the corner	4.18×10^{-6}	1.29

The near-plane rows are as quiet as the interior when the images are visited and approximately 350 000 times louder when they are not, while the interior is quiet in both columns. The invariance of the interior is what establishes that the masks are honest and that the difference measured is the plane rather than the selection: a module reporting only the left-hand column would pass equally well against a solver that had silently stopped mirroring.

Three further results are recorded because each answers a distinct question.

Completeness. Against a brute-force enumeration built over the stored particles and their three reflected copies, the traversal yields 457 458 weight-carrying directed pairs with a **deficit of zero and a surplus of zero**. Both directions matter: a deficit means the plane is still open, while a surplus means an image is visited twice, which reads as material concentrated on the plane rather than as a hole. The strict count is taken at $0.9h$ rather than at h , because an undeformed lattice at $h = 3\Delta x_0$ places whole shells of pairs at exactly $r = h$, and single precision in the kernel against double precision in the reference breaks those ties differently. That accounts for 4.31% of all pairs, every one of which lies in the weightless rim, and the rim membership is asserted per particle rather than excused.

Quarter against full. A second lattice, laid so that its rows fall exactly where the quarter model's rows and their images fall—at $\Delta x_0/2$ either side of a centre plane and out to the same distance, so that the two bodies share their outer free surfaces as well as their interiors—agrees with the quarter model on $\sum_j W_{ij} V_j$ to 2.0×10^{-6} , which is single-precision round-off, including 1.7×10^{-6} over the 960 particles lying within h of a plane. Sizing that outer extent wrongly by a single row is worth 40% and has nothing to do with mirroring; it was the first version of this check.

Axial momentum. Over 30 steps of a perturbed quarter block, $\sum_i m_i v_{x,i}$ drifts by exactly zero. The image pair forces are not antisymmetric within the stored set, since no reaction is applied to an image, but their x components cancel globally by the symmetry.

TRAP / OPERATIONAL NOTE: UNDER SYMMETRY PLANES, p_y AND p_z ARE NOT CONSERVATION STATEMENTS

The cut planes of a quarter model carry reactions, so the transverse momentum of the stored material is *supposed* to change, and a status line showing it drift is reporting the physics rather than a defect. The conservation statement is the one along the uncut direction, and it holds to round-off. This is the Cartesian reading of what Sec. 2.11 states about the radial momentum of a body of revolution, which is likewise a scalar flux and

not a conserved quantity. A useful side-signal does exist: on a deck symmetric under the exchange of y and z , the two transverse components must track one another, and on the three-dimensional impact deck they agree to within 2.4% at its committed duration of $2\ \mu\text{s}$. That signal degrades as the material fragments, because the two are then small residuals of large cancelling flows; it is informative early and meaningless late.

TRAP / OPERATIONAL NOTE: A QUARTER MODEL FILTERS THE FRAGMENTATION IT WAS BUILT TO SEE

Two symmetry planes admit only the azimuthal modes $m = 0, 4, 8, \dots$ and suppress $m = 1, 2, 3, 5, \dots$ by construction. Real hypervelocity debris clouds fragment at the low modes. Nothing warns about this and the resulting cloud looks entirely plausible: a quarter model is a valid solution of the *symmetrised* problem, and a sound cross-check of the axisymmetric formulation, but it is not a fragment-distribution prediction and nothing read off it should be quoted as one.

2.10 Time Integration and Adaptive Timestepping

2.10.1 The Symplectic Euler Update

The governing equations are integrated with the symplectic (semi-implicit) Euler scheme, in the kick–drift ordering

$$\mathbf{v}_i^{n+1} = \mathbf{v}_i^n + \Delta t \mathbf{a}_i^n, \quad (2.67)$$

$$\mathbf{x}_i^{n+1} = \mathbf{x}_i^n + \Delta t \mathbf{v}_i^{n+1}. \quad (2.68)$$

The position update consumes the *already updated* velocity, which is what makes the scheme symplectic and gives it the bounded, non-secular energy error characteristic of that class. It is nonetheless **first-order accurate**, and the discrete energy error it commits per step has a closed form that Sec. 2.11.2 derives and the solver measures. Under hourglass super-time-stepping the single call is split into kick, damp, and drift so that the stabiliser acts on the kicked velocity and the position is taken with what the stabiliser leaves (Sec. 2.7).

2.10.2 The Adaptive Timestep and the Clamp Chain

The adaptive time step is the minimum of an acoustic limit, a force limit, and a chain of scheme-specific caps, each of which can only lower it further:

$$\Delta t = \min \left(\text{CFL} \cdot \frac{h}{c_{\max}}, \text{CFL} \cdot \sqrt{\frac{h}{a_{\max}}}, \Delta t_{\text{JWL}}, \Delta t_{\text{MG}}, \Delta t_{\text{bulk}}, \Delta t_{\Pi}, \Delta t_{\text{HG}} \right), \quad (2.69)$$

where $c_{\max} = \max_i(c_i + \|\mathbf{v}_i\|)$ and $a_{\max} = \max_i \|\mathbf{a}_i\|$. In solid media the signal speed is floored at the dilatational wave speed $c_p = \sqrt{(K + \frac{4}{3}G)/\rho_0}$. The order of the chain is not arbitrary: the JWL products' acoustic limit is evaluated first because their sound speed swings by an order of magnitude over a detonation run, and the hourglass cap is evaluated last because super-time-stepping sizes its stage count from the Δt every preceding clamp has already reduced.

2.10.3 Where the Energy Budget Is Banked, and Why the Position Matters

Time integration is also where the energy budget of Sec. 2.11 is accumulated, and the placement of each accumulation inside the step is a correctness requirement rather than a matter of convenience.

Two conditions must hold simultaneously: the per-particle powers being banked must be multiplied by the Δt that the advection will actually take, and the velocities and accelerations those powers were built from must not yet have been overwritten by Eqs. (2.67)–(2.68).

In the **fluid solver** there is no offset between the force evaluation and the advection, so the banking sits immediately after the adaptive step and immediately before the kick. Banking earlier would multiply this step's power by the *previous* step's Δt ; on a run whose timestep moves by tens of per cent — which is what an adaptive acoustic-and-force CFL does the moment a dam-break front reaches a wall — that is a first-order error integrated over tens of thousands of steps.

In the **solid solver** the energy rate was accumulated by the *previous* step's force loop, and Δt still holds that step's value early in the substep, before Eq. (2.69) rewrites it. The banking therefore sits with the internal-energy update, near the top of the step. The offset is the convention on that solver, and adopting the fluid's placement there would pair this step's power with next step's timestep.

2.10.4 Diffusion Work: Accounting for the Regularisers

The δ -SPH and γ -SPH regularisers of Sec. 2.2 and Sec. 2.6 enter the volume evolution, not the momentum equation, so they perform no pairwise work and have no per-pair work conjugate to accumulate. They are nonetheless *dissipative*: diffusing mass across a pressure gradient converts stored compression energy into numerical entropy, and for a long time this appeared in the energy balance only as an unattributed residual — inferred from a snapshot difference rather than measured.

It is now path-integrated directly, as the work the diffusion does against thermodynamic pressure:

$$W_D = - \int_0^t \sum_i V_i p_i \frac{\mathcal{D}_i}{\rho_i} dt', \quad (2.70)$$

positive when the regulariser dissipates. Equation (2.70) makes no assumption about the equation of state and is therefore valid for the Tait, linear, polynomial Mie–Grüneisen, and JWL forms alike. It is accumulated inside the volume-evolution kernel, where $\mathcal{D}_i$ exists.

TRAP / OPERATIONAL NOTE: W_D Reads the Previous Step's Pressure

The accumulation of Eq. (2.70) happens in the volume-evolution kernel, but p_i is not written until the force kernel later in the same substep. The pressure multiplying $\mathcal{D}_i$ is therefore the value left by the *previous* step. This is the same one-step offset the shift and the deviatoric update already carry, and it is not an error — but it does make W_D a first-order quadrature rather than an exact path integral, and it is the reason W_D is reported as a measured channel rather than as an identity.

TRAP / OPERATIONAL NOTE: The Attribution Degrades With Time, Even Though the Total Does Not

The total energy $E(t)$ is a state function and is exact at every instant regardless of Δt . Its *apportionment* into channels is not: the viscous work is banked as a left-endpoint quadrature and is therefore first order. On `dambreak2d_dsph.json` out to $t = 0.6$ s, the measured viscous work reads -6.50% , -6.22% , -6.10% of E_0 at CFL 0.3, 0.15, 0.075, so the committed CFL overstates the true viscous loss by about 7%. Out to $t = 2.0$ s the same ladder reads -37.0% , -31.9% , -31.5% against a symplectic leftover of $+25\%$ at CFL 0.3: two large terms of opposite sign whose *sum* is right to a few per cent while neither is individually worth quoting. Refine the timestep before reading any single channel at late

time.

2.11 Energy and Momentum Conservation

sigmaSPH reports a closed energy balance and a total-momentum balance at the end of every run. This section defines both, distinguishes the quantities that are conserved state functions from those that are path-integrated measurements, and states the reasoning that fixes the placement of each term. The two solvers conserve *different* totals, and conflating them is the most frequent source of misinterpretation of the reported output.

2.11.1 Two Solvers, Two Totals

The weakly compressible fluid solver carries **no internal-energy variable**. The quantity it conserves is the sum of kinetic, stored elastic and gravitational potential energy,

$$E_{\text{fluid}} = \underbrace{\sum_i \frac{1}{2} m_i \|\mathbf{v}_i\|^2}_{\text{KE}} + \underbrace{\sum_i m_i e(\rho_i)}_{\text{SE}} + \underbrace{\sum_i (-m_i \mathbf{g} \cdot \mathbf{x}_i)}_{\text{PE}}. \quad (2.71)$$

The stored term SE is a necessary constituent of this total rather than an optional diagnostic. The Tait equation of state is barotropic — the pressure depends on the density alone — so the compression work is a state function of the density and is recovered in full upon expansion:

$$e(\rho) = \int_{\rho_0}^{\rho} \frac{p(r)}{r^2} dr. \quad (2.72)$$

The hypoelastic solid solver instead integrates an internal energy as the work conjugate of the pair forces it applies, and conserves

$$E_{\text{solid}} = \underbrace{\sum_i \frac{1}{2} m_i \|\mathbf{v}_i\|^2}_{\text{KE}} + \underbrace{\sum_i m_i e_{\text{int},i}}_{\text{IE}} + \underbrace{\text{HE}}_{\text{hourglass}} + \underbrace{\sum_i (-m_i \mathbf{g} \cdot \mathbf{x}_i)}_{\text{PE}} - \underbrace{W_{\text{left}}}_{\text{integrator}}. \quad (2.73)$$

The corresponding balance identities differ in one structural term:

$$E_{\text{fluid}}(t) - E_{\text{fluid}}(0) = W_{\text{visc}} - W_{\text{wall}} + W_{\text{left}} + R, \quad (2.74)$$

$$E_{\text{solid}}(t) - E_{\text{solid}}(0) = \quad \quad - W_{\text{wall}} \quad \quad + R. \quad (2.75)$$

TRAP / OPERATIONAL NOTE: The Solid Balance Contains No Viscous Term, and Adding One Double-Counts

Equation (2.75) omits W_{visc} by construction. The fluid solver must name its viscous loss as a path term because it possesses no reservoir in which to deposit the dissipated energy; in the solid solver the viscous heat is already contained in IE, placed there by the same pair force that removed it from KE, so the budget never observes it leave the total. Subtracting it a second time counts it twice. This structural difference is the reason that advancing one dam-break configuration through both solvers is the only reliable means of distinguishing a solver that conserves energy from one that merely fails to account for the energy it loses.

2.11.2 The Discrete Leftover of the Symplectic Integrator

A single step of Eqs. (2.67)–(2.68) does not advance the kinetic energy by the work of the applied force. Expanding the kick,

$$\frac{1}{2}m\|\mathbf{v} + \Delta t \mathbf{a}\|^2 - \frac{1}{2}m\|\mathbf{v}\|^2 = \underbrace{m\mathbf{v} \cdot \mathbf{a} \Delta t}_{\text{work of the force}} + \underbrace{\frac{1}{2}m\Delta t^2\|\mathbf{a}\|^2}_{\text{discrete leftover}}, \quad (2.76)$$

so the integrator introduces a surplus of order Δt^2 per step, which is first order per unit of physical time. **This surplus is neither a dissipation nor a defect**; it is an exact property of the discrete update, and the signature that identifies it is that halving Δt halves it. On `dambreak2d_dsph.json` the accumulated leftover halves at ratios of 2.08 and 2.07 across a CFL ladder of 0.3, 0.15 and 0.075, against the predicted value of 2.0.

The form evaluated by the solver removes the gravitational contribution:

$$W_{\text{left}} = \sum_i \frac{1}{2}m_i\Delta t^2 \left(\|\mathbf{f}_i\|^2 - \|\mathbf{g}\|^2 \right), \quad \mathbf{f}_i = \mathbf{a}_i - \mathbf{g}. \quad (2.77)$$

Equation (2.77) assumes a single uninterrupted kick. Under super-time-stepping the kick is split about the damper (Sec. 2.7.3), and the second half acts on a velocity the damper has already changed by $\Delta \mathbf{v}_{D,i}$. The two halves together then add

$$W_{\text{left}}^{\text{STS}} = \sum_i m_i \Delta \mathbf{v}_{D,i} \cdot \frac{1}{2} \Delta t \mathbf{a}_i \quad (2.78)$$

more kinetic energy than Eq. (2.77) accounts for. This cross term is banked into the same reservoir, which keeps Eq. (2.73) closed. It is negative wherever the damper opposes the acceleration. Without it, the residual on the impact configuration would read -0.9% of E_0 at CFL 0.3; with it, -0.001% .

TRAP / OPERATIONAL NOTE: Why $\|\mathbf{f}\|^2 - \|\mathbf{g}\|^2$ Rather Than $\|\mathbf{a}\|^2$

On a configuration without gravity the two expressions are bit-identical: $\mathbf{g} = \mathbf{0}$ gives $\mathbf{f} = \mathbf{a}$, and both the subtraction and the term $-\|\mathbf{g}\|^2$ are exact in floating-point arithmetic. In the presence of gravity the two expressions differ in **sign**. For a particle in free fall $\mathbf{f} = \mathbf{0}$, so the true leftover is $-\frac{1}{2}m\Delta t^2\|\mathbf{g}\|^2$ and the total must *gain* that amount for KE + IE + PE to remain flat, whereas the $\|\mathbf{a}\|^2$ expression subtracts the same magnitude. On the dam-break column the $\|\mathbf{a}\|^2$ expression therefore doubles the drift instead of removing it, by exactly a factor of two, as measured. The discrepancy is the gravitational potential, for which the solid solver maintains no separate reservoir, and which is where a dam-break configuration begins.

Neither solver adds the leftover correction into the internal energy state. Both solvers evaluate Eq. (2.77) and report it, and they differ only in where the resulting quantity is placed. The fluid solver leaves it on the right-hand side of Eq. (2.74), having no internal energy in which to hold it. The solid solver carries it *within* its total, as the term $-W_{\text{left}}$ of Eq. (2.73), entering negatively because it represents energy that the integrator added fictitiously to the kinetic energy rather than energy held by the material; this is the reason that Eq. (2.75) carries no leftover term of its own.

TRAP / OPERATIONAL NOTE: Why the Leftover Occupies a Reservoir Rather Than the State

Two bookkeeping conventions flatten the reported total by identical arithmetic. The convention adopted here accumulates Eq. (2.77) in a reservoir of its own and subtracts that reservoir in Eq. (2.73). The alternative subtracts the same quantity from the internal-energy rate of each particle, so that IE itself contains $-W_{\text{left}}$ and the total requires no explicit leftover term. The two agree on E_{solid} to round-off.

They do not agree on the meaning of e_{int} . Subtracting an integrator artefact from the state removes it from every particle, including particles that dissipate nothing and therefore possess nothing from which to subtract it, and the internal energy of those particles becomes negative. Under that convention $\sum_i m_i \min(e_{\text{int},i}, 0)$ measures -2.13% of E_0 on the dam-break configuration, distributed over 204 of 8100 particles, and -0.0709 , approximately 31% of E_0 , on the layered gelatin impact. No energy is created — the total remains conserved — but e_{int} is then a bookkeeping variable rather than an internal energy, and no temperature can be derived from it.

The reservoir convention leaves e_{int} a true path integral of the work performed by the pair forces, at no cost in conservation. Relative to the alternative, $\sum_i m_i \min(e_{\text{int},i}, 0)$ falls by a factor of 62 on the dam-break configuration, and to *exactly* zero on its shock-equation-of-state variant. Stability is unaffected on reactive configurations: on `detonation2d_axi.json` the reservoir convention reports a drift of $+0.271\%$ of E_0 against the alternative's $+0.291\%$.

One limitation survives. The centred ALE transport can still drive e_{int} below zero across a steep gradient, independently of the integrator leftover; on the layered gelatin impact the negative-energy deficit falls only by a factor of 1.6. The guard at the equation of state is therefore still required, and a derived temperature will sit at its floor on the affected particles.

TRAP / OPERATIONAL NOTE: The Stored Velocity Is the Half-Step Velocity, and Re-Timing the Energy Terms Does Not Remove the Leftover

The update of Eqs. (2.67)–(2.68) is kick–drift–kick leapfrog in which the stored velocity lies half a step behind the positions. A velocity-Verlet integrator, which would place velocities and positions at the same time level, therefore produces the same trajectory, and the leftover is not a consequence of the choice of integrator. What could be changed is the time level at which the energy terms are evaluated: reading them at the synchronised velocity $\mathbf{v} + \frac{1}{2}\Delta t \mathbf{a}$ with every force left on the stored velocity. Evaluating the solid stress power there would move the leftover into the wrong store. On `dambreak2d_solid_tait`, whose compression energy is a state function against which the internal energy can be checked, the gap $\text{IE} + W_{\text{visc}} - \text{SE} - W_D$ would change from $+2.3\%$ to -17.0% of E_0 , and $\sum_i m_i \min(e_{\text{int},i}, 0)$ from -0.6% to -6.6% . On that configuration the conservative share of the leftover is genuine excess kinetic energy, which no change of bookkeeping removes; the trapbox that follows shows that this is a property of the configuration, not of the scheme. Evaluating the fluid solver's viscous work at the exact midpoint would converge the W_{visc} channel at the committed CFL (-32.0% against a converged -31.4% of E_0 at $t = 2$ s, where the left-endpoint form reads -37.0%), at a lower tracking cost ($+13\%$ against $+18\%$ of the step time); the left-endpoint form is retained because the sum of the channels is already correct and the timestep ladder above already bounds the attribution. All of these figures are single runs.

TRAP / OPERATIONAL NOTE: The Leftover Is Real Energy Only Where Something Makes It So

Whether W_{left} is fictitious kinetic energy or mere bookkeeping can be measured, and the answer depends on the configuration. Reading every term at the synchronised velocity separates the leftover into two parts. The first is the drift of the mechanical energy (kinetic plus the recoverable store the state implies) net of every banked dissipation: this is *real* injection. The second is the drift of the priced internal energy away from that store, which is bookkeeping and telescopes away. Measured this way (`tools/centred_continuity_study.py`):

- **In free fall** the whole leftover is an artefact of reading kinetic energy at the half-step velocity. At the synchronised velocity, $\text{KE} + \text{PE}$ is conserved to 10^{-5} .
- **On a freely breathing Tait patch**, and on the free oscillating bar, the leftover grows to several times the mechanical energy while the mechanical energy holds to better than 1%. It is entirely bookkeeping.
- **On the committed dam break** it is real, 0.97–1.04 times W_{left} , converging at first order. The real part is produced where particles strike the reflecting wall, which clamps positions and reflects the half-step velocity. δ -SPH, the viscosity, gravity and the pressure clip all reduce it rather than produce it.
- **On an unconfined solid impact** it is real without any wall. It is produced at contact, and most of it came from the super-time-stepped damper rewriting the half-step velocity after the kick. That is the reason for the placement of Sec. 2.7.3, which removes 90% of it.

The common factor is an operator that rewrites the half-step velocity after the kick and banks its work at that level — the wall’s reflection, or the damper in its former position — acting during sudden contact loading. Time-centred alternatives to the integrator were measured as well: a kick–drift–kick predictor–corrector, and centring the geometry of the continuity and velocity-gradient operators. None was adopted. The predictor–corrector removes the leftover but amplifies undamped acoustic modes at every Δt , and centring the geometry changes little. The records are `reports/centred_continuity_2026-09-26.md` and the summary cited above.

2.11.3 Composition of the Internal-Energy Reservoir

For the solid solver, IE is a single integrated reservoir, but its constituents are not opaque: each term summed into it is also accumulated separately and reported on its own line, so that the fraction of the heat which is physical may always be separated from the fraction which is numerical. The reservoir is

$$\text{IE} = \underbrace{W_{\text{pair}}}_{\text{stress pair-force work}} + \underbrace{W_{\text{visc}}}_{\text{artificial viscosity}}, \quad (2.79)$$

and the end-of-run table presents it as a nested partition:

Reported line	Meaning
Internal Energy (IE)	The reservoir of Eq. (2.79); <i>in</i> the total
$\hookrightarrow$ Stress Pair-Force Work	Mechanical work of the Cauchy stress divergence
$\hookrightarrow$ Plastic Work (PW)	$\sum V \sigma_{vM} d\bar{\varepsilon}_p$ through the return map
$\hookrightarrow$ Viscous Work (VW)	Monaghan Π_{ij} , bulk q , and Morris shear heating
Integrator Leftover (LW)	Eq. (2.77); its own reservoir, in the total
Compression Energy (SE)	Recoverable elastic store; diagnostic, not in the total
Negative Internal Energy (NE)	$\sum m \min(e_{\text{int}}, 0)$; a state, not a budget term

Three entries in this table require particular care, each because it invites a specific misreading of the balance.

Plastic work is a share of IE rather than a store beside it. PW is path-integrated through the return map as $\sum V \sigma_{vM} d\bar{\varepsilon}_p$, with σ_{vM} evaluated on the *returned* deviator, that is, at the flow stress at which the increment was actually produced. This makes PW the fully implicit plastic work of the map itself rather than an independent quadrature of it, and it is physical, irreversible heat. The return map nevertheless lowers the deviator with which the stress divergence subsequently performs work, so these joules reach e_{int} through the ordinary pair force and are *already contained* in W_{pair} . Adding PW to the total counts it twice.

Compression energy is an independent estimate rather than a separate reservoir. For a hypoelastic material the recoverable strain energy is already contained in IE, because IE is the work conjugate of the *total* stress. SE is reported so that the difference $W_{\text{pair}} - \text{SE} - \text{PW}$ may be read as that portion of the mechanical work which was dissipated into numerical entropy by the δ -SPH mass diffusion and by the ALE shifting. For a strengthless material ($G = 0$) it vanishes identically.

Negative internal energy is reported but does not constitute a conservation failure. A negative specific energy would carry the product $\omega \rho e$, and with it the JWL and Mie–Grüneisen pressures, into a branch that has no physical content. That statement concerns the *pressure*, and the condition is accordingly enforced at the pressure: the equation-of-state accessor returns $\max(e_{\text{int}}, 0)$ at the six consumers for which the sign matters. The state itself is never clamped.

TRAP / OPERATIONAL NOTE: Clamping e_{int} at the State Would Constitute an Energy Source

The guard is placed at the consumer rather than at the state because clamping e_{int} to zero within the internal-energy update injects energy. The ALE transport re-creates the deficit on the following step, so the clamp banks the same shortfall repeatedly: on the shock-equation-of-state dam-break it injects +2.04% of E_0 over 0.6 s against a *standing* deficit of only −2.13%, converting a loss of 1.3% into a gain of 1.0%. Guarding the consumer instead leaves $\sum_i m_i e_{\text{int},i}$ a true path integral, and $\text{KE} + \text{IE} + \text{HE} + \text{PE}$ then closes to round-off on configurations whose equation of state reads the internal energy. The behavioural consequence is real and is correct: a particle whose energy has become negative must be repaid by positive work before its pressure leaves the floor.

2.11.4 Terms Outside the Reservoir

Three further channels are measured directly and lie outside IE.

Hourglass work (HE) is energy that the rate damper removed from the velocity field and deliberately did *not* deposit into e_{int} . Restoring it to the sum is what renders Eq. (2.73) a conserved quantity rather than a damped one. It vanishes identically whenever `Hourglass.alpha` is zero. Under super-time-stepping the damper is a standalone multi-stage operator rather than a pair force, so that no per-pair conjugate exists; HE is then measured by bracketing the operator

with kinetic-energy snapshots, which for an operator that writes nothing but $\mathbf{v}$ is precisely the definition of HE, and which makes the measurement *exact* where the explicit path's $\sum \mathbf{f} \cdot \mathbf{v} \Delta t$ is a first-order quadrature. Because the damper acts between the two half-kicks (Sec. 2.7.3), the snapshots are taken at the synchronised velocity, the same time level as every other term.

TRAP / OPERATIONAL NOTE: An Unmeasured HE Can Determine the Sign of the Entire Balance

The instrumentation of the damper under super-time-stepping is not a refinement of an already small term. If the multi-stage operator is left uninstrumented, HE is reported as identically zero on every super-time-stepped configuration while the damper continues to remove energy from the velocity field, and the magnitude of what goes unrecorded is of the same order as the balance itself. On the free axisymmetric impact the damper removes 4.5 J against a net loss of 3.8 J. On the layered gelatin stack HE amounts to 49% of E_0 , and omitting it reports a balance of -22.3% in place of $+26.8\%$: the sign of the reported energy balance is determined by this single term.

Boundary wall dissipation (WW) is accumulated at the reflecting wall, where the one-sided bounce removes exactly $\frac{1}{2}m(1 - c_f^2)v_n^2$ per impact, for any unit normal and therefore at a corner as well as at a face. The sign convention is that a *positive* value denotes energy removed, so the balance restores it. The coefficient c_f is `Domain.restitution`, with default 0.5; the value $c_f = 1$ gives a specular wall that conserves $\|\mathbf{v}\|$ exactly. The default is dissipative because the position clamp applied at the same wall is an energy *source* — displacing a particle onto the face plane compresses the local density without moving its neighbours — and no other term in the loop removes that contribution.

Diffusion work (W_D) is given by Eq. (2.70) and is discussed in Sec. 2.10.4.

Whatever remains after all of these contributions have been accounted for constitutes the residual R : the δ -SPH and γ -SPH density diffusions beyond what W_D captures, the particle shift together with its ALE transports, the wall's position clamp, and the discrepancy between the symmetric-volume pressure force and the variational one. In axisymmetry R additionally contains the measure gaps of Sec. 2.9, which arise because the pair-work conjugacy holds in the meridional-area measure whereas KE and IE are summed in the ring-mass measure; these are reported as separate instruments rather than as balance terms, since they represent energy that the total has *already* acquired.

2.11.5 Temperature, Derived From the Internal Energy

Temperature is derived from the internal energy, per particle:

$$T_i = T_{0,i} + \frac{e_{\text{int},i} - e_{\text{cold}}(\rho_i) - e_{0,i}}{C_{v,i}}. \quad (2.80)$$

Each subtraction removes a contribution that does not represent heat. The term $e_{\text{cold}}(\rho)$ is the recoverable compression energy of Eq. (2.72), extended to the polynomial branch of the Mie–Grüneisen equation of state; a compressed particle holds this energy without being hot. The term e_0 is the equation of state's *reference* internal energy, to which e_{int} is initialised and which would otherwise be read as a temperature at $t = 0$. The datum T_0 is the temperature from which the rise is measured. At $t = 0$ a particle occupies the reference density ρ_0 , so that $e_{\text{cold}} = 0$ and $e_{\text{int}} = e_0$, and $T = T_0$ *exactly*; the regression suite asserts this identity rather than approximating it.

Equation (2.80) is meaningful because e_{int} is a true path integral of the work performed by the pair forces. It therefore carries shock heating, delivered through the artificial viscosity, together with plastic work and elastic compression in a single quantity, which is the sum that a

hypervelocity impact requires. On a layered gelatin impact it yields a peak of 310.7 K against a datum of 293 K after 4000 steps.

The specific heat C_v is `Materials[] .specificHeat`, a thermodynamic property of the material. It is deliberately not a property of any strength model: were it held within the Johnson–Cook parameter block, a shock-equation-of-state target carrying a linear strength model, or no strength model at all, would possess no heat capacity and therefore no temperature — and that configuration is precisely the hypervelocity case of interest. A material for which $C_v = 0$ retains $T \equiv T_0$.

TRAP / OPERATIONAL NOTE: What Has No Temperature, and What χ Does Not Do

A JWL particle has no temperature. Detonation products possess no cold curve from which to integrate — their energy is chemical, released by the burn, and there is no reference state — so Eq. (2.80) is not evaluated and the particle retains T_0 . This is a statement about the model rather than an omission from it.

Johnson–Cook maintains no temperature field of its own. Its adiabatic increment $\Delta T = \chi \sigma_y \Delta \bar{\epsilon}_p / (\rho_0 C_p)$ is computed by the return map and **discarded**, because the same plastic heat already reaches e_{int} through the ordinary stress pair force — PW being a share of IE rather than a store beside it, as Sec. 2.11.3 sets out — so that crediting it a second time would double-count. The return map still *reads* T for thermal softening; it does not write it. The consequence is that the Taylor–Quinney fraction is implicitly $\chi = 1$: e_{int} holds the whole of the plastic work rather than the fraction χ of it, and the remainder is not modelled as stored dislocation energy. At $\chi = 0.9$ this overstates the plastic temperature rise by approximately 11%, which is a small price for having shock heating and viscous heating expressed in the same quantity.

Finally, e_{int} may still become slightly negative where the centred ALE transport undershoots a steep gradient, and Eq. (2.80) then reports a temperature below T_0 on the affected particles. The equation-of-state guard bounds the *pressure* consequence of this; it does not bound the reported temperature.

2.11.6 The Momentum Balance

A total-momentum balance is printed beside the energy table, and it is by far the shorter of the two because **momentum possesses no dissipation channel**. Every dissipative term in the scheme — both artificial viscosities, the hourglass damper, the J_2 return map and the particle shift — is pair-antisymmetric by construction and transports no momentum. Where the energy table must separate energy that was legitimately dissipated from energy that was spuriously created, the momentum table requires only a value and a datum, which makes $\sum_i m_i \mathbf{v}_i$ the sharper of the two instruments although it is the less informative one.

It is evaluated in `float64`, with m and $\mathbf{v}$ read together on every call and never cached across the run. Two denominators are offered, because the natural one is degenerate: the quotient $\% \text{ of } \|\mathbf{p}_0\|$ is the appropriate measure for a configuration containing a moving impactor, but it is exactly zero for any configuration beginning at rest and for any symmetric problem, so the gross momentum $\sum_i m_i \|\mathbf{v}_i\|$ is printed as a fallback; a drift amounting to 1% of it indicates that momentum equivalent to 1% of the material’s gross momentum has been supplied by no physical agent. Where an external agent does act, the table quantifies its contribution: with gravity active it reports the impulse Mgt owed over the elapsed time together with the residual that remains once this impulse has been removed, which renders the row verifiable rather than merely excused. On `dambreak2d_dsph.json` at 300 steps that residual is -1.03×10^{-3} against a momentum change of -3.27×10^3 , that is, three parts in 10^7 .

TRAP / OPERATIONAL NOTE: In Axisymmetry Only the Axial Row Is a Conservation Statement

With m denoting a ring mass per radian, p_x is the axial momentum of the entire body of revolution divided by 2π — the quantity that the conservative axial form exists to hold, and which is conserved to round-off. The radial row is **not** a component of the three-dimensional momentum vector; that component vanishes identically by symmetry, the outward momentum of every ring cancelling against the far side of the same ring. The quantity printed is $\sum_i m_i v_{r,i} = (1/2\pi) \int \rho v_r dV$, a scalar measure of net radial flux which is genuinely non-zero, is not conserved, and is driven directly by the hoop source $(\sigma_{rr} - \sigma_{\theta\theta})/r$. It grows as a crater opens and reverses as hoop tension draws the material back, and that sign reversal is physics rather than a leak. For the same reason $\|\mathbf{p}\|$ is deliberately not printed in axisymmetry: doing so would place the conserved p_x beneath the same square root as the unconserved radial flux and report their combination as a large violation.

3 Material Constitutive Models

3.1 Overview of Material Separation

In **sigmaSPH**, every physical property belongs exclusively to a named entry in the **Materials**[] card array. The simulation solver card (**Solver**) governs only the numerical regularisers (such as artificial viscosity, bulk shock viscosity, and hourglass damping).

A material entry defines its reference mass density ρ_0 , a volumetric equation of state (**eos**), and an optional solid strength model (**strength**). The presence of a **strength** card identifies the material as a solid; omitting it defines an inviscid fluid ($G = 0$).

3.2 Equations of State (EOS)

The volumetric response determines the hydrostatic thermodynamic pressure p from current density ρ (and specific internal energy e).

3.2.1 Linear Equation of State

The linear EOS represents acoustic linear compressibility and is standard for metallic media under moderate dynamic pressures:

$$p(\rho) = K \left(\frac{\rho}{\rho_0} - 1 \right), \quad (3.1)$$

where K is the bulk modulus. For solid materials declaring Young's modulus E and Poisson's ratio ν , the bulk modulus is derived analytically:

$$K = \frac{E}{3(1 - 2\nu)}. \quad (3.2)$$

Writing an explicit **bulkModulus** beside an elastic strength model is rejected as redundant. The linear bulk acoustic sound speed is constant:

$$c_0 = \sqrt{\frac{K}{\rho_0}}. \quad (3.3)$$

3.2.2 Tait Weakly Compressible Equation of State

For free-surface hydrodynamics, the Tait equation of state models a stiff, weakly compressible fluid:

$$p(\rho) = \frac{c_0^2 \rho_0}{\gamma_{\text{eos}}} \left[\left(\frac{\rho}{\rho_0} \right)^{\gamma_{\text{eos}}} - 1 \right], \quad (3.4)$$

where c_0 is the artificial reference sound speed (chosen to enforce Mach number $\text{Ma} \leq 0.10$) and γ_{eos} is the isentropic exponent (typically $\gamma_{\text{eos}} = 7$ for water). Setting $\gamma_{\text{eos}} = 1$ recovers the linear EOS identically. The local density-dependent sound speed evaluates to:

$$c(\rho) = c_0 \left(\frac{\rho}{\rho_0} \right)^{\frac{\gamma_{\text{eos}} - 1}{2}}. \quad (3.5)$$

3.2.3 Jones–Wilkins–Lee (JWL) Products Equation of State

High-explosive detonation products expand from megabar Chapman–Jouguet pressures down to atmospheric levels. The JWL equation of state defines pressure as a function of relative specific volume $V = \rho_0/\rho$ and specific internal energy per unit mass e :

$$p(V, e) = A \left(1 - \frac{\omega}{R_1 V}\right) \exp(-R_1 V) + B \left(1 - \frac{\omega}{R_2 V}\right) \exp(-R_2 V) + \frac{\omega \rho_0 e}{V}, \quad (3.6)$$

where A , B , R_1 , R_2 , and ω are empirical material calibration constants, and ρ_0 is the unreacted explosive reference density.

Specific Internal Energy Evolution

The internal energy per particle e_i evolves according to the first law of thermodynamics under mechanical $p dV$ work:

$$\frac{de_i}{dt} = -\frac{p_i}{\rho_i} \left(\frac{d\varepsilon_{\text{vol},i}}{dt} \right)_{\text{mech}} + \left(\frac{de_i}{dt} \right)_{\text{ALE}}, \quad (3.7)$$

where $(d\varepsilon_{\text{vol},i}/dt)_{\text{mech}} = \sum_j V_j (\mathbf{v}_i - \mathbf{v}_j) \cdot \nabla W_{ij}$ denotes the reversible mechanical volume change (excluding δ -SPH diffusion). The ALE transport increment follows Eq. (2.31).

3.2.4 Polynomial Mie–Grüneisen Equation of State

Under intense dynamic loading, high-velocity impact, and ballistic penetration, linear acoustic approximations ($p = K(\rho/\rho_0 - 1)$) fail to capture the steepening of compressive shock fronts and the substantial temperature and internal energy increments generated across shock discontinuities. The polynomial Mie–Grüneisen equation of state provides a robust, thermodynamically consistent description of condensed matter and soft tissue simulants (such as ballistic gelatin) under shock compression [2, 31].

Cubic Hugoniot Formulation

The thermodynamic pressure p is expressed as a cubic polynomial function of the volumetric compression ratio $\mu = \rho/\rho_0 - 1$, coupled to the internal energy per unit reference volume $E_{v01} = \rho_0 e$:

$$p(\mu, e) = C_0 + C_1 \mu + C_2 \mu^2 + C_3 \mu^3 + (C_4 + C_5 \mu) \rho_0 e, \quad \text{for } \mu \geq 0, \quad (3.8)$$

where ρ_0 is the reference density, e is the specific internal energy per unit mass, and C_0 through C_5 are polynomial constants. When the material is specified via standard shock Hugoniot properties under the linear shock velocity–particle velocity relation $U_s = c_0 + s u_p$, the polynomial coefficients are derived analytically [2]:

$$C_0 = P_0, \quad (3.9)$$

$$C_1 = \rho_0 c_0^2 - \frac{\Gamma_0}{2} P_0, \quad (3.10)$$

$$C_2 = (2s - 1) C_1, \quad (3.11)$$

$$C_3 = (s - 1)(3s - 1) C_1, \quad (3.12)$$

$$C_4 = \Gamma_0, \quad (3.13)$$

$$C_5 = \Gamma_0, \quad (3.14)$$

where c_0 is the bulk reference sound speed, s is the empirical linear Hugoniot slope, Γ_0 is the dimensionless reference Grüneisen parameter, and P_0 is the ambient reference pressure (defaulting to zero). At the zero-energy reference state ($\mu = 0, e = 0, P_0 = 0$), the initial bulk modulus evaluates to $K_0 = C_1 = \rho_0 c_0^2$.

Tensile Expansion Regime

In standard shock hydrocodes (such as AUTODYN and LS-DYNA), evaluating higher-order curvature terms ($C_2\mu^2$ and $C_3\mu^3$) in the expansion and rarefaction regime ($\mu < 0$) can generate unphysical non-monotonic stiffening or spurious negative pressures. When `linearExpansion` is active (the default), `sigmaSPH` truncates the nonlinear compressive terms in expansion:

$$p(\mu, e) = C_0 + C_1\mu + C_4\rho_0e, \quad \text{for } \mu < 0. \quad (3.15)$$

This formulation preserves the acoustic slope C_1 while preventing expansion instabilities.

Thermodynamic Sound Speed and Acoustic CFL Limit

The state-dependent isentropic sound speed $c(\rho, e) = \sqrt{(\partial p / \partial \rho)_s}$ is derived from the fundamental thermodynamic relation:

$$c^2(\rho, e) = \left(\frac{\partial p}{\partial \rho} \right)_e + \frac{p}{\rho^2} \left(\frac{\partial p}{\partial e} \right)_\rho. \quad (3.16)$$

Evaluating the partial derivatives from Eq. (3.8) and Eq. (3.15) yields:

$$\left(\frac{\partial p}{\partial \rho} \right)_e = \begin{cases} \frac{1}{\rho_0} [C_1 + 2C_2\mu + 3C_3\mu^2 + C_5\rho_0e], & \mu \geq 0, \\ \frac{C_1}{\rho_0}, & \mu < 0, \end{cases} \quad (3.17)$$

and

$$\left(\frac{\partial p}{\partial e} \right)_\rho = \begin{cases} (C_4 + C_5\mu)\rho_0, & \mu \geq 0, \\ C_4\rho_0, & \mu < 0. \end{cases} \quad (3.18)$$

At ambient conditions ($\mu = 0, e = 0, P_0 = 0$), this simplifies identically to $c = \sqrt{C_1/\rho_0} = c_0$. During the simulation, the local sound speed $c_i = c(\rho_i, e_i)$ is dynamically evaluated for every particle to enforce the explicit acoustic Courant–Friedrichs–Lewy (CFL) stability constraint:

$$\Delta t \leq C_{\text{CFL}} \frac{h}{\max_i (c_i + \|\mathbf{v}_i\|)}. \quad (3.19)$$

Internal Energy Coupling

Particles governed by the Mie–Grüneisen EOS track specific internal energy e_i , which evolves according to reversible volumetric work and irreversible plastic dissipation:

$$\frac{de_i}{dt} = -\frac{p_i}{\rho_i} \left(\frac{d\varepsilon_{\text{vol},i}}{dt} \right)_{\text{mech}} + \frac{\chi}{\rho_i} \mathbf{s}_i : \mathbf{D}_i^p + \left(\frac{de_i}{dt} \right)_{\text{ALE}}, \quad (3.20)$$

where χ is the Taylor–Quinney factor converting plastic work into heat, and $\mathbf{s}_i : \mathbf{D}_i^p$ is the rate of plastic dissipation. The pressure and the sound speed read the internal energy through the guard $\max(e_i, 0)$ (`e_for_eos`), so that a negative specific energy left by the centred ALE transport never takes the pressure through a branch without physics in it; the state e_i itself is integrated unclamped, as the work conjugate of the pair force (§2.11).

3.2.5 Holmquist–Johnson–Cook Compaction Equation of State (EOS004)

Every other solid in the library is a metal, whose volume change is elastic. Concrete is porous: its pores collapse under a few tens of megapascals and stay collapsed. The Holmquist–Johnson–Cook (HJC) model [17], used by Kim et al. [21] for all of their concrete targets, represents this with an irreversible compaction curve. It is one of three cards that make up the HJC model — `eos.type: "hjc"`, `strength.plasticity.model: "hjc"` (§3.3.6) and `damage.model: "hjc"` (§3.4.4) — which read each other’s constants and must be declared together; the configuration builder rejects any partial combination.

The Three-Phase Loading Curve

With the volumetric strain $\mu = \rho/\rho_0 - 1$, the loading envelope $P_{\text{env}}(\mu)$ has three phases:

$$P_{\text{env}}(\mu) = \begin{cases} K\mu, & \mu \leq \mu_c \quad (\text{OA, elastic}), \\ P_c + K_{\text{lock}}(\mu - \mu_c), & \mu_c < \mu \leq \mu_{pl} \quad (\text{AB, crushing}), \\ K_1\bar{\mu} + K_2\bar{\mu}^2 + K_3\bar{\mu}^3, & \mu > \mu_{pl} \quad (\text{BC, fully dense}), \end{cases} \quad (3.21)$$

with the crush point $(\mu_c, P_c) = (\text{crushStrain}, \text{crushPressure})$, the locking point $(\mu_{pl}, P_l) = (\text{lockStrain}, \text{lockPressure})$, and

$$K = \frac{P_c}{\mu_c}, \quad K_{\text{lock}} = \frac{P_l - P_c}{\mu_{pl} - \mu_c}, \quad \bar{\mu} = \frac{\mu - \mu_L}{1 + \mu_L}. \quad (3.22)$$

The elastic bulk modulus K is derived and not read. The strain μ_L from which the fully dense curve is measured is the locking strain of the *grain* density. It is derived by requiring the curve to be continuous at B: the host solves

$$K_1\bar{\mu}_B + K_2\bar{\mu}_B^2 + K_3\bar{\mu}_B^3 = P_l \quad (3.23)$$

for $\bar{\mu}_B$ by bisection, and sets $\mu_L = (\mu_{pl} - \bar{\mu}_B)/(1 + \bar{\mu}_B)$. On the 150 MPa concrete this gives $\mu_L = 0.0895$.

Unloading and the Compaction History

The curve is irreversible. Each particle carries the largest compression it has reached, μ_{max} , in a sorted per-particle field `hjc_mu_max` allocated only when an HJC material is declared. Loading and unloading are distinguished by μ against μ_{max} : a particle with $\mu \geq \mu_{\text{max}}$ is on the envelope and advances μ_{max} ; one with $\mu < \mu_{\text{max}}$ is on the unloading line through the point it left the envelope from,

$$P = P_{\text{env}}(\mu_{\text{max}}) - K_u(\mu_{\text{max}})(\mu_{\text{max}} - \mu), \quad (3.24)$$

with the unloading modulus interpolated between the elastic modulus at A and the fully dense modulus $K'_1 = K_1/(1 + \mu_L)$ at B:

$$K_u = \begin{cases} K, & \mu_{\text{max}} \leq \mu_c, \\ (1 - F)K + FK'_1, & F = \frac{\mu_{\text{max}} - \mu_c}{\mu_{pl} - \mu_c}, \quad \mu_c < \mu_{\text{max}} \leq \mu_{pl}, \\ K'_1, & \mu_{\text{max}} > \mu_{pl}. \end{cases} \quad (3.25)$$

The plastic volumetric strain is the zero-pressure intercept of the current unloading line,

$$\mu_p = \mu_{\text{max}} - \frac{P_{\text{env}}(\mu_{\text{max}})}{K_u(\mu_{\text{max}})}, \quad (3.26)$$

counted through AB only, since beyond B no porosity is left to crush. Because it is a function of μ_{max} alone, it needs no field of its own. The pressure is bounded below by the damaged tensile strength,

$$P \geq -T(1 - D), \quad (3.27)$$

with $T = \text{tensileStrength}$, and this cutoff is the whole of what damage does to an HJC pressure. The equation of state does not read the internal energy: the crush work is dissipation, and it reaches e_{int} through the pair force like any other work. It has no potential of density alone, so the compression-energy diagnostic is reported as incomplete on an HJC deck.

TRAP / OPERATIONAL NOTE: Four Departures from the Equations as Kim et al. Print Them

Each of the following is exercised by a check in `t40`.

1. **Where the dense curve starts.** The tabulated “locking volumetric strain” $U_{\text{lock}} = 0.1$ is the strain at B, μ_{pl} , and not the grain-density strain μ_L . Read as μ_L , it would make the pressure jump from P_l to zero at B; μ_L is therefore derived as above.
2. **Unloading in phase BC.** The printed “ $P = K_1 \bar{\mu}$ on unloading” does not depend on μ_{max} , so the pressure would jump the moment a dense particle begins to unload. Equation (3.24) passes through the point it leaves from.
3. **The dense unloading modulus** is $K'_1 = K_1/(1 + \mu_L)$ in both AB and BC. Using K_1 in AB and $K_1/(1 + \mu_L)$ in BC would make the unloaded pressure jump as μ_{max} crosses B; the difference is 8% on the 150 MPa set.
4. **The tensile strength branch** is continuous at every damage level (§3.3.6).

Loading against unloading is decided by μ against μ_{max} rather than by the plastic multiplier of Kim et al.’s eq. 31. The history criterion is exact for this equation of state and needs one stored scalar.

Sound Speed and Timestep

The static CFL floor is sized on the stiffest slope the curve reaches before full density, $\max(K, K'_1)$, and not on K : a crushed particle unloads at K'_1 , which on the 150 MPa set is 78.0 GPa against $K = 16.0$ GPa, giving a longitudinal wave speed

$$c_p = \sqrt{\frac{\max(K, K'_1) + \frac{4}{3}G}{\rho_0}} = 6.23 \text{ km/s} \quad (3.28)$$

rather than 3.99 km/s. The stiffening of the fully dense branch is followed per particle by `clamp_dt_hjc`, with $c^2 = \left[(1 + \mu) dP/d\mu + \frac{4}{3}G \right] / \rho$ evaluated on the tangent of Eq. (3.21) or (3.24).

3.3 Hypoelastic–Plastic Strength Formulation

Solid materials decompose the Cauchy stress tensor $\boldsymbol{\sigma}$ into a hydrostatic pressure p and a trace-free deviatoric stress tensor $\mathbf{s}$:

$$\boldsymbol{\sigma} = -p\mathbf{I} + \mathbf{s}, \quad \text{tr}(\mathbf{s}) = 0. \quad (3.29)$$

3.3.1 Kinematics and Velocity Gradients

The spatial velocity gradient tensor $\mathbf{L} \equiv \nabla \mathbf{v}$ is evaluated using the regularised inverse completeness matrix $\mathbf{L}_i$:

$$\nabla \mathbf{v}_i = \left[\sum_j V_j (\mathbf{v}_j - \mathbf{v}_i) \otimes \nabla W_{ij} \right] \mathbf{L}_i. \quad (3.30)$$

The rate of deformation tensor $\mathbf{D}$ and the spin (vorticity) tensor $\mathbf{W}$ are the symmetric and skew-symmetric parts of $\nabla \mathbf{v}$:

$$\mathbf{D} = \frac{1}{2} (\nabla \mathbf{v} + \nabla \mathbf{v}^T), \quad \mathbf{W} = \frac{1}{2} (\nabla \mathbf{v} - \nabla \mathbf{v}^T). \quad (3.31)$$

The deviatoric strain rate tensor is:

$$\dot{\mathbf{e}} \equiv \mathbf{D}' = \mathbf{D} - \frac{1}{3}\text{tr}(\mathbf{D})\mathbf{I}. \quad (3.32)$$

3.3.2 Objective Jaumann Stress Rate

To guarantee frame indifference under large rigid-body rotations, the deviatoric stress evolves according to the co-rotational Jaumann stress rate $\overset{\nabla}{\mathbf{s}}$:

$$\overset{\nabla}{\mathbf{s}} \equiv \dot{\mathbf{s}} - \mathbf{W}\mathbf{s} + \mathbf{s}\mathbf{W} = 2G\dot{\mathbf{e}}, \quad (3.33)$$

where $G = \frac{E}{2(1+\nu)}$ is the elastic shear modulus. Discretising over time step Δt yields the elastic trial stress $\mathbf{s}^{\text{trial}}$:

$$\mathbf{s}^{\text{trial}} = \mathbf{s}^n + [2G\dot{\mathbf{e}} + \mathbf{W}\mathbf{s}^n - \mathbf{s}^n\mathbf{W}] \Delta t. \quad (3.34)$$

To prevent numerical roundoff drift, $\mathbf{s}^{\text{trial}}$ is explicitly projected onto the symmetric trace-free subspace:

$$\mathbf{s}^{\text{trial}} \leftarrow \frac{1}{2} \left(\mathbf{s}^{\text{trial}} + (\mathbf{s}^{\text{trial}})^T \right) - \frac{1}{3}\text{tr}(\mathbf{s}^{\text{trial}})\mathbf{I}. \quad (3.35)$$

3.3.3 J_2 Linear Isotropic Hardening (MAT001)

Yielding is governed by the von Mises yield criterion with linear isotropic hardening. The von Mises equivalent stress is:

$$\sigma_{\text{eq}} = \sqrt{\frac{3}{2} \mathbf{s}^{\text{trial}} : \mathbf{s}^{\text{trial}}}. \quad (3.36)$$

The yield condition evaluates:

$$f(\mathbf{s}^{\text{trial}}, \bar{\varepsilon}_p) = \sigma_{\text{eq}} - \sigma_y(\bar{\varepsilon}_p) \leq 0, \quad (3.37)$$

where the current yield stress σ_y hardens with equivalent plastic strain $\bar{\varepsilon}_p$:

$$\sigma_y(\bar{\varepsilon}_p) = \sigma_{y0} + H\bar{\varepsilon}_p. \quad (3.38)$$

The plastic hardening modulus H is related to the tangent modulus $E_t = d\sigma/d\varepsilon$ by:

$$H = \frac{EE_t}{E - E_t}. \quad (3.39)$$

If $\sigma_{\text{eq}} > \sigma_y$, plastic yielding occurs. During the plastic increment $\Delta\bar{\varepsilon}_p$, the deviatoric stress relaxation $-3G\Delta\bar{\varepsilon}_p$ must match the expansion of the yield surface $+H\Delta\bar{\varepsilon}_p$. The radial-return plastic strain increment $\Delta\bar{\varepsilon}_p$ is solved analytically in a single step:

$$\Delta\bar{\varepsilon}_p = \frac{\sigma_{\text{eq}} - \sigma_y(\bar{\varepsilon}_p^n)}{3G + H}. \quad (3.40)$$

The deviatoric stress is returned radially to the updated yield surface:

$$\mathbf{s}^{n+1} = \left(1 - \frac{3G\Delta\bar{\varepsilon}_p}{\sigma_{\text{eq}}} \right) \mathbf{s}^{\text{trial}}, \quad \bar{\varepsilon}_p^{n+1} = \bar{\varepsilon}_p^n + \Delta\bar{\varepsilon}_p. \quad (3.41)$$

Because $0 \leq (\sigma_y + H\Delta\bar{\varepsilon}_p)/\sigma_{\text{eq}} < 1$, this closed-form projection is unconditionally stable.

3.3.4 Johnson–Cook Viscoplasticity (MAT002)

Under high-velocity impacts, penetrator dynamics, and explosive metal casing expansion, deformation occurs at extreme strain rates ($\dot{\varepsilon} > 10^3 \text{ s}^{-1}$) accompanied by substantial adiabatic heating. Under these conditions, the rate-independent linear hardening assumption overestimates flow resistance. The Johnson–Cook viscoplastic model [19] accounts for strain hardening, strain-rate sensitivity, and thermal softening via a multiplicative decomposition of the dynamic yield stress:

$$\sigma_y(\bar{\varepsilon}_p, \dot{\bar{\varepsilon}}_p, T) = \left(A + B\bar{\varepsilon}_p^n \right) \left[1 + C \ln \left(\max \left(1.0, \frac{\dot{\bar{\varepsilon}}_p}{\dot{\varepsilon}_0} \right) \right) \right] [1 - (T^*)^m], \quad (3.42)$$

where:

- A is the reference quasi-static yield stress at room temperature T_0 .
- B and n are the strain hardening modulus and exponent.
- C is the logarithmic strain-rate sensitivity coefficient.
- $\dot{\varepsilon}_0$ is the reference quasi-static strain rate (default: $1.0 \times 10^{-3} \text{ ms}^{-1} = 1.0 \text{ s}^{-1}$). The argument of the natural logarithm is clamped below by 1.0 so that low strain rates ($\dot{\bar{\varepsilon}}_p \leq \dot{\varepsilon}_0$) do not generate non-physical softening.
- T^* is the non-dimensional homologous temperature:

$$T^* = \text{clamp} \left(\frac{T - T_0}{T_m - T_0}, 0.0, 1.0 \right), \quad (3.43)$$

where T_0 is the reference room temperature (default: 293.0 K) and T_m is the melting temperature. When $T \geq T_m$, the thermal bracket vanishes ($1 - (T^*)^m = 0$), completely relaxing shear resistance. If $T_m \leq T_0$ (the default when omitted), thermal softening is inactive ($T^* \equiv 0$).

- m is the thermal softening exponent.

Newton–Raphson Radial Return Mapping

During explicit time integration over step Δt , the plastic strain rate is approximated as $\dot{\bar{\varepsilon}}_p = \Delta\bar{\varepsilon}_p/\Delta t$. Evaluating the homologous temperature at the start of the timestep ($T^* = T^{*,n}$), the discrete consistency condition requires:

$$g(\Delta\bar{\varepsilon}_p) \equiv \sigma_{\text{eq}} - 3G\Delta\bar{\varepsilon}_p - \sigma_y \left(\bar{\varepsilon}_p^n + \Delta\bar{\varepsilon}_p, \frac{\Delta\bar{\varepsilon}_p}{\Delta t}, T^{*,n} \right) = 0. \quad (3.44)$$

Because Eq. (3.42) contains non-integer power n and a logarithmic rate dependence, $g(\Delta\bar{\varepsilon}_p) = 0$ cannot be inverted analytically. The solver executes a local Newton–Raphson iteration on each GPU particle thread. The analytical derivative with respect to $\Delta\bar{\varepsilon}_p$ evaluates:

$$\frac{dg}{d\Delta\bar{\varepsilon}_p} = -3G - \left[\frac{\partial\sigma_y}{\partial\bar{\varepsilon}_p} + \frac{1}{\Delta t} \frac{\partial\sigma_y}{\partial\dot{\bar{\varepsilon}}_p} \right], \quad (3.45)$$

with:

$$\frac{\partial\sigma_y}{\partial\bar{\varepsilon}_p} = nB(\bar{\varepsilon}_p^n + \Delta\bar{\varepsilon}_p)^{n-1} \left[1 + C \ln \left(\frac{\Delta\bar{\varepsilon}_p}{\Delta t \dot{\varepsilon}_0} \right) \right] [1 - (T^*)^m], \quad (3.46)$$

$$\frac{1}{\Delta t} \frac{\partial\sigma_y}{\partial\dot{\bar{\varepsilon}}_p} = \left(A + B\bar{\varepsilon}_p^n \right) \frac{C}{\max(10^{-12}, \Delta\bar{\varepsilon}_p)} [1 - (T^*)^m]. \quad (3.47)$$

Initializing from the elastic predictor $\Delta\bar{\varepsilon}_p^{(0)} = \max\left(10^{-12}, \frac{\sigma_{\text{eq}} - \sigma_y^0}{3G}\right)$, the iteration converges quadratically:

$$\Delta\bar{\varepsilon}_p^{(k+1)} = \max\left(0.0, \Delta\bar{\varepsilon}_p^{(k)} - \frac{g(\Delta\bar{\varepsilon}_p^{(k)})}{g'(\Delta\bar{\varepsilon}_p^{(k)})}\right). \quad (3.48)$$

Convergence is achieved within 4 to 8 iterations across all practical engineering strain rates. Once converged, the deviatoric stress tensor is returned radially:

$$\mathbf{s}^{n+1} = \mathbf{s}^{\text{trial}} \frac{\sigma_{\text{eq}} - 3G\Delta\bar{\varepsilon}_p}{\sigma_{\text{eq}}}, \quad \bar{\varepsilon}_p^{n+1} = \bar{\varepsilon}_p^n + \Delta\bar{\varepsilon}_p. \quad (3.49)$$

Adiabatic Plastic Work Dissipation

Under shock and impact velocities, wave transit occurs over microseconds, rendering thermal diffusion negligible during the dynamic event. Plastic work is converted into heat adiabatically:

$$\rho_0 C_p \dot{T} = \chi \boldsymbol{\sigma} : \dot{\boldsymbol{\varepsilon}}_p = \chi \sigma_y \dot{\bar{\varepsilon}}_p, \quad (3.50)$$

where C_p is the specific heat capacity and χ is the Taylor–Quinney coefficient ($\chi = 0.90$ by default). Integrating over the plastic strain increment yields the temperature increment:

$$\Delta T = \frac{\chi}{\rho_0 C_p} \sigma_y^{n+1} \Delta\bar{\varepsilon}_p, \quad T^{n+1} = T^n + \Delta T. \quad (3.51)$$

The particle temperature T is stored as a persistent field (`temperature`) and is updated during the stress update and advected under ALE shifting corrections.

3.3.5 Hollomon Power-Law Hardening (Soft Tissue Surrogates)

Soft tissue surrogates such as ballistic gelatin (10% and 20% formulations) and synthetic hydrogels undergo substantial irreversible plastic shear deformation and cavitation under projectile impact, but thermal softening effects over the microsecond duration of cavitation expansion are negligible [2, 32]. Commercial hydrocodes (e.g. RADIOSS /MAT/LAW10 (HYD_JCOOK)) model this regime by omitting the thermal and strain-rate softening terms of the Johnson–Cook formulation, reducing the dynamic yield stress identically to the classical Hollomon power-law equation:

$$\sigma_y(\bar{\varepsilon}_p) = A + B \bar{\varepsilon}_p^n, \quad (3.52)$$

where A is the reference quasi-static initial yield stress (frequently designated as the *apparent yield stress* in ballistic penetration literature), B is the strain hardening modulus, and n is the dimensionless strain hardening exponent.

In `sigmaSPH`, this model is implemented without dedicated solver branching by parametrically uncoupling the general Johnson–Cook framework (§3.3.4): setting strain-rate sensitivity $C = 0.0$ and melting temperature $T_m = 0.0$ K reduces the rate and thermal multipliers in Eq. (3.42) strictly to unity. Setting $C_p = 0.0$ J/(kg · K) suppresses adiabatic heating, preserving constant ambient temperature. In scene decks, users can either select the model explicitly via `model: "hollomon"` (or `"power_law"`) under `Materials[] .strength.plasticity` with keys `A`, `B`, and `n`, or reference pre-calibrated library entries (§3.5).

Extreme Bulk-to-Shear Ratio and Hydrodynamic Decoupling

Ballistic gelatin combines a high volumetric shock stiffness governed by the polynomial Mie–Grüneisen EOS ($K_0 = \rho_0 c_0^2 \approx 2.38$ GPa) with an extremely small shear modulus ($E \approx 150$ – 282 kPa $\implies G \approx 51.7$ – 97.2 kPa), resulting in an extreme stiffness ratio:

$$\frac{K_0}{G} \approx 25\,000 \text{ to } 46\,000. \quad (3.53)$$

Because shear stiffness is four orders of magnitude smaller than shock compression stiffness, spurious zero-energy pairing and hourglass modes readily excite unless numerical regularisation is active. Users simulating ballistic gelatin must maintain both hourglass control (`Hourglass.alpha = 1.0`) and Particle Shifting Technology (`Shifting`) to prevent tensile and pairing lattice instabilities.

3.3.6 Holmquist–Johnson–Cook Pressure-Dependent Strength (MAT003)

The strength of concrete rises steeply with confinement, and a pulverised concrete keeps a frictional strength under compression. The HJC strength surface expresses both in terms of the normalised pressure $P^* = P/f_c$, the normalised tensile strength $T^* = T/f_c$ and the damage D of §3.4.4:

$$\sigma^* = \min \{ \sigma_0^* R(\dot{\varepsilon}), S_{\max} \}, \quad \sigma_0^* = \begin{cases} A(1 - D) + B(P^*)^N, & P^* \geq 0, \\ \max \left\{ 0, A(1 - D) + A \frac{P^*}{T^*} \right\}, & P^* < 0, \end{cases} \quad (3.54)$$

with the rate factor

$$R(\dot{\varepsilon}) = 1 + C \ln \dot{\varepsilon}^*, \quad \dot{\varepsilon}^* = \max \left(1, \frac{\dot{\varepsilon}_{\text{eff}}}{\dot{\varepsilon}_0} \right), \quad \dot{\varepsilon}_{\text{eff}} = \sqrt{\frac{2}{3}} \dot{\varepsilon} : \dot{\varepsilon}, \quad (3.55)$$

and the yield stress $\sigma_y = f_c \sigma^*$. The constants are `fc`, `A`, `B`, `N`, `C`, `Smax` and `eps0_dot` on the plasticity card; T is read from the equation-of-state card.

The cap $S_{\max}$ applies to the rate-scaled strength on both branches. The tensile branch of Eq. (3.54) departs from the form printed by Kim et al., $AP^*/(T^*(1 - D)) + A$, which meets the compressive branch $A(1 - D)$ at $P^* = 0$ only when $D = 0$. The form used here is continuous at every D , vanishes at the tensile cutoff $P^* = -T^*(1 - D)$ of Eq. (3.27), and coincides with the printed branch at $D = 0$.

Return map. There is no hardening in $\bar{\varepsilon}_p$, so the return is the one-shot radial return of perfect plasticity, Kim et al.’s eq. 22: $\Delta \bar{\varepsilon}_p = (\sigma_{\text{VM}} - \sigma_y)/(3G)$, evaluated by the shared `j2_radial_return` at zero hardening modulus. It reads the pressure stored at the end of the previous step, the same one-step offset the rest of the substep carries. Where σ_y itself is zero, at the tensile cutoff, the deviator is returned to the origin.

TRAP / OPERATIONAL NOTE: A Fully Damaged Concrete Particle Keeps Its Compressive Strength

Unlike a damaged metal (§3.4), an HJC particle at $D = 1$ is *not* returned to the origin. Pulverised concrete retains the frictional strength $B(P^*)^N$ under compression, which is the point of the model; only the cohesive term $A(1 - D)$ is removed. The strength card takes `shearModulus` alone: `youngsModulus` and `poissonRatio` are rejected on it, because the bulk modulus is the equation of state’s K and a second statement of it could only contradict the first.

3.4 Continuum Damage Mechanics and Dynamic Spallation

Under hypervelocity impact, high-explosive detonation, and intense laser shock loading, high-amplitude compressive shock waves propagate through metallic targets. Upon reaching unconstrained rear boundaries or free surfaces, these compressive pulses reflect as relief rarefaction fans. When release waves originating from opposed free boundaries intersect in the interior of a target, their constructive interference drives the local volumetric state into intense triaxial hydrostatic

tension ($\sigma_h = -p > 0$). In physical engineering solids, micro-voids nucleate at inclusions and grain boundaries, expand rapidly, and coalesce into macroscopic spall fracture surfaces.

3.4.1 The Tensile Bridging Trap in Standard SPH

In classical updated Lagrangian SPH, pairwise momentum interaction forces are governed by the symmetric pressure gradient:

$$\mathbf{a}_i^{\text{pressure}} = - \sum_j V_j \left(\frac{p_i}{\rho_i} + \frac{p_j}{\rho_j} \right) \nabla W_{ij}. \quad (3.56)$$

Under tensile expansion where $\rho < \rho_0$, standard volumetric equations of state (such as linear acoustic or polynomial Mie–Grüneisen relations) return negative pressures ($p < 0$). In standard SPH, this negative pressure generates a powerful pairwise attractive force between receding particles whose smoothing kernels overlap. Across nascent fracture planes, negative pressures reach several gigapascals (e.g. -3.5 GPa in aluminium or -5.0 GPa in tantalum), acting as an unphysical “elastic glue” that artificially tethers separated fragments together. Real voids cannot open, cracks cannot propagate, and detached spall layers remain trapped in artificial tensile bridging.

Furthermore, heuristic remedies such as reference-configuration neighbor bond tracking (peridynamics) are disqualified in **sigmaSPH**: storing 55 to 122 reference neighbor indices per particle incurs prohibitive GPU memory overhead (+1.2 to 4.7 GB for large decks), violates counting sort memory locality, and catastrophically conflicts with Particle Shifting Technology (PST, §2.4), where regularisation shifts of $2\text{--}6\%\Delta x_0$ exceed brittle failure strains on step 1, spontaneously shattering bonds in intact material.

3.4.2 Unilateral Pressure Decomposition and Threshold Spall

sigmaSPH resolves the tensile bridging defect within the Eulerian updated Lagrangian framework through a continuum damage formulation paired with a unilateral pressure projection. A scalar continuum damage variable $D_i \in [0, 1]$ is allocated per particle using existing sorted scratch memory buffers, requiring zero auxiliary memory and incurring zero sorting overhead.

Threshold Spall Cutoff (`spallPressure`): In the simple threshold spall model (`model: "threshold"`), damage initiates instantaneously once triaxial hydrostatic tension $-p_{\text{raw}}$ crosses the critical spall strength p_{spall} :

$$D_i \leftarrow 1.0 \quad \text{if } -p_{\text{raw}} > p_{\text{spall}}. \quad (3.57)$$

Once initiated, damage evolution is irreversible (D_i remains 1.0).

Unilateral Volumetric Pressure Law: The effective pressure p_{eff} delivered to the momentum conservation kernel is decomposed unilaterally into compressive and tensile projections:

$$p_{\text{comp}} = \max(p_{\text{raw}}, 0.0), \quad p_{\text{tens}} = \min(p_{\text{raw}}, 0.0), \quad (3.58)$$

$$p_{\text{eff}} = p_{\text{comp}} + (1 - D_i)p_{\text{tens}}. \quad (3.59)$$

- **Intact State ($D = 0$):** $p_{\text{eff}} = p_{\text{raw}}$, preserving full elastic and shock acoustic stiffness in both compression and tension.
- **Damaged State ($D = 1$):** The tensile projection is eliminated entirely ($p_{\text{eff}} = \max(p_{\text{raw}}, 0.0) \geq 0$). As particles pull apart, tensile attraction drops identically to zero, allowing crack planes to open freely without artificial cohesive resistance.

- **Fragment Contact and Re-compression:** If damaged fragments subsequently collide ($p_{\text{raw}} > 0$), $p_{\text{eff}} = p_{\text{raw}}$ restores 100% of compressive stiffness, preventing unphysical particle interpenetration upon contact.

Deviatoric Quenching and Hourglass Gating: Coincident with volumetric separation, deviatoric shear stress is quenched across damaged particles:

$$\mathbf{s}_{\text{eff}} = (1 - D_i)\mathbf{s}_i. \quad (3.60)$$

Additionally, the pairwise velocity-difference hourglass damper is gated by the pair damage:

$$\mathbf{a}_{\text{hg},ij} \leftarrow (1 - \max(D_i, D_j)) \mathbf{a}_{\text{hg},ij}, \quad (3.61)$$

preventing numerical dampers from penalising the legitimate physical separation velocity of flying spall scabs.

3.4.3 Microstructural Void Nucleation and Growth (Cocks–Ashby Model)

To model ductile spallation driven by micro-void nucleation, growth, and coalescence under dynamic triaxial stress states, **sigmaSPH** implements the microstructural constitutive formulation of Qamar et al. (2025) [27], uniting Chu–Needleman cumulative normal nucleation [9] with Cocks–Ashby void growth [11].

Chu–Needleman Cumulative Normal Nucleation

Void nucleation is governed by the historical peak hydrostatic tension $\sigma_h^* = \max_{0 \leq \tau \leq t}(-p(\tau))$, tracked per particle in `ps.sigma_h_peak`. The nucleated void volume fraction f_n follows a cumulative Gaussian distribution:

$$f_n(\sigma_h^*) = \frac{f_{n0}}{2} \left[1 + \operatorname{erf} \left(\frac{\sigma_h^* - \sigma_{hM}}{\sigma_{hS}\sqrt{2}} \right) \right], \quad (3.62)$$

where f_{n0} is the maximum nucleation void fraction, σ_{hM} is the mean nucleation tensile stress, and σ_{hS} is the standard deviation. Because Taichi lacks a native CUDA device function for $\operatorname{erf}(x)$, a high-precision rational approximation (Abramowitz & Stegun 7.1.26) executing entirely within GPU registers is implemented in `material_models/DAM001_cocks_ashby.py`, guaranteeing maximum absolute error $< 1.7 \times 10^{-7}$ without branching or iterative latency.

Cocks–Ashby Void Growth Law

Total void volume fraction (porosity) $f \in [0, f_{\text{max}}]$ evolves via nucleation, deviatoric plastic growth, and triaxial hydrostatic growth:

$$\dot{f} = \dot{f}_n + \dot{f}_e + \dot{f}_s. \quad (3.63)$$

The deviatoric plastic void growth rate $\dot{f}_e$ is driven by matrix shear flow:

$$\dot{f}_e = c_1 \sinh \left[c_2 \left(\frac{m - 0.5}{m + 0.5} \right) \frac{\sigma_h}{\tau_f} \right] \left[\frac{1}{(1 - f^*)^m} - (1 - f^*) \right] \dot{\epsilon}^p, \quad (3.64)$$

where $\tau_f = \sigma_y/\sqrt{3}$ is the dynamic shear flow stress, m is the strain-rate sensitivity exponent, and f^* is the effective porosity ($f^* = f$ under tension $\sigma_h > 0$, and $f^* = \max(0, f - f_n)$ under compression $\sigma_h \leq 0$).

Under extreme triaxiality ($\sigma_h/\tau_f \gg 1$), deviatoric shear strain is suppressed ($\dot{\epsilon}^p \approx 0$). To model triaxial cavitation, the hydrostatic void growth rate $\dot{f}_s$ is active for $\sigma_h > 0$:

$$\dot{f}_s = c_4 \left(\frac{f^*}{m+1} \right) \left(c_5 \frac{\sigma_h}{\tau_f} \right)^m \dot{\epsilon}_{\text{eff}}, \quad (3.65)$$

where $c_4 = 1.0$, $c_5 = 0.1$, and $\dot{\epsilon}_{\text{eff}}$ is the effective matrix strain rate:

$$\dot{\epsilon}_{\text{eff}} = \dot{\epsilon}^p + \frac{\dot{f}_e + \dot{f}_s}{1-f}. \quad (3.66)$$

Closed-Form GPU Resolution: To bypass expensive iterative implicit solves on the GPU, the algebraic coupling between $\dot{f}_s$ and $\dot{\epsilon}_{\text{eff}}$ is solved in exact closed form:

$$\kappa = \frac{c_4 f^*}{(1-f)(m+1)} \left(c_5 \frac{\sigma_h}{\tau_f} \right)^m, \quad \dot{\epsilon}_{\text{eff}} = \frac{\dot{\epsilon}^p + \dot{f}_e/(1-f)}{1 - \min(\kappa, 0.95)}. \quad (3.67)$$

This formulation evaluates $\dot{f}_s$ deterministically in a single forward kernel evaluation.

Continuum Degradation and ALE Porosity Transport

As voids expand, the material softens according to a continuous degradation factor:

$$s_d = 1 - \tanh(a_1 f), \quad D = 1 - s_d = \tanh(a_1 f), \quad (3.68)$$

where $a_1 = 25.0$. The degradation factor scales the plastic flow stress $\sigma_y \leftarrow s_d \sigma_y$ and effective shear modulus $G_{\text{eff}} = s_d G$. Effective pressure follows the unilateral law of Eq. (3.59).

When Particle Shifting Technology (PST) is active, the intensive scalar porosity field f is convected with the particle shifting velocity:

$$\left(\frac{df_i}{dt} \right)_{\text{ALE}} = (\delta \mathbf{u}_i^{\text{shift}} \cdot \nabla) f_i, \quad (3.69)$$

preventing numerical shifting regularisation from dispersing the sharp spall fracture interface.

3.4.4 Holmquist–Johnson–Cook Cumulative Damage (DAM002)

Concrete under impact fails by accumulated crushing and shear as much as by tension. The HJC damage accumulates both the equivalent plastic strain and the plastic volumetric strain against a pressure-dependent strain to failure:

$$\Delta D = \frac{\Delta \bar{\epsilon}_p + \max(\Delta \mu_p, 0)}{\max\{D_1(P^* + T^*)^{D_2}, \varepsilon_{f,\min}\}}, \quad D \in [0, 1], \quad (3.70)$$

with D_1 , D_2 and $\varepsilon_{f,\min} = \text{EFMIN}$ on the damage card. $\Delta \bar{\epsilon}_p$ is the return map's own production in the step, not the change of `eps_plastic`, which the ALE transport also moves. $\Delta \mu_p$ is the increase of the plastic volumetric strain of §3.2.5 as $\mu_{\max}$ advances; only positive increments count. μ_p is non-decreasing where $K'_1 > K$, which holds for the 150 and 200 MPa sets and not for the 20 and 44 MPa sets ($K_1 = 17 \text{ GPa}$ against $K \approx 17\text{--}20 \text{ GPa}$).

The damage is advanced in the stress-acceleration pass, together with $\mu_{\max}$, and the tension cutoff of Eq. (3.27) is applied there with the damage of the same step. A material with a Grady–Kipp tension card replaces that cutoff, and the accrual of D in tension, with the flaw model of §3.4.5. An HJC material **bypasses the unilateral pressure decomposition** of Eq. (3.59): the cutoff $-T(1-D)$ already removes the tension, and applying both would degrade it by $(1-D)^2$. Under PST with `Shifting.aLEHistory`, D and $\mu_{\max}$ of HJC particles are ALE-transported like the porosity (§3.4.3), over same-material pairs only, so that a metal neighbour's spall damage is never read as concrete damage.

TRAP / OPERATIONAL NOTE: Crushing Alone Takes the 150 MPa Concrete to Full Damage, and EFMIN Is Not in the Paper

With the published constants, compaction alone brings the 150 MPa concrete to $D = 1$ by the time it reaches the locking point: $\mu_p(B) = 0.090$ against a failure strain of 0.01–0.2 along the way. That is a property of the constants, not of the implementation. Kim et al. do not quote $\varepsilon_{f,\min}$; the library carries Holmquist et al.'s 0.01. At low confinement, $P^* + T^* \rightarrow 0$, the failure strain is that floor, and it is the state of the plate around an impact crater once the pressure has relaxed. On the axisymmetric Chocron deck (`hvi_axi_al_concrete150_chocron_mm_GPa_ms.json`, the impact of §5.5 without the tension card) it controls the damaged diameter: 102 mm at 0.01 against 74 mm at 0.05 at 1 mm spacing, around the experiment's 80 mm. The library keeps 0.01, and the deck is *not* tuned to 80 mm.

3.4.5 Grady–Kipp Tensile Damage with Weibull Flaws (DAM003)

Why HJC Alone Cannot Fragment a Target

The HJC model of §3.2.5–§3.4.4 treats tension in three coupled ways, and together they fail a brittle target as a block rather than as a set of fragments.

- The pressure is cut off at $-T(1 - D)$. With $K = 16$ GPa and $T = 7.59$ MPa on the 150 MPa concrete, that cutoff is reached at a volumetric dilation of $T/K = 0.05\%$.
- At the cutoff the tension branch of the strength surface, $A(1 - D) + AP^*/T^*$, is exactly zero, so the return map sends the deviator to the origin in every step.
- The strain to failure there is the floor $\varepsilon_{f,\min} = 0.01$ of Eq. (3.70), because $P^* + T^* \leq 0$.

Every particle of the material carries the same T . A stress wave therefore brings whole regions to the cutoff at the same time, and nothing distinguishes one particle from its neighbour. On a 2 mm quarter of the Chocron impact (§5.5) run to 1.1 ms, HJC alone leaves 97.6% of the plate at $D \geq 0.99$, and the largest connected piece of intact concrete is 0.08% of the plate. That is a powder, not a fragment distribution.

A brittle solid fails differently. Its weakest flaws open first, the cracks they grow relieve the stress around them, and the material between the cracks survives as fragments. Where the loading is fast, many flaws open before any of them can relieve the others, and the fragments are small. Two ingredients carry that behaviour: a tensile strength that differs from particle to particle, and a crack that needs time to grow. The `damage.tension` card adds both to an HJC material, in the SPH form of Benz and Asphaug [5, 6] of the flaw model of Grady and Kipp [15]. HJC keeps compression; tension is governed by the flaws. A material without the card behaves exactly as in §3.4.4.

The Flaw Population

The number of flaws per unit volume that are active at a tensile strain ε follows the Weibull law

$$n(\varepsilon) = k \varepsilon^m. \quad (3.71)$$

The flaws of each material are drawn once, on the host, over all of its particles. Flaw $j = 1, 2, \dots$ has the activation strain

$$\varepsilon_j = \left(\frac{j}{kV_{\text{tot}}} \right)^{1/m}, \quad (3.72)$$

and is given to a particle chosen at random with probability V_i/V_{tot} , $V_i = m_i/\rho_0$, until every particle holds at least one flaw. That takes about $N \ln N$ flaws for N particles. A particle stores

three numbers of its share: its lowest threshold $\varepsilon_{\min,i}$, its highest $\varepsilon_{\max,i}$ and its count n_i . Between the two, the number of its active flaws is interpolated on the same power law:

$$n_{\text{act},i}(\varepsilon) = \begin{cases} 0, & \varepsilon < \varepsilon_{\min,i}, \\ 1 + (n_i - 1) \frac{\varepsilon^m - \varepsilon_{\min,i}^m}{\varepsilon_{\max,i}^m - \varepsilon_{\min,i}^m}, & \varepsilon_{\min,i} \leq \varepsilon < \varepsilon_{\max,i}, \\ n_i, & \varepsilon \geq \varepsilon_{\max,i}. \end{cases} \quad (3.73)$$

The three numbers are sorted per-particle fields (**flaw_eps_min**, **flaw_eps_max**, **flaw_n**). They are the only material constants in the solver that vary within a material.

The constant k is not given directly but through a reference pair: the quasi-static tensile strength σ_{ref} of a specimen of volume V_{ref} , taken as the strength of that specimen's weakest flaw,

$$k V_{\text{ref}} \left(\frac{\sigma_{\text{ref}}}{E} \right)^m = 1, \quad E = \frac{9KG}{3K + G}. \quad (3.74)$$

σ_{ref} defaults to the eos **tensileStrength**, so the quoted T keeps its meaning at the size it was measured on. **referenceVolume** has no default, because it carries the deck's length unit. K is the HJC elastic modulus and G the stated shear modulus, which give $E = 42.6$ GPa on the 150 MPa concrete. A particle is smaller than the specimen, and so stronger on average by the Weibull size effect $(V_{\text{ref}}/V)^{1/m}$. With the 150×300 mm cylinder (5.30×10^6 mm³) as reference and $m = 8$, a 2 mm particle's weakest flaw lies at 23 / 39 / 49 MPa (1% / median / 99%), and the weakest flaw of the whole quarter plate at 10 MPa.

Activation and Growth

The strain that activates flaws is read off the undamaged stress:

$$\varepsilon_i = \frac{\max(\sigma_{1,i}, 0)}{E}, \quad \sigma_{1,i} = \lambda_{\max}(-p_{\text{raw},i} \mathbf{I} + \mathbf{s}_i), \quad (3.75)$$

where p_{raw} is the equation of state's pressure before any damage acts on it and $\mathbf{s}_i$ is the stored deviator, which is the effective (undamaged) one. The deviator enters through the principal value, which is how tension combined with shear activates flaws. The tensile damage D_t then grows as

$$\frac{dD_{t,i}^{1/3}}{dt} = n_{\text{act},i}^{1/3} \frac{c_g}{R_s}, \quad c_g = f_{c_g} c_l, \quad c_l = \sqrt{\frac{K + \frac{4}{3}G}{\rho_0}}, \quad (3.76)$$

with R_s the particle radius and $f_{c_g} = \text{crackSpeedFactor}$, 0.4 by default. D_t never decreases. The cube root on n_{act} is Grady and Kipp's crack-volume argument: n_{act} cracks of radius $c_g t$ in a particle of radius R_s occupy the fraction $n_{\text{act}}(c_g t/R_s)^3$ of it. In a volume where n_{act} grows as ε^m , this gives the continuum rate dependence of the tensile strength, $\sigma \propto \dot{\varepsilon}^{3/(m+3)}$; a growth rate linear in n_{act} would give $\dot{\varepsilon}^{1/(m+1)}$ instead. A particle holds only about $\ln N$ flaws, so that continuum limit is not reached inside one particle. An ensemble of independent 2 mm particles at $m = 8$ (test T41, with the cap below) is rate-independent below about 100 s^{-1} at a mean strength of 35.6 MPa, with local exponents rising to 0.4–0.5 between 3×10^3 and $3 \times 10^4 \text{ s}^{-1}$.

Benz and Asphaug limit the damage to the share of active flaws, $D_t \leq n_{\text{act}}/n$. That cap is the card's **flawCap**, true by default. With **flawCap: false**, a particle with any active flaw grows to $D_t = 1$ within $R_s/(n_{\text{act}}^{1/3} c_g)$, about $0.6 \mu\text{s}$ at 2 mm.

Coupling to HJC

On a material with the card, the solver applies the tensile damage as follows.

1. **Pressure.** The stored pressure is p_{raw} in compression and $(1 - D_c)(1 - D_t)p_{\text{raw}}$ in tension, with D_c the HJC damage. There is no cutoff at $-T$: the flaws set the tensile strength, with its size and rate effects.
2. **Transmitted deviator.** In the pair force and the axisymmetric source term, a particle whose stored pressure is not positive transmits $(1 - D_t)\mathbf{s}$; in compression it transmits $\mathbf{s}$. In uniaxial tension two thirds of the tensile principal stress is deviatoric, so a coupling that degraded only the pressure would leave a cracked particle carrying most of its load. On the 2 mm Chocron quarter such a coupling produces no localisation at all, and 95% of the plate ends as rubble. A crack that has closed again carries shear up to the friction BP^*N of the HJC surface.
3. **Strength surface.** The surface of Eq. (3.54) reads the combined damage $D = 1 - (1 - D_c)(1 - D_t)$, and its tension branch is flat at the cohesion: the pressure is floored at zero before the surface is evaluated. The branch falling linearly to zero at the cutoff is what leaves a particle with no shear strength at all; tension combined with shear reaches the flaws through σ_1 instead.
4. **HJC damage.** D_c accrues by Eq. (3.70) only while $p_{\text{raw}} > 0$, that is, through crushing and shear under confinement, the regime HJC is calibrated in.
5. **Density diffusion across cracks.** The δ -SPH diffusion of a pair is switched off when the more damaged of the two particles has $D_t \geq 0.99$ (§2.2.1). Without that cut, the low density of the bulking rubble in a crack diffuses into the intact crack faces, puts them in tension and widens every crack until the intact cells between cracks are gone: on the 2 mm quarter the rubble share would grow from 35% at 0.1 ms to 85% at 1.1 ms. With the cut, the damage pattern at 0.1 ms is the pattern at 1.1 ms.
6. **ALE transport.** Under PST with `aleHistory`, D_t is transported with D_c and μ_{max} . The flaws are not transported, because they belong to the material.

The output carries `damage_t` and `flaw_eps_min`, and `Output.colorField: "damageTension"` colours the particles by D_t .

TRAP / OPERATIONAL NOTE: Hiding the Crack Opening from the Crack Faces Stops Cracks from Growing

A crack in SPH is a band about a kernel wide, not a discontinuity, and every neighbour sum across it reads its opening as a smooth dilation of the particles on its faces. It is tempting to remove the cracked bonds from the continuity sum and the velocity gradient as well as from the diffusion. That also removes the strain concentration at the crack tip, which is a partially damaged particle, and the cracks stop propagating. On the 2 mm quarter, leaving opening cracked bonds out of both sums gives 14% rubble and a plate that remains one body (86% of the mass). Weighting every bond by $1 - D_t$ in addition makes the opening and closing halves of a vibration unequal, which ratchets the rubble into compression and raises the HJC crush damage from 11% to 39% of the plate. Only the density diffusion is cut.

TRAP / OPERATIONAL NOTE: The Chocron Deck Runs `flawCap: false`, a Departure from Benz and Asphaug

With the cap, a particle in a crack band whose stress the crack has already relieved stops activating flaws and stays partially damaged, $D_t \approx 0.6$ – 0.95 . The crack network forms, but the material across it remains connected. On the 2 mm quarter the plate then stays one

body holding 80% of its mass at the 0.99 fragment threshold, and the number of fragments swings from 1 to 46 as that threshold is lowered from 0.99 to 0.5. Without the cap, cracks separate the material on either side and the plate breaks into sectors between radial cracks. The price is that crack growth is threshold-controlled, and the measured rate dependence above was obtained with the cap on.

TRAP / OPERATIONAL NOTE: Crack Width Is Set by the Smoothing Length

The crack bands are two to five particles wide. At $\Delta x_0 = 2$ mm the fully damaged share of the Chocron plate is therefore 54.5%, and it is a property of the resolution rather than of the concrete. Everything smaller than about ten particles is rubble. The axisymmetric form assigns flaws per ring, since a particle's volume is its ring's, so a fragment there is a ring. The quarter model mirrors every flaw across its symmetry planes, so a fragment meeting a plane has its shape there forced.

3.4.6 Johnson–Cook Fracture with Regularised Softening (DAM004)

The spall models above are driven by hydrostatic tension and answer whether a plate spalls. They do not answer what a tensile bar asks: at which plastic strain a ductile metal fractures under the stress state it is in. That is the purpose of the fracture criterion of Johnson and Cook [20], the usual companion of the Johnson–Cook strength model. `damage.model: "johnson_cook"` adds it, with a regularised softening branch after initiation.

Initiation

An accumulator ω grows with the plastic strain produced by the return map,

$$\omega = \sum \frac{\Delta \varepsilon_p}{\varepsilon_f}, \quad \varepsilon_f = \max\left([D_1 + D_2 \exp(D_3 \eta)] [1 + D_4 \ln \max(1, \dot{\varepsilon}_p / \dot{\varepsilon}_0)] [1 + D_5 T^*], \text{EFMIN}\right), \quad (3.77)$$

with $\eta = -p/\sigma_{vM}$ the stress triaxiality, formed from the undamaged EOS pressure and the returned deviator. $\dot{\varepsilon}_0$ and the homologous temperature T^* are those of the material's Johnson–Cook plasticity card, which the fracture card therefore requires. $\Delta \varepsilon_p$ is the production of the return map, not the change of `eps_plastic`, which the ALE transport also moves. While $\omega < 1$, the stress is not touched: the criterion is uncoupled.

Regularised Softening

Setting $D = 1$ as soon as $\omega = 1$ would make the energy a crack consumes proportional to the volume of the particles it crosses, and hence to Δx_0 ; the post-peak branch of a force–displacement curve could not converge. After initiation, D therefore grows with the plastic *displacement* across a particle,

$$\Delta D = \frac{L_c \Delta \varepsilon_p}{u_f}, \quad L_c = \Delta x_0, \quad (3.78)$$

so that $D = 1$ is reached after an opening u_f (`failureDisplacement`) whatever the spacing, and a crack dissipates about $\sigma_y u_f / 2$ per unit area [16]. The part of the increment that carries ω past 1 already softens. D then acts as for every metal: the Johnson–Cook yield surface is scaled by $1 - D$, tension is removed by the unilateral split of Eq. (3.59), the deviator is returned to the origin at $D = 1$, and the hourglass damper of a pair is scaled by $1 - \max(D_i, D_j)$. Compression is kept, so nothing is eroded. Under PST with `aleHistory`, ω and D are ALE-transported over same-material pairs.

TRAP / OPERATIONAL NOTE: The Regularisation Assumes a Band One L_c Wide

Equation (3.78) makes the dissipated energy independent of Δx_0 only if the band in which the plastic strain localises is L_c wide. In SPH that band is set by the kernel, $h = 3\Delta x_0$ at `supportRadiusFactor: 6`, so the energy is resolution-independent only up to that constant factor, and only as long as the band keeps scaling with h .

TRAP / OPERATIONAL NOTE: The Compression-Energy Cross-Check Overstates a Failed Body

The budget's elastic-store cross-check `SE` is evaluated from the density with the undamaged EOS. A damaged particle that dilates without carrying tension therefore appears as stored energy. On the axisymmetric dogbone, `SE` grows from about 0.1 J to about 100 J after separation, while the energy balance itself still closes. Read `SE` only before failure.

3.5 The Material Constants Library

To eliminate repetitive parameter declaration and prevent unit errors across simulation decks, `sigmaSPH` provides a centralized material constants library in `material_models/material_library.py`.

3.5.1 Explicit Dynamics Unit Convention (mm/ms/GPa)

The library is parameterized in the consistent explicit dynamics unit system summarised in Table 3.1. In this system, velocities are numerically identical to m/s (1 mm/ms = 1 m/s), and stresses and elastic moduli are specified directly in gigapascals (GPa).

Table 3.1: Consistent explicit dynamics unit system used in the material constants library.

Physical Quantity	Library Unit	SI Equivalent
Length	mm	10^{-3} m
Time	ms	10^{-3} s
Mass	kg	1 kg
Velocity	mm/ms	1 m/s
Stress / Pressure / Moduli	GPa	10^9 N/m ² = 1 kg/(mm · ms ²)
Density (ρ)	kg/mm ³	10^9 kg/m ³ (e.g. 8.96×10^{-6} kg/mm ³ = 8960 kg/m ³)
Specific Internal Energy (e)	J/kg	1 (mm/ms) ² = 1 m ² /s ²
Specific Heat Capacity (C_p)	J/(kg · K)	1 (mm/ms) ² /K
Strain Rate ($\dot{\epsilon}$)	ms ⁻¹	10^3 s ⁻¹ (e.g. 1.0×10^{-3} ms ⁻¹ = 1.0 s ⁻¹)

3.5.2 Calibrated Material Parameters

Table 3.2 lists the calibrated constants for the four core structural metallic materials carried on the linear equation of state, Table 3.4 the separate aluminium entry calibrated on the shock Hugoniot for hypervelocity impact, and Table 3.3 the calibrated properties for soft tissue simulants (10% and 20% ballistic gelatin) under shock and high-strain-rate impact [2, 28].

Al 2017-T4 on the Shock Hugoniot

The library entry `aluminium` carries the linear equation of state, which is the correct volumetric model over the elastic and moderately plastic range and the wrong one at shock strengths

Table 3.2: Calibrated physical, shock Hugoniot, and viscoplastic constants for core engineering metals [19]. The reference strain rate $\dot{\epsilon}_0 = 0.001 \text{ ms}^{-1}$ is 1 s^{-1} .

Parameter	Copper (copper)	Aluminium (aluminium)	Steel (construction_ steel)	Stainless (stainless_ steel)
density0 [kg/mm ³]	8.96×10^{-6}	2.70×10^{-6}	7.85×10^{-6}	7.90×10^{-6}
youngsModulus [GPa]	124.0	69.0	210.0	193.0
poissonRatio	0.34	0.33	0.30	0.30
<i>Johnson–Cook Viscoplasticity (MAT002)</i>				
A (yieldStress) [GPa]	0.090	0.324	0.350	0.310
B [GPa]	0.292	0.114	0.275	1.000
n	0.31	0.42	0.36	0.65
C	0.025	0.002	0.022	0.020
$\dot{\epsilon}_0$ (eps0_dot) [ms ⁻¹]	0.001	0.001	0.001	0.001
T ₀ [K]	293.0	293.0	293.0	293.0
T _m [K]	1356.0	925.0	1780.0	1720.0
m	1.09	1.34	1.00	1.00
C _p [J/(kg · K)]	383.0	896.0	486.0	500.0
χ (chi)	0.90	0.90	0.90	0.90
<i>Fallback Linear Hardening Constants (MAT001)</i>				
σ _{y0} [GPa]	0.090	0.324	0.350	0.310
H (hardeningModulus) [GPa]	1.00	0.50	1.50	1.00

Table 3.3: Calibrated physical, shock Hugoniot, and viscoplastic constants for ballistic gelatin [2].

Parameter	10% Ballistic Gelatin (gelatin_10)	20% Ballistic Gelatin (gelatin_20)
density0 [kg/mm ³]	1.03×10^{-6} (1030 kg/m ³)	1.06×10^{-6} (1060 kg/m ³)
youngsModulus [GPa]	1.50×10^{-4} (150 kPa)	2.82×10^{-4} (282 kPa)
poissonRatio	0.45	0.45
<i>Polynomial Mie–Grüneisen Equation of State (EOS003)</i>		
c ₀ (c0) [mm/ms]	1.52 (1520 m/s)	1.52 (1520 m/s)
s (s)	1.80	1.87
Γ ₀ (gamma0)	0.17	0.17
P ₀ (P0) [GPa]	0.0	0.0
e ₀ (e0) [mm ² /ms ²]	0.0	0.0
<i>Hollomon Power-Law Hardening / Johnson–Cook (MAT002)</i>		
A (yieldStress) [GPa]	5.2×10^{-4} (520 kPa)	3.5×10^{-3} (3500 kPa)
B [GPa]	1.0×10^{-5} (10 kPa)	1.0×10^{-5} (10 kPa)
n	0.10	0.10
C	0.0	0.0
$\dot{\epsilon}_0$ (eps0_dot) [ms ⁻¹]	0.001 (1 s ⁻¹)	0.001 (1 s ⁻¹)
T ₀ [K]	293.0	293.0
T _m [K]	0.0 (thermal softening disabled)	0.0 (thermal softening disabled)
m	1.00	1.00
C _p [J/(kg · K)]	0.0	0.0
χ (chi)	0.90	0.90

where the Hugoniot curves away from a straight line in p - μ . The separate entry `al2017_t4` exists for that regime: it is the same class of alloy placed on the polynomial Mie–Grüneisen equation of state of Sec. 3.2.4, entered through the physical shock constants c_0 and s of the linear $U_s = c_0 + su_p$ fit, from which the solver forms the polynomial coefficients itself. Its constants are given in Table 3.4.

Table 3.4: Constants of the `al2017_t4` library entry. The equation-of-state half is a genuine Al 2017 shock fit; the strength half is a substitution, discussed below.

Parameter	Value	Remark
<code>density0</code> [kg/mm ³]	2.79×10^{-6}	2790 kg/m ³
<code>youngsModulus</code> [GPa]	72.4	present only to fix G
<code>poissonRatio</code>	0.33	present only to fix G
<code>specificHeat</code> [J/(kg · K)]	875.0	C_v for Eq. (2.80)
<i>Polynomial Mie–Grüneisen Equation of State (EOS003)</i>		
c_0 (<code>c0</code>) [mm/ms]	5328.0	bulk sound speed
s (<code>s</code>)	1.34	Hugoniot slope
Γ_0 (<code>gamma0</code>)	2.0	
P_0 (<code>P0</code>) [GPa]	0.0	
e_0 (<code>e0</code>) [mm ² /ms ²]	0.0	
$K_0 = C_1 = \rho_0 c_0^2$ [GPa]	79.20	derived
$C_2 = (2s - 1)C_1$ [GPa]	133.06	derived
$C_3 = (s - 1)(3s - 1)C_1$ [GPa]	81.32	derived
<i>Johnson–Cook Viscoplasticity (MAT002), substituted from Al 2024-T351</i>		
A (<code>yieldStress</code>) [GPa]	0.265	
B [GPa]	0.426	
n	0.34	
C	0.015	
$\dot{\epsilon}_0$ (<code>eps0_dot</code>) [ms ⁻¹]	0.001	1 s ⁻¹
T_0 [K]	293.0	
T_m [K]	775.0	
m	1.00	
C_p [J/(kg · K)]	875.0	
χ (<code>chi</code>)	0.90	implicitly 1; see Sec. 2.11.5
<i>Fallback Linear Hardening Constants (MAT001)</i>		
σ_{y0} [GPa]	0.265	
H (<code>hardeningModulus</code>) [GPa]	0.60	

Two entries in Table 3.4 are not transcribed constants and must not be read as though they were.

The elastic pair does not set the bulk modulus. Young’s modulus and Poisson’s ratio appear in the entry solely to fix the shear modulus $G = E/[2(1 + \nu)] = 27.22$ GPa, which the hypoelastic deviatoric update requires and which the equation of state does not supply. The bulk response is taken entirely from the shock fit, $K_0 = \rho_0 c_0^2 = 79.20$ GPa, and that value overrides whatever the declared pair would imply. The pair is therefore not required to reproduce it, and the apparent inconsistency between $E/[3(1 - 2\nu)] = 70.98$ GPa and $K_0 = 79.20$ GPa is a property of the entry rather than an error in it.

The Johnson–Cook constants belong to Al 2024-T351. Al 2017-T4 has no well-attested open-literature Johnson–Cook calibration, and substituting the 2024-T351 set is the standard workaround in the hypervelocity impact literature. The substitution biases the projectile’s resistance to deformation and to breakup slightly *high*. Nothing in the solver marks a material card as carrying borrowed constants, so the bias is invisible at run time and is recorded here and in the library source instead.

Pure Tantalum on the Shock Hugoniot with Cocks–Ashby Spall Damage

The library entry `tantalum` provides a complete, physically validated parameter set for pure polycrystalline tantalum (Ta) subjected to hypervelocity impact, shock compression, and dynamic spallation [27]. It couples the polynomial Mie–Grüneisen shock equation of state with Johnson–Cook viscoplasticity and the Cocks–Ashby dynamic void growth model described in §3.4.3. Its calibrated constants are summarised in Table 3.5.

Holmquist–Johnson–Cook Concretes and the Aluminium of Kim et al.

Four library entries carry the concrete constants of Kim et al. [21]: `concrete_hjc_150` from their Table 5 (after Chen et al. [7]), the target of the Chocron impact, and `concrete_hjc_20_44` and `_200` from their Table 8 (after Hu et al. [18]). Each declares the three HJC cards of §3.2.5, §3.3.6 and §3.4.4 together. The megapascal values of the source tables are divided by 10^3 . The constants are listed in Table 3.6, and the lower block gives what the host derives from them.

The entry `aluminium_kim2025` is the aluminium of Kim et al.’s Table 2, used there for every aluminium projectile and plate: Johnson–Cook strength (§3.3.4) on the polynomial Mie–Grüneisen equation of state (§3.2.4), the cubic expansion of the Hugoniot the paper writes in its referenced form. Its constants are $\rho_0 = 2.71 \times 10^{-6} \text{ kg/mm}^3$; $c_0 = 5300 \text{ mm/ms}$, $s = 1.5$, $\Gamma_0 = 1.7$, so that $K_0 = \rho_0 c_0^2 = 76.1 \text{ GPa}$; $A = 0.175 \text{ GPa}$, $B = 0.380 \text{ GPa}$, $n = 0.34$, $C = 0.0015$, $\dot{\epsilon}_0 = 0.001 \text{ ms}^{-1}$, $T_0 = 273 \text{ K}$, $T_m = 775 \text{ K}$, $m = 1.0$, $\chi = 0.9$. The paper gives the shear modulus $G = 27.6 \text{ GPa}$ and no elastic pair; the entry declares $E = 73.416 \text{ GPa}$ at $\nu = 0.33$, which reproduces that G , and the bulk response is the equation of state’s. The specific heat, $875 \text{ J/(kg} \cdot \text{K)}$, is not in the paper; it is the handbook value for aluminium alloys and only converts heat into the temperature Johnson–Cook softens with.

3.5.3 Material Aliases

Material identifiers in scene decks are normalized case-insensitively and whitespace-stripped via `normalize_material_name()`. The library accepts any of the synonyms listed in Table 3.7.

3.6 Programmed Burn Detonation Mechanics

Detonation in energetic materials is handled via a programmed Huygens lighting-time burn front rather than an expensive stiff chemical kinetics solver.

3.6.1 Lighting Front Arrival Time

Given one or more discrete detonator locations $\mathbf{x}_{\text{det},k}$ initiating at times $t_{\text{det},k}$, the detonation front travels at the constant Chapman–Jouguet detonation velocity D_{CJ} . The arrival time $t_{\text{light},i}$ at particle reference position $\mathbf{x}_{0,i}$ is computed once at initialization:

$$t_{\text{light},i} = \min_k \left(t_{\text{det},k} + \frac{\|\mathbf{x}_{0,i} - \mathbf{x}_{\text{det},k}\|}{D_{\text{CJ}}} \right). \quad (3.79)$$

3.6.2 Burn Fraction and Pressure Gating

As the detonation front sweeps across the explosive block, the burn fraction $F_i(t)$ transitions smoothly from 0 (unreacted solid) to 1 (fully reacted gas):

$$F_i(t) = \text{clamp} \left(\frac{t - t_{\text{light},i}}{\Delta t_{\text{burn}}}, 0, 1 \right), \quad \Delta t_{\text{burn}} = \frac{2h}{D_{\text{CJ}}}. \quad (3.80)$$

Table 3.5: Constants of the `tantalum` library entry calibrated for hypervelocity impact and spallation [27].

Parameter	Value	Remark
<code>density0</code> [kg/mm ³]	1.669×10^{-5}	16 690 kg/m ³
<code>youngsModulus</code> [GPa]	185.7	sets initial shear modulus $G_0 = 69.0$ GPa
<code>poissonRatio</code>	0.345	
<code>specificHeat</code> [J/(kg · K)]	140.0	C_v for Eq. (2.80)
<i>Polynomial Mie–Grüneisen Equation of State (EOS003)</i>		
<code>c0</code> (<code>c0</code>) [mm/ms]	3410.0	bulk reference sound speed (3410 m/s)
<code>s</code> (<code>s</code>)	1.20	linear Hugoniot slope
Γ_0 (<code>gamma0</code>)	1.60	reference Grüneisen coefficient
P_0 (<code>P0</code>) [GPa]	0.0	ambient reference pressure
e_0 (<code>e0</code>) [mm ² /ms ²]	0.0	reference internal energy
$K_0 = \rho_0 c_0^2$ [GPa]	194.06	<i>derived</i> initial bulk modulus
$C_2 = (2s - 1)C_1$ [GPa]	271.68	<i>derived</i> polynomial coefficient
$C_3 = (s - 1)(3s - 1)C_1$ [GPa]	100.91	<i>derived</i> polynomial coefficient
<i>Johnson–Cook Viscoplasticity (MAT002)</i>		
A (<code>yieldStress</code>) [GPa]	0.733	quasi-static initial yield stress (733 MPa)
B [GPa]	0.540	strain hardening modulus (540 MPa)
n	0.28	strain hardening exponent
C	0.054	strain-rate sensitivity coefficient
$\dot{\epsilon}_0$ (<code>eps0_dot</code>) [ms ⁻¹]	0.001	reference strain rate (1 s ⁻¹)
T_0 [K]	293.0	reference ambient temperature
T_m [K]	3269.0	melting temperature
m	0.44	thermal softening exponent
C_p [J/(kg · K)]	140.0	specific heat capacity
<i>Cocks–Ashby Dynamic Spall Damage (§3.4.3)</i>		
c_1	1.5	deviatoric plastic void growth coefficient
c_2	0.72	deviatoric plastic void growth coefficient
c_4	1.0	hydrostatic void cavitation growth coefficient
c_5	0.1	hydrostatic void cavitation growth coefficient
a_1	25.0	damage degradation softening coefficient
f_{n0}	1.0×10^{-4}	Chu–Needleman nucleation amplitude
σ_{hM} [GPa]	0.778	mean nucleation hydrostatic tension (778 MPa)
σ_{hS} [GPa]	0.0389	standard deviation of nucleation tension (38.9 MPa)
m	4.0	rate sensitivity exponent in void growth
$f_{\max}$	0.50	maximum porosity before complete cohesion loss

Table 3.6: Constants of the four Holmquist–Johnson–Cook concrete entries [21]. Rows below the second rule are derived by `derive_hjc_eos`, not declared.

Parameter	_150	_20	_44	_200
density0 [10^{-6} kg/mm ³]	2.70	2.40	2.40	2.50
shearModulus G [GPa]	20.16	12.5	14.86	24.3
<i>Equation of state (EOS004)</i>				
crushPressure P_c [GPa]	0.016	0.0069	0.0147	0.0667
crushStrain μ_c	0.001	0.00041	0.00074	0.0021
lockPressure P_l [GPa]	0.8	0.8	0.8	0.8
lockStrain μ_{pl}	0.1	0.1	0.1	0.1
K_1, K_2, K_3 [GPa]	85, -171, 208	17, 38, 29.8	17, 38, 29.8	85, -171, 208
tensileStrength T [GPa]	0.00759	0.0022	0.004	0.02
<i>Strength (MAT003)</i>				
fc [GPa]	0.15	0.02	0.044	0.2
A, B, N	0.79, 1.6, 0.61	0.79, 1.6, 0.61	0.79, 1.6, 0.61	0.3, 1.73, 0.79
$C, S_{\max}$	0.007, 7.0	0.007, 7.0	0.007, 7.0	0.007, 7.0
eps0_dot [ms ⁻¹]	0.001	0.001	0.001	0.001
<i>Damage (DAM002)</i>				
D_1, D_2, EFMIN	0.04, 1.0, 0.01	0.04, 1.0, 0.01	0.04, 1.0, 0.01	0.04, 1.0, 0.01
$K = P_c/\mu_c$ [GPa]	16.00	16.83	19.86	31.76
K_{lock} [GPa]	7.92	7.96	7.91	7.49
μ_L (grain-density locking strain)	0.0895	0.0548	0.0548	0.0895
$K'_1 = K_1/(1 + \mu_L)$ [GPa]	78.01	16.12	16.12	78.01
$\mu_p(B)$	0.0898	0.0504	0.0504	0.0898
c_p on $\max(K, K'_1)$ [km/s]	6.23	3.74	4.07	6.65

Table 3.7: Recognized material aliases in `material_models/material_library.py`.

Standard Library Key	Recognized Synonyms and Aliases
aluminium	"aluminum", "al", "al6061", "al6061-t6"
al2017_t4	"al2017", "al2017-t4", "al2017t4", "aluminium_2017", "aluminum_2017"
construction_steel	"steel", "structural_steel", "mild_steel", "s355"
copper	"cu", "ofhc_copper"
stainless_steel	"ss304", "304", "304_stainless", "304_stainless_steel", "aisi304", "aisi_304", "ss316", "316", "ss316l", "316l", "stainless"
tantalum	"ta", "pure_tantalum"
gelatine_10	"gelatine_10", "gelatin10", "ballistic_gelatin_10"
gelatine_20	"gelatine_20", "gelatin20", "ballistic_gelatin_20"

The effective pressure delivered to the momentum equations is gated by the burn fraction:

$$p_i = F_i(t) \cdot p_{\text{JWL}}(\rho_i, e_i). \quad (3.81)$$

Particles start with initial specific detonation energy $e_{i,0} = E_0/\rho_0$, which is converted into thermodynamic expansion work as $F_i(t) \rightarrow 1$.

3.7 Kinematic Runaway Limiters and Particle Erosion

Under extreme shear localization or explosive spalling, numerical velocity runaways and unbounded grid distension can occur. **sigmaSPH** provides kinematic limiters to keep unstable simulations bounded.

3.7.1 Affine-Corrected Neighbourhood Velocity Limiter

To bound local outlier velocities without damping rigid-body motion or homogeneous affine deformation, the expected velocity $\mathbf{v}_{\text{pred},i}$ is estimated from the affine-extrapolated velocities of neighbouring particles:

$$\mathbf{v}_{\text{pred},i} = \frac{\sum_j W_{ij} V_j \left[\mathbf{v}_j + \mathbf{G}_j^{(-i)} (\mathbf{x}_i - \mathbf{x}_j) \right]}{\sum_j W_{ij} V_j}. \quad (3.82)$$

Here $\mathbf{G}_j^{(-i)}$ is the neighbour's velocity gradient with particle i 's self-contribution removed:

$$\mathbf{G}_j^{(-i)} = \nabla \mathbf{v}_j + V_i (\mathbf{v}_i - \mathbf{v}_j) \otimes \nabla W_{ij}. \quad (3.83)$$

If the non-affine residual exceeds the velocity tolerance u_{target} :

$$\|\mathbf{v}_i - \mathbf{v}_{\text{pred},i}\| \leq u_{\text{target}}. \quad (3.84)$$

The limiter is strictly dissipative: adjustments are discarded if they would increase $\|\mathbf{v}_i\|$.

3.7.2 Deformation Jacobian Bounds and Particle Erosion

The deformation Jacobian $J = V/V_0 = \rho_0/\rho$ tracks local volume expansion and compression. The **Erosion** card configures bounding bounds $[J_{\min}, J_{\max}]$ and maximum equivalent plastic strain $\bar{\epsilon}_{p,\max}$:

- **Mode "cap":** Clamps J_i into $[J_{\min}, J_{\max}]$, bounding tensile spall pressure to $p \geq K(1/J_{\max} - 1)$.
- **Mode "erode":** Completely removes the particle from all neighbour interactions and freezes its state once $J_i < J_{\min}$, $J_i > J_{\max}$, or $\bar{\epsilon}_{p,i} > \bar{\epsilon}_{p,\max}$.

4 Scene File Keyword Reference

4.1 Scene File Architecture and Keyword Card Philosophy

sigmaSPH scene decks are formatted in JSON using a card-based architecture directly inspired by modern explicit hydrocode formats (such as LS-DYNA keywords and Abaqus input cards). Rather than relying on a flat namespace where parameters can conflict or default silently, options are partitioned into distinct, top-level keyword cards:

```
{
  "Domain":      { ... },      // Box dimensions, spacing, 2D/3D kinematic
    ↳ constraints
  "Time":        { ... },      // CFL coefficients, duration, step cap
  "Solver":      { ... },      // Solver selection and numerical regularisers
  "Load":        { ... },      // Gravity and boundary-condition time ramp
  "Diffusion":   { ... },      // Delta-SPH and gamma-SPH diffusion controls
  "Shifting":    { ... },      // Particle Shifting Technology (PST)
  "Hourglass":   { ... },      // Kinematic zero-energy mode control (solid only)
  "VelocityLimit": { ... },    // Runaway velocity limiters (solid only)
  "Erosion":     { ... },      // Jacobian clamping and particle deletion (solid
    ↳ only)
  "Output":      { ... },      // Trajectory formats, intervals, and rendering

  "Materials":   [ ... ],      // Physical constants, EOS, and constitutive
    ↳ strength
  "FluidBlocks": [ ... ],      // Initial fluid particle lattice regions
  "SolidBlocks": [ ... ],      // Initial solid particle lattice regions
  "Constraints": [ ... ],      // Prescribed displacement boundary conditions
  "Detonators":  [ ... ],      // Programmed burn initiation points (JWL only)

  "ParticlesFile": "path/to/particles.nc", // Optional AMBER-NetCDF preprocessed
    ↳ state
  "Configuration": { ... }      // Flat canonical override layer
}
```

4.1.1 Card Layout Guarantees

The scene parser (`config_builder.SimConfig`) enforces strict syntactic integrity:

1. **Rejection of Unknown Keys:** Any key not declared in the parameter registry (`params.py`) triggers an immediate fatal error with an automated “did you mean” suggestion. Silent defaults for mistyped options are impossible.
2. **No Inert Sub-Parameters:** Nested options reside strictly inside sub-cards. For instance, adaptive hourglass parameters live in `Hourglass.adaptive`, and yield properties reside in `Materials[].strength.plasticity`. The presence of the sub-card acts as the operational switch.
3. **Prohibition of Derived Redundancy:** A parameter that is derived cannot be redundantly specified. If an elastic strength model defines Young’s modulus and Poisson’s ratio,

setting `eos.bulkModulus` raises an error. The scene-wide sound speed c_0 and reference density ρ_0 are derived from declared materials and cannot be entered manually.

4. **Strict Separation of Material Physics:** The `Solver` card carries only numerical knobs (e.g., artificial viscosity, hourglass damping). Physical moduli, densities, and yield limits reside solely within `Materials[]`.

4.2 Declaring Materials and Utilizing the Material Library

In `sigmaSPH`, all physical material properties reside exclusively in the top-level `Materials[]` list. To eliminate repetitive specification of standard engineering constants and protect against unit transcription errors, `config_builder.py` automatically integrates with the material constants library (`material_models/material_library.py`, §3.5).

When parsing `Materials[]`, `config_builder.SimConfig` applies recursive dictionary expansion (`expand_material_preset()`) whenever an entry references a library preset. This provides five convenient declaration patterns:

4.2.1 Pattern 1: Zero-Boilerplate Pure Library Reference

When the entry's `name` matches a recognized material name or alias (such as `"copper"`, `"aluminium"`, or `"construction_steel"`), and mandatory cards such as `density0`, `eos`, or `strength` are omitted, the solver populates the complete material card directly from the library preset:

```
"Materials": [
  {"name": "copper"},
  {"name": "aluminium"}
]
```

Simulation blocks (`SolidBlocks[]` or `FluidBlocks[]`) assign these properties simply by setting `"material": "copper"` or `"material": "aluminium"`. All density, equation of state, elasticity, and default Johnson–Cook viscoplasticity parameters are automatically loaded in consistent mm/ms/GPa units.

4.2.2 Pattern 2: Custom Material Identifier with Library Inheritance

In multi-component impact or ballistics problems, components often carry functional roles (such as `"projectile"`, `"casing"`, or `"target"`). Setting the `library` key allows an entry with an arbitrary `name` to inherit baseline properties from a specified library preset:

```
"Materials": [
  {"name": "impactor", "library": "copper"},
  {"name": "target", "library": "aluminium"}
]
```

Lattice blocks reference `"material": "impactor"` and `"material": "target"`, inheriting the full calibrated constants of OFHC copper and 6061-T6 aluminium respectively.

4.2.3 Pattern 3: Switching Constitutive Model to Linear Hardening

By default, the material library configures metals with the Johnson–Cook viscoplastic model (`model: "johnson_cook"`). If an analysis requires classical rate-independent J_2 plasticity with linear isotropic hardening (MAT001), the user switches the model via the `model` key within `strength.plasticity`:

```
"Materials": [
  {
    "name": "aluminium",
    "strength": {
      "plasticity": {
        "model": "linear"
      }
    }
  }
]
```

The recursive merger sets `model: "linear"` while automatically retaining the library preset's calibrated linear yield stress ($\sigma_{y0} = 0.324$ GPa) and plastic hardening modulus ($H = 0.50$ GPa). The user may also supply explicit `yieldStress` and `hardeningModulus` values to override these defaults.

4.2.4 Pattern 4: Granular Parameter Overrides

Users can override any individual physical or constitutive parameter without having to re-declare the remainder of the material definition. Only the modified keys need to appear in the deck:

```
"Materials": [
  {
    "name": "copper",
    "strength": {
      "plasticity": {
        "A": 0.120,
        "B": 0.350
      }
    }
  }
]
```

In this configuration, copper retains its baseline reference density (8.96×10^{-6} kg/mm³), elastic moduli ($E = 124$ GPa, $\nu = 0.34$), rate sensitivity ($C = 0.025$), and thermal softening constants ($T_m = 1356$ K), while yielding at an elevated initial stress $A = 0.120$ GPa and hardening with $B = 0.350$ GPa.

4.2.5 Pattern 5: Suppressing Plasticity (Purely Elastic Solid)

To simulate a material purely elastically without plastic deformation, the `plasticity` sub-card is explicitly set to `null`:

```
"Materials": [
  {
    "name": "aluminium",
    "strength": {
      "plasticity": null
    }
  }
]
```

The resulting material carries copper or aluminium elasticity and acoustic wave speeds, but bypasses the plastic yield condition and radial return map entirely.

4.3 Laying Blocks, Carving Regions, and Placing Bodies

Initial material is declared as axis-aligned lattice blocks, `SolidBlocks[]` and `FluidBlocks[]`, each of which lays a regular lattice of spacing $\Delta x_0 = 2\text{Domain.particleRadius}$ between its `start` and `end` corners. A block may additionally carry a `shape` object, which carves a region out of that lattice and discards whatever falls outside it; the primitives and their keys are tabulated in §4.4.3. Bodies that are not boxes are therefore built in two steps — a block that is large enough, and a region that cuts it down — and the two steps have separate responsibilities, which is the source of the traps below.

TRAP / OPERATIONAL NOTE: `shape: "circle"` IN A 3D BLOCK IS A CYLINDER, NOT A SPHERE

The `circle` region tests the x and y coordinates and ignores z entirely. The consequence is that one key describes three different bodies in the three kinematic modes: a disk in a single plane-strain layer, a *cylinder along z* in a full three-dimensional lattice, and a sphere in an axisymmetric deck, where the meridional disk is revolved about the axis. An axisymmetric deck therefore writes a spherical projectile as a `circle` and is correct to do so, and a three-dimensional deck that copies it silently receives a cylinder. Use `sphere` in three dimensions.

TRAP / OPERATIONAL NOTE: `center` DEFAULTS TO THE CENTROID OF THE LATTICE, NOT TO THE BODY

Where `center` is omitted the region is centred on the centroid of the lattice actually laid inside the block. That is the right default for a body centred in its own box and the wrong one for everything else — most sharply in a half or quarter symmetry model, whose body is centred on the cut plane or on the cut corner while the lattice centroid sits in the middle of the retained fraction. The region then carves away nearly all of the block without any error being raised: the case on record took a revolved dogbone from 10 417 particles down to 14. Write `center` out explicitly whenever the body is not centred on its block.

TRAP / OPERATIONAL NOTE: TWO BODIES LAID CLOSER THAN h ARE ALREADY IN CONTACT AT $t = 0$

The gap between two blocks is written in the scene file in units of length and reads naturally as a lattice spacing, but the range over which particles interact is the support radius h , which at the house `Domain.supportRadiusFactor` of 6.0 is $3\Delta x_0$. A projectile laid one Δx_0 clear of a target plate is thus a *third of a support radius* away, and the two bodies exchange forces on the very first step across whatever velocity discontinuity the deck gave them — 6700 mm/ms on the hypervelocity decks. Nothing warns about it and the run looks entirely normal. Placing the projectile at $2h$ instead lets it arrive ballistically and generates the contact from the dynamics rather than imposing it in the initial condition. Either is defensible; it should be a choice, and the two are not interchangeable when results from the decks are compared.

4.3.1 Decks Carrying a Symmetry Reduction

A deck declaring `Domain.axisymmetric` or `Domain.symmetryPlanes` (§2.9) models a fraction of its body, and the rest of the body is supplied at run time as mirror images of the stored particles. The scene file has to be written so that those images fit, and the conventions below all follow from that.

TRAP / OPERATIONAL NOTE: LAY THE FIRST ROW AT $\Delta x_0/2$ FROM THE CUT

Particles are placed at **half the interparticle spacing** from the axis and from every symmetry plane: the block `start` coordinate on each cut is `Domain.particleRadius =  $\Delta x_0/2$` , *not* 0. A row on the cut is counted on top of its own image; a row further out leaves a gap the images cannot fill. The reasoning, a figure, and the load-time checks are in §2.9.2.

Every particle must have a non-negative coordinate on each cut, which is checked and is fatal; the offset of the innermost row is printed at build time and warned about when it lies outside $[\frac{1}{4}, \frac{3}{4}] \Delta x_0$. The reactions carried by the cuts mean that the transverse momentum components are no longer conservation statements, so a status line showing them drift is reporting the physics rather than a defect.

Axisymmetric Decks

An axisymmetric deck lays a single meridional layer: x is the axis of revolution, y is the radius $r \geq 0$, and the layer is revolved about x . With `particleRadius = 0.125` the lattice starts at $y = 0.125$:

```
"Domain": { "start": [0,0,0], "end": [70,40,2.25], "particleRadius": 0.125,
            "supportRadiusFactor": 6.0, "axisymmetric": true },
"SolidBlocks": [
  { "objectId": 0, "start": [40.0, 0.125, 0.0], "end": [60.01, 20.01, 0.001], ... },
  { "objectId": 1, "start": [20.0, 0.125, 0.0], "end": [30.01, 5.01, 0.001], ...,
    "shape": {"type": "circle", "radius": 5.0, "center": [25.0, 0.0, 0.0]} }
]
```

- **$z = 0$ exactly, and one layer.** Both are checked at load. The layer is revolved, not extruded, so there is no thickness for it to sit in the middle of — unlike a plane-strain block, which is conventionally placed at mid-domain. The block above therefore runs from $z = 0$ to $z = 0.001$.
- **A circle centred on the axis is a sphere.** The half-disk left in the half-plane revolves into one, and a rectangular block becomes a solid cylinder. The `center` of the circle is written on the axis, $y = 0$, not on the first row.
- **A tensile specimen uses `dogboneaxi`, not `dogbone`.** The `dogbone` profile runs its long axis along y , which here is the radius, so revolving it produces a *tube* and not a bar — with a reasonable particle count and no complaint. The validator therefore refuses `dogbone` and `dogbone3d` in an axisymmetric deck and `dogboneaxi` outside one. In `dogboneaxi` the long axis is x , the radius is measured from the axis at $y = 0$ and *not* from `center`, and the widths are diameters.
- **The $y = 0$ face is not a wall.** The axis is a mirror, and the padding clamp leaves it and the z faces alone; the remaining faces behave as in any other deck.
- **Constraints boxes must straddle $z = 0$.** The selection is an inclusive box test on all three components and the layer sits at $z = 0$ exactly, so the box must contain $z = 0$; padding it to either side is the safe form, and `DogBoneAxi.grip_box` does this for the dogbone grips. The constrained components are "xy", the meridional pair, which is what "xyz" means for a body of revolution.
- Everything else — **Materials**, **Hourglass**, **Shifting**, **plasticity** — reads exactly as in a plane-strain deck.

Cartesian Symmetry Planes

`Domain.symmetryPlanes` is "y" for a half model cut at $y = 0$ or "yz" for a quarter model cut at $y = 0$ and $z = 0$. The planes sit at the coordinate origin, and every retained body is laid in the positive octant starting half a spacing off each plane. The quarter hypervelocity deck, with `particleRadius = 0.025`, reads:

```
"Domain": { "particleRadius": 0.025, "supportRadiusFactor": 6.0,
            "symmetryPlanes": "yz", "boundary": "open" },
"SolidBlocks": [
  { "objectId": 0, "start": [-3.2, 0.025, 0.025], "end": [3.2, 3.201, 3.201], ...,
    "shape": {"type": "sphere", "radius": 3.175, "center": [0.0, 0.0, 0.0]} },
  { "objectId": 1, "start": [4.0, 0.025, 0.025], "end": [6.0, 10.001, 10.001], ...,
    "shape": {"type": "cylinder", "radius": 10.0, "axis": "x", "center": [5.0, 0.0,
      ↪ 0.0]} }
]
```

- **Write center out explicitly, on the cut.** The body of a quarter model is centred on the corner where the two planes meet, while the default `center` is the centroid of the retained lattice; left to the default, the region carves away almost the whole block (trap above).
- **A full three-dimensional domain is required.** The key is rejected together with `Domain.axisymmetric`, which already mirrors about $y = 0$, and with `Domain.planeStrain`, which leaves no third dimension to cut.
- **There is no plane at $x = 0$.** The impact axis is x in every deck, so the cuts run along the direction of travel.
- **A quarter model is a symmetrised problem.** It suppresses every fragmentation mode other than $m = 0, 4, 8, \dots$; see §2.9.6 before quoting a debris distribution from one.

One point of housekeeping is met almost exclusively on decks that carry a symmetry reduction, because those are the decks large enough to be worth framing: under `Domain.boundary: "open"` the box may be shrunk freely to frame the interactive view, or left out of the deck altogether, because it bounds nothing that the solver does. The trap accompanying §1.3 gives the measurement, and records that this manual once asserted the opposite.

4.4 Keyword Cards and Parameter Tables

The following tables are generated directly from the authoritative parameter registry in `params.py`.

4.4.1 Master Keyword and Card Summary

Table 4.1 provides an executive directory of all top-level cards and data blocks supported in `sigmaSPH` scene decks. Each card links directly to its comprehensive specification and parameter description in the subsequent sections.

Keyword Card	Status	Physical and Algorithmic Scope
Domain	Required	Box dimensions, particle spacing Δx_0 , and 2D/3D kinematic constraints.
Time	Required	Acoustic and acceleration CFL, total duration, and step count limits.
Solver	Required	Solver selection (DSPH vs. SOLID), bulk viscosity, and artificial viscosity.
Materials	Required	Physical definitions: reference density, equations of state, and constitutive strength.
FluidBlocks	Optional	Initial particle lattices instantiated as weakly compressible fluids.
SolidBlocks	Optional	Initial particle lattices instantiated as hypoelastic-plastic solids.
Load	Optional	Body forces (gravity) and displacement boundary-condition ramp parameters.
Diffusion	Optional	Numerical density (δ -SPH) and low-Mach pressure (γ -SPH) diffusion.
Shifting	Optional	Particle Shifting Technology (PST) for regularising particle disorder.
Hourglass	Solid only	Kinematic zero-energy mode damper and super-time-stepping (STS) acceleration.
VelocityLimit	Solid only	Affine-corrected neighbourhood prediction and absolute runaway velocity caps.
Erosion	Solid only	Jacobian ($J = V/V_0$) clamping and equivalent plastic strain erosion limits.
Constraints	Solid only	Prescribed displacement boundary conditions and fixed boundary grips.
Burn	JWL only	Programmed burn detonation front width and release kinematics.
Detonators	JWL only	Spatial initiation locations and detonation trigger times.
Output	Optional	AMBER NetCDF, Extended-XYZ, PLY, OBJ dumps, and visualization settings.
Configuration	Optional	Flat legacy parameter override layer applied on top of card definitions.

Table 4.1: Structured directory of scene-file keyword cards and data blocks.

4.4.2 Keyword Card Specifications

Domain Card

The simulation domain bounding box, spatial discretisation spacing, and dimensional configuration (3D Cartesian, 2D plane strain, or 2D axisymmetry).

Key	Type	Default	Description
start	vec3	[0.0, 0.0, 0.0]	[x,y,z] lower corner of the simulation box (optional under boundless open boundaries).
end	vec3	[1.0, 1.0, 1.0]	[x,y,z] upper corner (optional under boundless open boundaries; required under 'reflect'). padding = support_radius is clamped at each face.
particleRadius	float	required	$\Delta x_0 = 2 * \text{particleRadius}$ is the lattice spacing.
supportRadius	float	4.0	$h = \text{particleRadius} * \text{factor}$. 4.0 ($= 2 \Delta x_0$) is the fluid default; solid scenes should use 6.0 ($= 3 \Delta x_0$). Larger is NOT better – 3/4/5 Δx_0 reach 0.300/0.280/0.220 strain before failing (8).
planeStrain	bool	false	Single XY layer, Z motion suppressed. Needs a one-particle-thick block and a domain Z extent of about 3h.
axisymmetric	bool	false	Single XY layer read as a MERIDIONAL half-plane revolved about the x axis: x is axial, y is the radius $r \geq 0$, $z = 0$. Mutually exclusive with planeStrain. Every particle must have $y \geq 0$ and $z = 0$ (11).

Key	Type	Default	Description
<code>symmetryPlanes</code>	<code>str</code>	<code>""</code>	Cartesian symmetry planes, as a subset of "yz" – "y" for a half model cut at $y = 0$, "yz" for a quarter model cut at $y = 0$ and $z = 0$. The planes sit at the coordinate origin, which is where <code>Domain.start</code> is. Neighbour sums visit the MIRROR IMAGES of the stored material across each plane, and the corner image across both, exactly as axisymmetry does at the axis, so the plane does not read as a free surface (20; 14.3 for the argument). Lay the lattice from $\Delta x_0/2$ off each plane, as an axisymmetric deck does off the axis. Requires a full 3D domain: rejected with axisymmetric (which already mirrors about $y = 0$) and with <code>planeStrain</code> (which pins z). An $x = 0$ plane is not offered.
<code>axisRadiusMin</code>	<code>float</code>	<code>0.5</code>	Floor on the radius the $1/r$ geometric terms are evaluated at, in units of Δx_0 . 0.5 is the half-spacing a lattice laid from $y = \Delta x_0/2$ never goes below, and it is what keeps the hoop stiffness inside the acoustic CFL (11.3).
<code>conservativeAxial Momentum</code>	<code>bool</code>	<code>true</code>	Carry the PAIR radius $(r_i + r_j)/2$ in every axial pair force and let the pair sum generate the <code>sigma_xr/r</code> source, instead of carrying particle i 's own r_i and adding the source separately. The two agree in the continuum limit; the first conserves the axial ring momentum $\sum m v_x$ to round-off and the second does not (11.4). Radial is unaffected either way. <code>false</code> restores the pre-adoption form, which is what every number recorded before this existed was taken with.

Key	Type	Default	Description
conservativeRadial Force	bool	false	Carry the PAIR radius $(r_i + r_j)/2$ in every radial pair force as well, and reduce the explicit geometric source from $(\sigma_{rr} - \sigma_{tt})/r$ to $-\sigma_{tt}/r$, the pair sum now generating the σ_{rr}/r half of it. What this buys is NOT momentum – the net radial ring momentum of a body of revolution is identically zero by symmetry, so there is nothing there to conserve – but ENERGY. With both components carrying the pair radius, $m_i a_i$ is pair-antisymmetric, so the pair force’s work telescopes in the RING measure m and not merely in the meridional-plane measure mw, which is the +30 J of the free copper-aluminium impact’s remaining +18.7% energy gain. It also adds the hoop stress power $\sigma_{tt} v_r/r$ to the energy equation, which that equation was missing outright, and with both the deck goes from +18.7% to -4.1% – what its plane-strain twin reads (16.5). **It is nevertheless default false and should stay that way for axis-sensitive work, because the accuracy it trades away is not small.** The legacy form gets the uniform-stress cancellation out of $\sum_j w_j \text{grad } W = 0$, which a lattice satisfies to $1e-5$; this form needs $\sum_j w_j (r_j - r_i) \text{grad } W = e_r$, which is first-order kernel consistency and holds only to a few per cent with the bare gradients the stress divergence uses. On t21’s patch test under a pure hydrostat, $a /(\sigma /(\rho h))$ goes from $1.6e-5$ to $2.2e-2$ in the interior and from $1.6e-5$ to $2.9e-1$ on the axis; at the 10 GPa pressures of an impact deck the interior figure is a spurious radial force of order tens of per cent of the real one. Also costs a stabiliser dt cap near the axis, exactly as conservativeAxialMomentum does.

Key	Type	Default	Description
axialImageRadius	str	"ring"	Which radius the conservative AXIAL rescaling reads for a MIRROR IMAGE neighbour, and it is a straight trade between two things that cannot both be had (16.6). 'ring' (the default) reads the image's ring radius +r_j: a ring mass is positive and an image is the same physical ring as its original, and this is exactly what makes $r_i \lambda_{ij} = r_j \lambda_{ji}$ hold for image pairs, which is the identity the axial RING MOMENTUM telescopes on. 'signed' reads the coordinate the image actually SITS at, -r_j, which is the radius the geometric expansion behind the rescaling is written in. Measured on t21: 'ring' gives a ring momentum drift of 2.9e-08 (round-off, 4e+05 times better than the legacy form) and a relative error of 1.14e-01 in the generated $\sigma_{xr}/(\rho r)$ source ON THE AXIS, against 2.6e-03 in the interior; 'signed' gives 2.6e-03 on the axis, no worse than the interior, and a drift of 7.8e-04, only 33 times better than legacy. Choose 'signed' when near-axis accuracy matters more than the momentum balance and you have checked that it does for your problem; the default is 'ring' because 14.4 exists to secure that balance. Applies to the stress pair force only – Monaghan's Π_{ij} and the hourglass damper generate no geometric source, so signing them would cost conservation and buy nothing. <i>Choices: "ring", "signed".</i>
boundary	str	"open"	'open': boundless domain where particles are not bound and move freely across space via the sparse hashed neighbour search. 'reflect': rigid bouncing walls placed one support radius inside the bounding box; only makes sense in combination with specified Domain.start and Domain.end. Committed scenes under data/scenes/ are pinned to 'reflect' so recorded numbers do not move under the new default. <i>Choices: "open", "reflect".</i>
restitution	float	0.5	Coefficient of restitution c_f of the bouncing wall: $v := (1 + c_f)(v \cdot n)n$, applied only to a particle whose velocity still points OUT of the domain, so the outgoing normal speed is c_f times the incoming one and the tangential components are untouched. In [0, 1]. The default 0.5 removes 75% of the normal kinetic energy per hit, which answers the position clamp that accompanies the bounce: teleporting a particle onto the wall plane injects energy through the density field and nothing else takes it out. 1.0 is the specular wall, which conserves $\ \mathbf{v}\ $ exactly at a face and at a corner and is what a deck that never touches its boundary wants; 0 absorbs the normal component entirely (18).

Time Card

Timestep controls, acoustic and acceleration CFL factors, simulation termination criteria, and step limits.

Key	Type	Default	Description
CFL	float	required	Courant number for the acoustic and force limits.
stepSize	float	—	Explicit dt override; also becomes the ceiling dt_max. Adaptive if absent.
duration	float	—	Stop time in seconds.
maxSteps	int	—	Hard cap on render cycles. A run that hits it looks exactly like one that finished – see 11.
stepsPerRenderUpdate	int	—	Solver substeps between GUI frames.
pairSoundSpeed	str	"samematerial"	Which pairs the finite-difference sound speed $c^2 = dp / d\rho $ of the acoustic dt limit is taken over (6.9). 'sameMaterial' (the default) skips pairs whose K or ρ_0 differ, across which the quotient compares two different EOS and is not a sound speed at all; 'off' drops the estimate entirely, which for a hypoelastic material is provably the same run; 'all' is the pre-2026-09-15 behaviour and is what every measurement taken before that date was taken with. <i>Choices:</i> "all", "samematerial", "off".

Solver Card

Selection of the physical solver (fluid vs. hypoelastic-plastic solid) and numerical regularisation parameters. Note that physical material properties do not reside here; they belong exclusively in `Materials[]`.

Key	Type	Default	Description
type	str	required	Which solver runs: 'deltaPlusSPH' (the weakly compressible fluid solver) or 'hypoElastic' (the solid solver). Discriminates this card. It is a SCENE-wide choice because the two are different classes; whether an individual material is a solid or a fluid is decided by whether its <code>Materials[]</code> entry declares a strength model. <i>Choices:</i> "deltaPlusSPH", "hypoElastic".
viscosity	float	required	Morris shear viscosity. In a solid scene this is a damper, not physics.
allowNegativePressure	bool	false	Permit $p < 0$. Required wherever the material is in tension (every solid scene, and the rotating patch). In a violent free-surface flow it is actively dangerous – it cost the dambreak two runs in three (10.3).
stiffness	float	50000.0	Only used to initialise the rotating-patch Fourier pressure field. An initial condition, not a material constant, which is why it is still on this card.
dampingCoefficient	float	0.0	Velocity-proportional damping for dynamic relaxation.
objectiveRate	str	"jaumann"	'none' drops the spin terms. Exists so t11 can show they are load-bearing; do not use it. <i>Choices:</i> "jaumann", "none".
kernelCorrection	bool	false	Correct grad v with the regularised L. Needed only when a body with free surfaces rotates significantly (6.1).
lEigTol	float	0.1	Eigenvalue guard for that correction. Must sit below a convex corner's $\lambda = 0.254$ and above degeneracy.

Key	Type	Default	Description
trackViscousWork	bool	true	Accumulate the work done by the artificial viscosity, reported as the VW column of the energy budget (16) and as <code>solver.compute_viscous_energy()</code> . On BOTH solvers, but it means something different on each, and the difference is the whole of section 16.3. In the fluid solver it is a path term of the budget: that solver has no internal energy, so the joules it measures have left the total and must be named or they appear as an unexplained residual. In the solid solver the same joules are already inside IE – the energy equation is the work conjugate of the pair force that was applied (13.3) – so it is a diagnostic rather than a term, and subtracting it would count it twice. It is reported there so that the two solvers can be compared on one deck, which is how the dambreak comparison establishes that they dissipate the same amount and differ only in whether they keep it. It costs one f32 read-modify-write per PAIR on the hot loop – measured at 7% of the step time on the 2D fluid dambreak and 20% on the solid one, both at 8100 particles – which is why it can be turned off; set false on a production run that is not being measured. Inert where the term it measures does not exist: <code>viscosity: 0</code> on the fluid side, and <code>viscosity: 0</code> with no Monaghan <code>alpha/beta</code> on the solid side. Nothing is compiled in either case.

Solver.bulkViscosity Sub-Card Quadratic von Neumann–Richtmyer shock viscosity $q = C_Q \rho l^2 (\nabla \cdot \mathbf{v})^2$ added to the pressure strictly under compressive volumetric strain ($\nabla \cdot \mathbf{v} < 0$). Provides shock capturing without damping post-shock elastic oscillations.

Key	Type	Default	Description
quadratic	float	0.0	C_Q of the von Neumann-Richtmyer shock viscosity $q = C_Q \rho l^2 (\nabla \cdot \mathbf{v})^2$, added to the pressure ONLY where $\nabla \cdot \mathbf{v} < 0$. Unlike <code>viscosity</code> (linear in the velocity difference, active everywhere) this is quadratic and confined to compression, so it captures a shock without damping the elastic response around it. 1.0-2.0 is the classical range; 0 disables it (6.7).
lengthScale	str	" Δx_0 "	Which length l is: ' Δx_0 ' is the lattice spacing, the zone size of the classical definition and the default; 'h' is the kernel support, which at the solid supportRadiusFactor of 6 ($= 3 \Delta x_0$) makes the same C_Q NINE times stronger. Quote C_Q with the length it was calibrated against (6.7). Choices: " Δx_0 ", "h".

Solver.artificialViscosity Sub-Card Monaghan pairwise artificial viscosity $\Pi_{ij} = (-\alpha \bar{c} \mu_{ij} + \beta \mu_{ij}^2) / \bar{\rho}$, acting strictly on approaching particle pairs ($\mathbf{v}_{ij} \cdot \mathbf{r}_{ij} < 0$).

Key	Type	Default	Description
alpha	float	0.0	alpha of Monaghan's $\Pi_{ij} = (-\alpha\bar{c}\mu_{ij} + \beta\mu_{ij}^2)/\bar{\rho}$, the LINEAR half, applied only to a pair that is approaching ($\mathbf{v}_{ij} \cdot \mathbf{r}_{ij} < 0$). Same dissipation as <code>viscosity</code> above but sign-gated, so it does not damp the elastic wave either side of the shock. 0.5-1.0 is the classical range; 0 disables both halves (6.8).
beta	float	0.0	beta of the same Π_{ij} , the QUADRATIC half. Unlike <code>bulkViscosity</code> (which is quadratic in a kernel-smoothed $\text{div } \mathbf{v}$, per particle) this one grows as a single PAIR closes, which is what stops interpenetration at a contact. Classically 2 alpha; 0 leaves the linear half alone (6.8).
epsilon	float	0.01	epsilon of $\mu_{ij} = h\mathbf{v}_{ij} \cdot \mathbf{r}_{ij} / (\ \mathbf{r}_{ij}\ ^2 + \epsilon h^2)$, which keeps μ finite as a pair closes. 0.01 is the standard value and the one the Morris <code>viscosity</code> already uses; raising it softens the close-range limit.

Burn Card

Kinematic parameters governing the programmed burn model (§15). Controls the numerical front smearing width and compression-release behaviour. Initiation points reside in `Detonators`; detonation wave speed resides in `Materials[] .eos`.

Key	Type	Default	Description
width	float	1.5	b of $F1 = (t - t_l) D / (b L)$: how many lengths L the prescribed front takes to open the burn fraction from 0 to 1. b = 1.5 with $L = \Delta x_0$ reproduces the classical zone-based form exactly, which burns a zone in 1.5 transit times. A quoted b is meaningless without the L it was quoted against (13).
lengthScale	str	" Δx_0 "	Which length L is. ' Δx_0 ' is the lattice spacing and the default; 'h' is the kernel support, which at the solid supportRadiusFactor of 6 is THREE times Δx_0 and so makes the same b a three times wider front. Same trap as <code>bulkViscosity.lengthScale</code> , and the same reason the key exists. <i>Choices</i> : " Δx_0 ", "h".
volumeBurn	bool	true	Whether the compression half $F2 = (1 - V) / (1 - V_{CJ})$ is taken alongside the lighting-time half, $F = \max(F1, F2)$. It lights material that the arriving wave has already compressed even where the straight-line lighting time says it should not be lit yet, which is what keeps a converging or a reflected front from running ahead of its own clock. It does NOT make the model predictive: neither half can fail to burn (13).

Load Card

External acceleration loads (gravity) and displacement-boundary-condition time ramp parameters.

Key	Type	Default	Description
gravitation	vec3	required	[gx, gy, gz] body force.

Key	Type	Default	Description
rampTime	float	0.0	Smoothstep ramp of the prescribed displacement, then hold. A step in grip velocity would ring the specimen for the rest of the run (6.3).

Diffusion Card

Explicit numerical density and pressure diffusion (Molteni-Colagrossi δ -SPH and Collé γ -SPH). Both share a unified stability budget ($\delta + \gamma \leq 1/3$).

Key	Type	Default	Description
delta	float	0.0	delta coefficient of the Molteni-Colagrossi density diffusion. Shares one explicit-diffusion budget with the gamma term (10.2).
pairs	str	"samematerial"	Which pairs the Molteni-Colagrossi density diffusion is taken over. 'sameMaterial' (the default) skips pairs whose material differs, across which $\rho_j - \rho_i$ is a property of the INTERFACE and not an error in the density field, so the term stops being a regulariser and becomes a source. Measured on the cylinder test: across a 4.9x density jump the ungated term drives the detonation products to $V = 0.24$ and 200 GPa, against $V = 0.83$ and 26 GPa with the gate on, where V_{CJ} is 0.74 and p_{CJ} is 37 (13.6). If set to 'all', a warning banner is printed. Choices: "all", "samematerial".

Diffusion.gamma Sub-Card Collé et al. (2019) low-Mach Riemann-based pressure flux stabilisation Γ_{ij} .

Key	Type	Default	Description
coefficient	float	0.0	gamma-SPH low-Mach flux stabilisation. delta = 0.1 is worth gamma = 0.4; the measured upper bound moves from about 1.2 at delta = 0 to about 0.8 at delta = 0.1 (10.2).
momentum	bool	false	Apply the same correction in the advective momentum flux. NOT the dissipative half (10.2).
renormN	bool	false	Take n_{ij} from the renormalised gradient rather than the bare radial direction. Off because $L_i \neq L_j$ costs the pair term its antisymmetry.
clampedP	bool	false	Let the negative-pressure clamp through to Γ_{ij} . Off by default: the clamp is a statement about the pressure FORCE, and Gamma uses pressure only as a density proxy. true reproduces the 13.4 tables.

Shifting Card

Particle Shifting Technology (PST) parameters. Omitting this card completely disables particle shifting. Controls whether Sun et al. (2017) displacement shifting or Collé et al. (2019) velocity shifting is employed.

Key	Type	Default	Description
enabled	bool	true	MANDATORY for solids – without it the specimen carries essentially no load (6.4). Omitting the whole Shifting card turns shifting off.
mode	str	"sun"	'sun' = the delta+-SPH position correction; 'colle' = the Colle et al. (2019) 4.3 shifting VELOCITY, $dr = dv*dt$. Every solid scene uses 'colle' (10). <i>Choices</i> : "sun", "colle".
maMode	str	"local"	Reference Mach number: 'local' uses $\ v_i\ $, 'global' uses $\max \ v\ $, 'fixed' uses maFixed. Solid scenes use 'fixed' – a quasi-static specimen has no velocity scale to key on. <i>Choices</i> : "local", "global", "fixed".
maFixed	float	0.1	The fixed amplitude. NOT transferable between modes: 0.02 under 'sun' and 0.05 under 'colle' are the same regularisation (10.3).
R	float	0.2	Monaghan tensile correction R.
n	int	4	Monaghan tensile correction exponent.
maxShift	float	0.05	Per-step displacement limiter, in units of Δx_0 . Not optional: the raw shift grows without bound with local disorder.
lambdaLo	float	0.4 / 0.2	Below this lambda, no shift. Defaults to 0.4 under 'sun' and the paper's 0.2 under 'colle'.
lambdaHi	float	0.75	Above this lambda, the full shift; between LO and HI, tangential only.
normal	str	"gradlam"	Surface normal for the tangential projection: $\langle \nabla \lambda \rangle$, or the eigenvector of E belonging to lambda_min, which compute_lam already has. <i>Choices</i> : "gradlam", "eigvec".
clamp	str	derived	'displacement' caps $\ \delta r\ $ at $\maxShift \cdot \Delta x_0$, 'velocity' caps $\ \delta v\ $ at $m_\delta \ v_i\ $, 'both' applies both. Defaults to 'both' for 'colle' except under maMode 'fixed' – the velocity clamp goes to ZERO, not slack, where the material is at rest. <i>Choices</i> : "displacement", "velocity", "both".
mDelta	float	0.4	The velocity clamp fraction, Colle's m_delta. Only read when clamp is 'velocity' or 'both'.
aleMomentum	bool	true	ALE transport of momentum (5).
aleHistory	bool	true	ALE transport of sigma_dev, $\bar{\epsilon}_p$ lastic AND the internal energy e_{int} (5). The third was omitted from this line for as long as it existed; it is the one with no bound on it, and 16.10 measures the operator amplifying e at ≈ 1.0015 per step on a strengthless free-surface deck.
aleDiffusion	float	1.0	Upwind stabilisation of the ALE transport of e_{int} , in units of the numerical diffusivity of first-order upwinding: xi = 1.0 adds exactly $ \delta u \Delta x_0 / 2$, which is what upwinding the centred operator would have cost. The centred difference under one-stage explicit Euler is unconditionally unstable at $\sqrt{1 + (k \nu)^2}$ per step (5), so this is not optional; 0.0 restores the unstabilised operator and is a diagnostic ablation, not a setting. Self-scaling on the shift, so it vanishes identically where nothing is shifted and needs no material constant. Applies to e_{int} alone – sigma_dev and $\bar{\epsilon}_p$ lastic run the same difference form and are bounded by the yield surface and by their clamp instead (5).

Key	Type	Default	Description
aleEnergy Conservative	bool	false	Make the ALE advection of e_{int} conserve Sum m e by a global fix-up each step. The advection re-samples e at the shifted position while the mass stays put, so its sum is a discrete Sum m (dr . grad e), which is not zero; on the 3D Chocron quarter it is +4.5% of E0, made at the shock (6.11, 19). With this on, each particle's advective rate a_i is replaced by a_i - a_i S/G, with S = Sum m a and G = Sum m a over the step, so that the mass-weighted sum is zero and the correction lands only on particles that are being advected, in proportion to how much. The diffusion is pairwise conservative already and is left alone. The correction is banked as ie_ale_fix. Off by default, as a per-deck option: with fixed masses it cannot also be consistent (a uniform shift through a gradient really does change Sum m e, which t29 checks), and on the gelatin stack it feeds on itself (e_neg -1.7% -> -47%). On in the Chocron, Al 2017 and Qamar decks, where it removes +5 to +13% of E0 with the observables unchanged (reports/ale_energy_transport_2026-09-26.md).
lEigTollLam	float	0.02	Eigenvalue guard for $\langle \nabla \lambda \rangle$, where the 1/e amplification is wanted.

Hourglass Card

Kinematic zero-energy mode suppression for solid mechanics. Mandatory for all solid calculations to prevent non-affine shear locking and grid pairing.

Key	Type	Default	Description
alpha	float	0.0	MANDATORY for solids. Damps the non-affine part of the relative velocity. 2.0 reaches 26% strain, 4.0 is needed for the dogbone; it also sets dt (9.1).
saturate	bool	false	Prototype: integrate the damping exactly over the step. LIFTS THE dt CAP AND DESTROYS THE ANSWER – kept as evidence (9.1).

Hourglass.adaptive Sub-Card Adaptive hourglass damping parameter scaling based on measured non-affine velocity magnitude.

Note: The mere presence of this sub-card synthesises HOURGLASS_ADAPTIVE.

Key	Type	Default	Description
alphaMin	float	0.25	Floor for the adaptive controller; HOURGLASS_ALPHA is the ceiling.
uTarget	float	0.05	Target non-affine velocity, m/s. 0.15 is the largest that kept the answer.
updateEvery	int	10	Steps between controller updates.
rise	float	1.25	Per-update rise limit, deliberately asymmetric with fall.
fall	float	0.99	Per-update fall limit.

Hourglass.superTimeStepping Sub-Card Runge–Kutta–Legendre (RKL1) super-time-stepping sub-cycling of the hourglass damper, relaxing the explicit stability timestep restriction by up to 5× to 11×.

Note: The mere presence of this sub-card synthesises `HOURLASS_STS`.

Key	Type	Default	Description
mode	str	"subcycle"	'subcycle' takes n plain forward-Euler damper steps of dt/n inside one hydro step, which reproduces the one-stage damping EXACTLY at the same damper work (9.2). 'rkl' uses the s-stage Runge-Kutta-Legendre polynomial instead; it is kept as evidence – it is stable at every dt it claims and still destroys the specimen, because it is built to advance a diffusion accurately and the damper's job is to annihilate. <i>Choices:</i> "subcycle", "rkl".
stages	int	0	Number of sub-steps (subcycle) or RKL1 stages (rkl). 0 auto-sizes: the smallest count that lets dt reach the limit the rest of the scheme sets.
refreshGradV	bool	true	Recompute grad v after the hydro kick so the affine reference the damper subtracts matches the field it damps. Off is the control for whether a stale reference is what a long super-step costs.
safety	float	0.8	Fraction of the RKL1 stability interval $s^2 + s$ that dt may use. 0.8 is the same 80% the one-stage cap keeps of its own interval of 2.

VelocityLimit Card

Runaway velocity limiters (affine-corrected neighbourhood prediction and absolute velocity capping) to bound catastrophic non-physical acceleration.

Key	Type	Default	Description
cap	float	0.0	Hard ceiling on $\ \mathbf{v}\ $, m/s. Deliberately not defaulted from c_p : only the scene knows what 'too fast' means. Does NOT redistribute (6.5).

VelocityLimit.limiter Sub-Card Affine-corrected neighbourhood prediction limiter bounding non-affine particle velocity deviations.

Note: The mere presence of this sub-card synthesises `VELOCITY_LIMIT`.

Key	Type	Default	Description
u	float	—	Cap on how far a particle's velocity may depart from what its own neighbourhood predicts, m/s. Defaults to $0.02 c_p$, about 2x above the worst healthy excursion measured (6.5).
conserve	bool	true	Hand the removed momentum back to the donor's own neighbourhood, so the limiter conserves $\sum(m \mathbf{v})$ and $\sum(m \mathbf{x})$ exactly. Off is the non-conservative control.

Erosion Card

Deformation limiters acting on the deformation Jacobian $J = V\rho_0/m$, providing clamping bounds or particle deletion upon excessive distortion or plastic strain.

Key	Type	Default	Description
mode	str	"off"	'cap' clamps $J = V \rho_0/m$ back into $[\text{minVolumeRatio}, \text{maxVolumeRatio}]$ and leaves the particle in the simulation; 'erode' takes the particle out of every neighbour sum and freezes it. 'off' is bit-identical to no card. <i>Choices:</i> "off", "cap", "erode".
minVolumeRatio	float	0.3	Lower bound on V/V_0 . Below it the particle is capped or eroded. The linear EOS makes this a pressure bound too: $p \leq K(1/J_{\min} - 1)$.
maxVolumeRatio	float	1.5	Upper bound on V/V_0 , i.e. the tensile side. $p \geq K(1/J_{\max} - 1)$; with no spall model this is what stops the tension growing without limit.
maxPlasticStrain	float	0.0	Equivalent plastic strain at which a particle is eroded, 0 = never. Only meaningful under mode 'erode': capping $\bar{\epsilon}_p$ would freeze the yield surface, which hardening-wise is the opposite of failure.

Output Card

Diagnostic, visualization, trajectory dump intervals, and camera framing options.

Key	Type	Default	Description
txt	bool	false	Extended-XYZ dumps dump_XXXX.xyz, readable by OVITO. Every object is written, and an object_id column says which block each particle came from – in a multi-block scene that is the only thing that tells impactor from target once the two have interpenetrated.
netcdf	bool	false	Binary AMBER-NetCDF trajectory <scene>.nc, one growing file for the whole run rather than a file per frame, readable by OVITO Basic, at $\approx 45\%$ of the size of the extended-XYZ dump and ≈ 5 ms per frame against ≈ 66 ms. The per-particle fields are the four lists at the top of netcdf_writer.py and are NOT the XYZ dump's columns: position, velocity, object_id, density, pressure and lam always; for a solid also sigma_xx, sigma_yy, sigma_zz, sigma_xy, von_mises, $\bar{\epsilon}_p$ lastic, eroded, e_{int} and temperature, plus sigma_xz and sigma_yz in 3D, burn_f on a JWL deck, and damage and porosity if used by active materials. The out-of-plane shears are left out under planeStrain and axisymmetric because a single layer makes them identically zero; sigma_zz is not, and is written in 2D too – it is the hoop stress in axisymmetry. Needs the netCDF4 package; see section 1.
deformation Gradient	bool	false	Allocate and integrate the deformation gradient F , exposing it as ps.F . Off by default: nothing in the constitutive update reads F – it is integrated from grad v purely as a diagnostic – so a run that does not ask for it saves a 3x3 field per particle, a second one for its rate, and the matrix product that advances it every step. Turn it on for a test or a tools/ study that measures a logarithmic strain $\log F_{yy}$.
ply	bool	false	PLY point clouds.

Key	Type	Default	Description
obj	bool	false	Rigid-body OBJs.
frame	bool	false	PNG screenshots of the GGUI window, one every Output.interval render cycles. Because it screenshots the window rather than rendering offscreen, it is REFUSED under run_simulation.py -no-gui rather than quietly producing nothing; a headless run wanting a trajectory wants Output.netcdf.
constraints	bool	—	Record force and displacement on each constraint at regular intervals to disk (<scene>_constraints.csv). Active by default for solid simulations with constraints.
constraints Interval	int	1	Timestep interval for recording constraint force and displacement. Default is 1 (every timestep) for smooth force-displacement curves.
interval	int	—	Render cycles between dumps, counted from zero, so the FIRST frame of every particle dump – netcdf, txt and ply alike – is the state at $t = 0$ and not the state after the first render cycle. That frame is the reference configuration an OVITO displacement modifier measures against. It is the state as LAID, though: positions, velocities, volumes, density, temperature and object ids are the initial condition, while the fields the solver derives inside a step – pressure, lam, the stress components, von Mises – are still zero there, because no kernel has run on it yet (18). The PNG stream under Output.frame is a screenshot of the window and so cannot start before the first render; image N shows one render cycle later than data frame N.
colorField	str	—	Defaults to vonMises for solids and pressure for fluids. burnFraction and internalEnergy exist only in a scene that declares a JWL material; damage reflects the spallation continuum damage state (0=intact, 1=failed); porosity reflects void volume fraction f; damageTension is the Grady-Kipp tensile damage D_t of an HJC material with a tension card; jcOmega is the Johnson-Cook fracture initiation accumulator omega, on [0, 1], of a material with damage model 'johnson_cook'; burnFraction is on a fixed [0, 1] scale so a front reads as a front. <i>Choices:</i> "pressure", "divergence", "vonMises", "epsPlastic", "burnFraction", "internalEnergy", "damage", "porosity", "damageTension", "jcOmega".
cameraMode	str	"iso"	'front' looks straight at the XY plane and frames the domain. <i>Choices:</i> "iso", "front".
invisibleObjects	list	—	Object ids to hide.

Materials Card Array

Declaration of an individual material: physical reference density, equation of state, and optional solid constitutive strength model. Referenced by block definitions via the **material** key.

Every material is declared with a unique **name**, a reference density **density0**, an equation of state **eos**, and an optional constitutive **strength** card.

Key	Type	Default	Description
name	str	required	Unique identifier referenced by block material keys.
specificHeat	float	0.0	Specific heat capacity of the material, a THERMODYNAMIC property and not a property of any strength model. It is what converts the thermal part of the internal energy into a temperature: $T = T_0 + (e_{\text{int}} - e_{\text{cold}}(\rho) - e_0)/C_v$, evaluated per particle in <code>update_temperature</code> (6.2.2). Units are the deck's own – $[\text{J}/(\text{kg.K})]$ and $[(\text{mm}/\text{ms})^2/\text{K}]$ are numerically the same number, so a handbook value needs no conversion between an SI deck and a $\text{mm}/\text{ms}/\text{GPa}$ one. Left at 0 the temperature stays at its initial value and nothing thermal happens, which is the right default for a deck that does not care. It used to live at <code>strength.plasticity.Cp</code> , which made it unavailable to a shock-EOS target with a linear strength model or none at all; that location is still accepted as a deprecated alias.
density0	float	required	Reference density ρ_0 .
delta	float	—	Molteni-Colagrossi density diffusion (delta-SPH) coefficient for this material. If omitted, inherits from the global <code>Diffusion.delta</code> (or 0.0). Setting <code>delta = 0.0</code> for gaseous detonation products avoids spurious mass diffusion into vacuum.
library	str	—	Name of a library preset to populate default physical constants for this material (e.g. 'copper', 'aluminium', 'construction_steel').

Materials[].eos**** Equation of state governing volumetric response: **linear** ($p = K(\rho/\rho_0 - 1)$), **tait** (weakly compressible fluid), **jwl** (Jones–Wilkins–Lee detonation products), or **mie_gruneisen** (polynomial Mie–Grüneisen shock EOS).

Key	Type	Default	Description
type	str	required	Pressure branch, and REQUIRED on every Materials[] entry. 'linear' is $p = K(\rho/\rho_0 - 1)$, what every metal here is on. 'tait' is $p = S((\rho/\rho_0)^\gamma - 1)$ with $S = c_0^2 \rho_0 / \gamma$, the weakly compressible fluid EOS; at $\gamma = 1$ it is identically the linear branch. 'jwl' is the Jones-Wilkins-Lee equation of state for detonation products, $p = p(\rho, e)$, which reads a specific internal energy and therefore switches the energy equation on (13). 'mie_gruneisen' is the polynomial Mie-Grüneisen equation of state for condensed solids and biological tissue (gelatine.pdf). 'hjc' is the Holmquist-Johnson-Cook three-phase compaction EOS for concrete (6.11), $p = p(\mu, \mu_{\text{max}})$, which remembers the largest compression reached. <i>Choices:</i> "linear", "tait", "jwl", "mie_gruneisen", "polynomial_mie_gruneisen", "mg", "hjc".
bulkModulus	float	required (linear)	K of the linear branch, and legal ONLY on a material with no strength model. Where there is one, $K = E/(3(1-2\nu))$ is DERIVED from it and stating K here could only contradict that.
c_0	float	required (tait, mie_gruneisen)	Numerical speed of sound of the Tait branch, or physical bulk sound speed of the Mie-Grüneisen branch, at the reference density.
exponent	float	required (tait)	Tait exponent gamma. 7 for water; 1 makes the branch identical to 'linear'.
A	float	required (jwl)	A of $p = A(1 - w/(R1 V))\exp(-R1 V) + B(1 - w/(R2 V))\exp(-R2 V) + w \rho e$, with $V = \rho_0/\rho$. A pressure, so it is in the deck's own pressure unit – published JWL fits are usually quoted in Mbar, which is 100 GPa.
B	float	required (jwl)	B of the same expression; the second, softer exponential.
R1	float	required (jwl)	R1, dimensionless. Typically 4-5.
R2	float	required (jwl)	R2, dimensionless. Typically 1-2.
omega	float	required (jwl)	omega, the Grueneisen coefficient of the products and the only place the internal energy enters. As $V \rightarrow \infty$ the exponentials vanish and the EOS becomes the ideal gas with $\gamma = 1 + \omega$.
e0	float	required (jwl)	Initial specific internal energy e0. For JWL: detonation energy per unit MASS, $e0 = E0/\rho_0$, where E0 is the energy per unit initial volume that a published fit quotes; for Mie-Grüneisen: initial specific internal energy (default: 0.0).
detonationVelocity	float	required (jwl)	D, the measured detonation velocity. Under a programmed burn this is not a result but an input: it sets the lighting time of every particle, $t_l = \ \mathbf{x}_0 - \mathbf{x}_{\text{det}}\ /D$, and the rate at which the burn fraction opens.
cjDensity	float	required (jwl)	ρ_{CJ} , the density at the Chapman-Jouguet state. Only the volume half of the burn reads it, through $V_{\text{CJ}} = \rho_0/\rho_{\text{CJ}}$.

Key	Type	Default	Description
vMin	float	0.1	Floor on $V = \rho_0/\rho$. The $w/(R V)$ terms diverge to minus infinity as $V \rightarrow 0$, which is outside the fit's validity rather than physics; this bounds the branch instead of letting a single over-compressed particle produce an unbounded negative pressure. Inert in a healthy run.
s	float	required (mie_gruneisen)	Linear shock Hugoniot slope parameter s relating shock velocity to particle velocity: $U_s = c_0 + s^*$ up.
gamma0	float	required (mie_gruneisen)	Dimensionless reference Grüneisen parameter Gamma0.
Gamma0	float	—	Alias for gamma0.
gruneisen	float	—	Alias for gamma0.
P0	float	0.0	Reference initial pressure P0 (default: 0.0).
p0	float	—	Alias for P0.
linearExpansion	bool	true	Whether to suppress C2 and C3 terms in expansion ($\mu < 0$) following standard hydrocode conventions (default: True).
C0	float	—	Direct polynomial coefficient C0 (pressure unit).
C1	float	—	Direct polynomial coefficient C1 (pressure unit).
C2	float	—	Direct polynomial coefficient C2 (pressure unit).
C3	float	—	Direct polynomial coefficient C3 (pressure unit).
C4	float	—	Direct polynomial coefficient C4 (dimensionless).
C5	float	—	Direct polynomial coefficient C5 (dimensionless).
crushPressure	float	—	HJC: P_crush, the end of the elastic phase OA. With crushStrain it sets the elastic bulk modulus $K = P_crush/\mu_crush$.
crushStrain	float	—	HJC: μ_crush , the volumetric strain $\mu = \rho/\rho_0 - 1$ at P_crush.
lockPressure	float	—	HJC: P_lock, the pressure at which the pores are closed (point B).
lockStrain	float	—	HJC: the volumetric strain at P_lock, μ_plock – what the parameter tables call U_lock. The grain-density strain μ_lock that the dense curve is measured from is DERIVED from it by continuity at B, not read (6.11).
K1	float	—	HJC: K1 of the fully dense curve $P = K1 x + K2 x^2 + K3 x^3$; also the unloading modulus of compacted material.
K2	float	—	HJC: K2 of the fully dense curve (may be negative).
K3	float	—	HJC: K3 of the fully dense curve.
tensileStrength	float	—	HJC: tensile strength T (positive). The pressure is cut off at $-T(1 - D)$, and $T^* = T/f_c$ enters the strength surface and the damage law.
spallPressure	float	0.0	Tensile hydrostatic spall cutoff (positive value, in pressure units, e.g. 1.2 GPa). When hydrostatic tension $-P$ exceeds this threshold, continuum damage D evolves towards 1.0, unilaterally degrading tensile pressure and deviatoric stress to model physical spallation and separation without artificial tensile bridging.

Materials[].strength Deviatoric stress response and elasticity. The presence of this card marks the material as a solid; omitting it defines an inviscid fluid ($G = 0$).

Key	Type	Default	Description
type	str	required (with strength)	The strength model. 'hypoElastic' is the only one there is. **Present means the material is a solid**: it has a shear modulus, carries a deviatoric stress and may yield. An entry with no strength card is a FLUID, described by its equation of state alone ($G = 0$), which is what a JWLL material has always been. <i>Choices</i> : "hypoElastic".
youngsModulus	float	required (with strength)	E. $G = E/(2(1+\nu))$, the bulk modulus $K = E/(3(1-2\nu))$ of the linear EOS, and the P-wave speed $c_p = \sqrt{(K+4G/3)/\rho_0}$ are all derived from it; c_p is what the CFL uses.
poissonRatio	float	required (with strength)	ν .
shearModulus	float	—	G, legal ONLY on a Holmquist-Johnson-Cook material (plasticity model 'hjc'), which states its shear modulus directly because its bulk modulus is not a free constant: $K = \text{crushPressure}/\text{crushStrain}$ comes from the eos card, and youngsModulus and poissonRatio beside it could only contradict that (6.11).

Materials[].strength.plasticity J_2 von Mises plasticity with linear isotropic hardening (MAT001) or Johnson–Cook viscoplasticity (MAT002). The presence of this sub-card activates plastic yield. Selected via model ("linear" or "johnson_cook"). For linear hardening, specify yieldStress *or* yieldStrain, and hardeningModulus *or* tangentModulusRatio. For Johnson–Cook, parameters A, B, n, C, eps0_dot, T0, Tm, m, Cp, and chi govern rate sensitivity, adiabatic heating, and thermal softening.

Key	Type	Default	Description
model	str	"linear"	Plasticity model. 'linear' for J2 linear isotropic hardening (MAT001), 'johnson_cook' for Johnson-Cook viscoplasticity (MAT002), 'hollomon' / 'power_law' for Hollomon power-law hardening ($\sigma_y = A + B * \bar{\epsilon}_p^n$), or 'hjc' for the Holmquist-Johnson-Cook concrete surface (MAT003, 6.11), which must be paired with eos.type 'hjc' and damage.model 'hjc'. <i>Choices</i> : "linear", "johnson_cook", "hollomon", "power_law", "hjc".
yieldStress	float	—	σ_{y0} , in the deck's pressure unit. Give this OR yieldStrain, not both.
yieldStrain	float	—	The UNIAXIAL elastic strain at yield; $\sigma_{y0} = E * \epsilon_y$. A plane-strain specimen does not yield there – it yields at $\sigma_{yy} = 1.125 \sigma_{y0}$ at $\nu = 0.3$ (6.2).
hardeningModulus	float	—	$H = d \sigma_y / d \bar{\epsilon}_p$. Give this OR tangentModulusRatio. 0 is perfect plasticity.
tangentModulusRatio	float	—	E_t/E instead of H; since strains add, $H = E r/(1-r)$. Must be in $[0, 1)$.
A	float	—	Johnson-Cook initial yield stress A in the deck's pressure unit (GPa). Under 'hjc': the normalised cohesive strength A (dimensionless), as are B and C.
B	float	—	Johnson-Cook strain hardening modulus B in the deck's pressure unit (GPa).

Key	Type	Default	Description
n	float	—	Johnson-Cook strain hardening exponent n (dimensionless).
C	float	—	Johnson-Cook strain rate sensitivity coefficient C (dimensionless).
eps0_dot	float	0.001	Johnson-Cook reference strain rate eps0_dot in the deck's time unit (ms^{-1}). Default: $0.001 \text{ ms}^{-1} = 1.0 \text{ s}^{-1}$.
T0	float	293.0	Johnson-Cook reference room temperature T0 in Kelvin (default: 293.0 K).
Tm	float	0.0	Johnson-Cook melting temperature Tm in Kelvin. When $T_m > T_0$, thermal softening is active.
m	float	1.0	Johnson-Cook thermal softening exponent m (dimensionless).
Cp	float	0.0	Specific heat capacity Cp in $\text{J}/(\text{kg}\cdot\text{K}) = (\text{mm}/\text{ms})^2/\text{K}$ for adiabatic plastic work heating.
chi	float	0.9	Taylor-Quinney coefficient chi: fraction of plastic work converted to heat (default: 0.9).
f_c	float	—	HJC: unconfined compressive strength f_c, which normalises the strength $\sigma^* = \sigma_y/f_c$ and the pressure $P^* = P/f_c$.
N	float	—	HJC: pressure hardening exponent N of $\sigma^* = [A(1 - D) + B P^{*N}](1 + C \ln \text{eps_dot}^*)$.
Smax	float	—	HJC: normalised maximum strength S_max, the cap on σ^* .

Materials[].damage

Key	Type	Default	Description
model	str	"none"	Continuum damage model type: 'none', 'threshold', 'cocks_ashby', 'hjc' (the Holmquist-Johnson-Cook cumulative law, 6.11, which requires the HJC eos and plasticity as well), or 'johnson_cook' (the Johnson-Cook fracture criterion with regularised softening, 6.12, which requires Johnson-Cook plasticity). <i>Choices:</i> "none", "threshold", "spall", "cocks_ashby", "qamar", "hjc", "johnson_cook", "jc".
spallPressure	float	0.0	Tensile spall pressure cutoff [GPa].
c1	float	1.5	Cocks-Ashby deviatoric plastic void growth coefficient c1.
c2	float	0.72	Cocks-Ashby deviatoric plastic void growth coefficient c2.
c4	float	1.0	Cocks-Ashby hydrostatic void growth coefficient c4.
c5	float	0.1	Cocks-Ashby hydrostatic void growth coefficient c5.
a1	float	25.0	Continuum damage degradation coefficient a1 in $\text{sd} = 1 - \tanh(a1 * f)$.
fn0	float	0.0001	Chu-Needleman nucleation volume fraction fn0.
sigma_hM	float	0.778	Chu-Needleman mean nucleation hydrostatic tension [GPa].
sigma_hS	float	0.0389	Chu-Needleman standard deviation of nucleation tension [GPa].
m	float	4.0	Rate sensitivity exponent m in Cocks-Ashby void growth.
f_max	float	0.5	Maximum allowable porosity f_max before complete cohesion loss.

Key	Type	Default	Description
rateMode	str	"effective"	Cocks-Ashby void growth strain rate coupling: 'effective', 'volumetric_fs', 'volumetric', 'unconstrained', 'unconstrained_stress'. <i>Choices:</i> "effective", "qamar", "volumetric_fs", "volumetric", "unconstrained", "unconstrained_stress".
D1	float	—	HJC: D1 of the strain to failure $\bar{\epsilon}_p \hat{f} + \mu_p \hat{f} = D1 (P^* + T^*)^{D2}$. Johnson-Cook: D1 of $\epsilon_f = [D1 + D2 \exp(D3 \eta)][1 + D4 \ln \epsilon_{\dot{p}}][1 + D5 T^*]$, $\eta = -p/\sigma_v M$ the stress triaxiality.
D2	float	—	HJC: D2 of the same expression. Johnson-Cook: D2, the amplitude of the triaxiality term.
EFMIN	float	0.01	HJC: floor on the strain to failure, which stops a small increment near the tensile cutoff from failing the material outright. 0.01 is Holmquist et al.'s value; Kim et al. do not quote one. Johnson-Cook: the same floor on ϵ_f , default 0 there (no floor beyond a guard against division by zero).
D3	float	—	Johnson-Cook failure: D3, the triaxiality exponent of ϵ_f , normally negative so that the strain to failure falls as the triaxiality rises.
D4	float	0.0	Johnson-Cook failure: D4 of the rate term $1 + D4 \ln \max(1, \epsilon_{\dot{p}}/\epsilon_{\dot{p}0})$, $\epsilon_{\dot{p}0}$ the plasticity card's reference rate.
D5	float	0.0	Johnson-Cook failure: D5 of the temperature term $1 + D5 T^*$, T^* the plasticity card's homologous temperature.
failure Displacement	float	—	Johnson-Cook failure: the plastic displacement u_f over which the damage goes from 0 to 1 once the accumulator $\omega = \sum d \bar{\epsilon}_p / \epsilon_f$ has reached 1, $dD = \Delta x_0 d \bar{\epsilon}_p / u_f$. Regularises the softening, so that the energy a crack dissipates, $\approx \sigma_y u_f / 2$ per unit area, does not scale with the particle spacing (6.12). Required, in the deck's length unit.

Materials[].damage.tension

Key	Type	Default	Description
model	str	—	Tension damage of an HJC concrete (DAM003, 6.11): 'gradyKipp', Weibull flaws per particle (Benz-Asphaug assignment) that grow at a finite crack speed. With it, the HJC cutoff $-T(1 - D)$ and the EFMIN route to damage in tension are replaced; HJC keeps compression. Legal only on a damage.model 'hjc' card. <i>Choices:</i> "gradyKipp".
m	float	8.0	Weibull modulus m of $n(\epsilon_p) = k \epsilon_p^m$: the scatter of the flaw strengths, and with it the size effect $(V_{\text{ref}}/V)^{1/m}$ and the rate dependence of the tensile strength. 8 is an assumption from the 5-12 range quoted for concrete, not a measured value.
referenceStrength	float	—	Quasi-static tensile strength of a specimen of referenceVolume, which fixes k through $k V_{\text{ref}} (\sigma_{\text{ref}}/E)^m = 1$. Defaults to the eos tensileStrength.

Key	Type	Default	Description
referenceVolume	float	—	Volume of the specimen referenceStrength was measured on, in the deck's length unit cubed. Required: a default would carry a unit. A 150 x 300 mm cylinder is $5.30e6 \text{ mm}^3$.
crackSpeedFactor	float	0.4	Crack growth speed as a fraction of the longitudinal wave speed $\sqrt{(K + 4G/3)/\rho_0}$ of the intact material.
flawCap	bool	true	Benz and Asphaug's cap $D_t \leq n_{\text{act}}/n$: a particle cannot be more damaged than the share of its flaws that are active. With it, crack bands stay partially damaged ($D_t \approx 0.6-0.95$) and the material across them stays connected; false lets any active flaw grow its particle to $D_t = 1$, so cracks separate fragments. The Chocron Grady-Kipp deck runs false (CODE_DESCRIPTION 6.11).
seed	int	0	Seed of the flaw assignment. The same seed, particle set and constants give the same flaws; a different seed is a different sample of the same concrete.

4.4.3 Lattice Blocks, Constraints, and Detonators

FluidBlocks and SolidBlocks

Particle lattice regions are defined via arrays of JSON objects under **FluidBlocks** or **SolidBlocks**. Each block instantiates a regular Cartesian lattice discretized with spacing Δx_0 .

Key	Type	Default	Description
objectId	int	required	Unique integer tag assigned to all particles in the block.
material	str	required	Name of the material entry declared in Materials[] .
start	vec3	required	Minimum bounding corner coordinates $[x_{\min}, y_{\min}, z_{\min}]$.
end	vec3	required	Maximum bounding corner coordinates $[x_{\max}, y_{\max}, z_{\max}]$.
velocity	vec3	$[0, 0, 0]$	Initial uniform velocity vector $[v_x, v_y, v_z]$.
translation	vec3	$[0, 0, 0]$	Rigid translation applied to particle lattice coordinates.
scale	vec3	$[1, 1, 1]$	Spatial scaling factors along coordinate axes.
density	float	<i>derived</i>	Initial density override (defaults to material density0).
color	vec3	$[0.2, 0.6, 1.0]$	RGB display rendering color vector.
rotVelocity	vec3	$[0, 0, 0]$	Initial angular velocity vector ω about the origin.
divVelocity	float	0.0	Initial isotropic divergence velocity field.
shape	object	<i>none</i>	Geometric shape descriptor for carving non-box geometries.

Shape Carving

A block may define an optional **shape** object, which carves a region out of the lattice laid inside the block's bounding box; particles falling outside the region are discarded. The accepted primitives are drawn from the registry, so this list cannot drift from the validator:

- "circle" — tests x and y only: a disk in one layer, a cylinder along z in a 3D block, and a sphere when revolved about the axis of an axisymmetric deck. Required keys: **radius**.
- "cylinder" — radial distance measured in the two components that are not **axis**; its length is the block's business, not the region's. Required keys: **radius**.
- "dogbone" — flat tensile coupon, carved in the xy plane. Required keys: **tabWidth**, **tabHeight**, **gaugeWidth**, **radius**.
- "dogbone3d" — the same coupon revolved about its own long axis. Required keys: **tabWidth**, **tabHeight**, **gaugeWidth**, **radius**.
- "dogboneaxi" — the meridional half-section of that same revolved coupon, which is what an axisymmetric deck lays: the long axis is x and y is the radius, measured from the axis at $y = 0$ rather than from **center**. Required keys: **tabWidth**, **tabHeight**, **gaugeWidth**, **radius**.
- "sphere" — $\|\mathbf{p} - \mathbf{c}\| \leq r$, a genuine ball. Required keys: **radius**.

Key	Type	Default	Description
type	str	required	Geometric primitive, one of the names listed above.
radius	float	required	Outer radius. On the three dogbone shapes it is instead the fillet radius of the shoulder.

Key	Type	Default	Description
center	vec3	<i>lattice centroid</i>	Centre of the primitive. Give this explicitly whenever the body is not centred on the block: the default is the centroid of the lattice actually laid, which in a half or quarter symmetry model sits nowhere near the cut corner the body is centred on, and the region then carves away almost everything.
axis	str	"x"	Longitudinal axis of a cylinder ("x", "y" or "z"). Only the two off-axis components of center are read. Accepted by "cylinder" only.
tabWidth	float	required	Width of the gripped tab of a dogbone coupon. Accepted by "dogbone", "dogbone3d", "dogboneaxi" only.
tabHeight	float	required	Height of the gripped tab of a dogbone coupon. Accepted by "dogbone", "dogbone3d", "dogboneaxi" only.
gaugeWidth	float	required	Width of the reduced gauge section of a dogbone coupon. Accepted by "dogbone", "dogbone3d", "dogboneaxi" only.

Constraints (Displacement Boundary Conditions)

Displacement boundary conditions are applied exclusively to solid materials via the **Constraints** array:

Key	Type	Default	Description
objectId	int	required	Target block object identifier to be constrained.
direction	str	required	Axis along which displacement occurs ("x", "y", "z").
displacement	float	required	Total peak displacement Δu achieved after the ramp time.
grip	str	required	Grip location descriptor (e.g., "xMin", "xMax", "all").
gripLength	float	0.0	Spatial thickness along the grip axis wherein particles are constrained.

Detonators (Programmed Burn Initiation)

High-explosive materials governed by the JWL equation of state require at least one initiation point specified within the **Detonators** array:

Key	Type	Default	Description
position	vec3	required	Detonator center coordinate vector $[x, y, z]$.
time	float	0.0	Ignition trigger time t_{det} at which detonation wave initiates.
material	str	<i>none</i>	Optional target JWL explosive material name (defaults to all JWL materials).

Configuration and ParticlesFile

Two additional deck keys exist at the top level:

- **Configuration:** A flat dictionary of canonical parameter names (e.g., "PST_MAX_SHIFT": 0.05). This is applied as an override layer *on top of* all keyword cards. It is primarily utilized by regression tests and automated parameter studies.

- **ParticlesFile:** Path string pointing to an external AMBER NetCDF particle initialization file. Bypasses grid block generation and directly populates particle coordinates and initial states.

5 Scene Catalog and Verification Benchmarks

5.1 Overview

This chapter documents the benchmark validation cases, experimental comparison decks, and example engineering simulations provided with the `sigmaSPH` solver in `data/scenes/`. Each entry presents the continuum physical problem, geometric discretisation, material constitutive models, initial and boundary conditions, numerical regularisation parameters, and measured computational execution performance across modern GPU hardware.

5.2 Hypervelocity Impact in 3D Quarter Symmetry (`hvi_3d_quarter_al2017.json`)

The simulation deck `data/scenes/hvi_3d_quarter_al2017.json` (documented in technical records and `CODE_DESCRIPTION.md` as `hvi_3d_quarter_al2017_sphere_on_disc_mm_GPa_ms.json`) models the hypervelocity impact of a solid spherical aluminium projectile striking a thin circular aluminium plate at 6.7 km/s (6700 m/s). This configuration represents the canonical Whipple bumper shield problem standard in spacecraft orbital debris protection and terminal ballistics literature.

5.2.1 Physical Problem and Whipple Shield Mechanics

When a meteoroid or orbital debris fragment strikes a thin spacecraft bumper sheet at velocities well in excess of the bulk acoustic wave speed ($v_0 > c_0$), intense planar shock waves propagate backward into the projectile and forward into the target plate. Peak shock pressures reach the megabar regime (> 100 GPa), substantially exceeding the Hugoniot Elastic Limit (HEL) and theoretical shear resistance of metallic alloys.

As compressive shock waves reflect from the free surfaces of the projectile and the rear face of the target, release rarefaction fans unload the shocked material into severe triaxial tension. This rapid unloading causes shock heating, extensive plastic shear deformation, hydrodynamic melting, dynamic cavitation, and spallation, disintegrating both the projectile and the perforated target plug into an expanding spray of fine liquid droplets and solid fragments known as a debris cloud. The engineering function of the Whipple shield is to disperse this impactor kinetic energy over a wide spatial footprint, mitigating catastrophic concentrated breach of underlying pressure vessels.

5.2.2 Geometric Discretisation and Quarter-Symmetry Reduction

Simulating a hypervelocity impact in full three-dimensional Cartesian space demands tens of millions of particles to resolve thin shock discontinuities and microscopic spall fragments. Because the spherical projectile and circular target disc share rotational symmetry about the impact axis, the physical problem is symmetric across Cartesian reflection planes.

`sigmaSPH` models this geometry as a **quarter-symmetry model** using the card setting `Domain.symmetryPlanes: "yz"`. The computational domain discretises only the positive octant (x along the impact axis, $y \geq 0$, $z \geq 0$), bounded by two orthogonal symmetry cut planes at $y = 0$ and $z = 0$. Missing particle interactions across the cuts are evaluated dynamically

through the in-kernel mirror traversal described in §2.9.4, avoiding both artificial ghost particle allocation and spurious boundary truncation forces.

Lattice Spacing and Particle Counts: The simulation is discretised with a nominal particle radius `Domain.particleRadius` = 0.025 mm, establishing an initial uniform inter-particle lattice spacing of:

$$\Delta x_0 = 2 \cdot \text{particleRadius} = 0.05 \text{ mm} \quad (50 \mu\text{m}). \quad (5.1)$$

In strict accordance with the lattice placement rule established in §2.9.2, the initial particle lattice starts at half a spacing off each cut plane ($y_{\text{start}} = z_{\text{start}} = 0.025 \text{ mm} = \Delta x_0/2$), ensuring that the stored particles and their mirror reflections form a continuous, uninterrupted regular lattice across $y = 0$ and $z = 0$.

The geometric components comprise:

- **Spherical Projectile:** A sphere of radius $R = 3.175 \text{ mm}$ (diameter 6.35 mm, corresponding to a standard quarter-inch sphere), centered at the origin $[0.0, 0.0, 0.0]$. Cut across two planes, the retained quarter-sphere contains exactly:

$$N_{\text{sphere}} = 268\,043 \text{ particles}. \quad (5.2)$$

(A full 360° sphere at this resolution would require 1 072 172 particles).

- **Target Disc:** A cylindrical bumper plate of radius $R = 10.0 \text{ mm}$ (diameter 20.0 mm) and thickness 2.0 mm, aligned along the impact axis with its front face at $x = 4.0 \text{ mm}$ and rear face at $x = 6.0 \text{ mm}$ (centered at $[5.0, 0.0, 0.0]$). Cut across the two symmetry planes, the retained quarter-disc contains exactly:

$$N_{\text{target}} = 1\,256\,760 \text{ particles}. \quad (5.3)$$

(A full 360° target disc would require 5 027 040 particles).

- **Total Computational State:** The total active particle count allocated and integrated in GPU device memory is:

$$N_{\text{total}} = N_{\text{sphere}} + N_{\text{target}} = 268\,043 + 1\,256\,760 = 1\,524\,803 \text{ particles}. \quad (5.4)$$

Modeling this physical body in a full three-dimensional domain without symmetry reduction would exceed 6.1×10^6 particles, requiring over 4.6 GB of sorted field memory.

5.2.3 Initial Conditions and Ballistic Standoff

The projectile particles are initialized with a uniform translational velocity along the positive x -axis:

$$\mathbf{v}_{\text{sphere}}(t = 0) = \begin{bmatrix} 6700.0 \\ 0.0 \\ 0.0 \end{bmatrix} \text{ mm/ms} = \begin{bmatrix} 6700.0 \\ 0.0 \\ 0.0 \end{bmatrix} \text{ m/s}. \quad (5.5)$$

The target plate is initialized completely at rest ($\mathbf{v}_{\text{target}}(t = 0) = \mathbf{0}$). Gravitational body loads are set to zero.

Initial Separation and Avoidance of Startup Transients: The forwardmost tip of the spherical projectile is located at $x = +3.175 \text{ mm}$, while the impact face of the target plate begins at $x = 4.000 \text{ mm}$. The physical standoff distance between the two bodies is:

$$\Delta x_{\text{standoff}} = 4.000 \text{ mm} - 3.175 \text{ mm} = 0.825 \text{ mm}. \quad (5.6)$$

At the declared support factor `Domain.supportRadiusFactor = 6.0`, the smoothing kernel compact support radius is $h = 3\Delta x_0 = 0.15$ mm. The initial clearance between the bodies satisfies:

$$\frac{\Delta x_{\text{standoff}}}{h} = \frac{0.825 \text{ mm}}{0.15 \text{ mm}} = 5.5 h. \quad (5.7)$$

Because this initial standoff is substantially larger than $2h$, neither body resides within the compact support neighborhood of the other at $t = 0$. The projectile travels ballistically for approximately $0.123 \mu\text{s}$ before mechanical interaction commences. This deliberate standoff eliminates the severe unphysical startup transients that occur when bodies are laid within $1h$ of each other, where initial velocity discontinuities (6700 m/s) are injected directly into kernel gradient summations on step 0.

Domain Framing: The domain configuration specifies `boundary: "open"`, dropping `Domain.start` and `Domain.end`. As established in §1.3, under open boundary conditions the linear hashed neighbor search is completely unbounded; the bounding box plays no physical role in neighbor interaction and serves only to establish visual camera framing during interactive rendering.

5.2.4 Material Constitutive Models

Both components are composed of Al 2017-T4 aluminium alloy referenced through the material constants library entry `al2017_t4` (§3.5).

Polynomial Mie–Grüneisen Equation of State (EOS003)

At an impact velocity of 6.7 km/s, linear acoustic approximations ($p = K\mu$) severely fail. Both projectile and target evaluate pressure from the polynomial Mie–Grüneisen shock equation of state (§3.2.4), calibrated against experimental linear shock Hugoniot data $U_s = c_0 + su_p$:

$$\rho_0 = 2.79 \times 10^{-6} \text{ kg/mm}^3 \quad (2790 \text{ kg/m}^3), \quad (5.8)$$

$$c_0 = 5328.0 \text{ mm/ms} \quad (5328 \text{ m/s}), \quad (5.9)$$

$$s = 1.34, \quad (5.10)$$

$$\Gamma_0 = 2.0. \quad (5.11)$$

Under shock compression ($\mu = \rho/\rho_0 - 1 \geq 0$), the Hugoniot polynomial coefficients evaluate to $C_0 = 0$, $C_1 = \rho_0 c_0^2 = 79.20$ GPa, $C_2 = (2s - 1)C_1 = 133.06$ GPa, $C_3 = (s - 1)(3s - 1)C_1 = 81.32$ GPa, and $C_4 = C_5 = \Gamma_0 = 2.0$. In the rarefaction expansion regime ($\mu < 0$), linear expansion truncation ($C_2 = C_3 = 0$) prevents spurious non-physical stiffening under tension.

Crucially, the deck sets `allowNegativePressure: true`, allowing physical tensile release waves and dynamic spallation without an artificial zero-pressure clamp. The specific heat capacity $C_v = 875.0 \text{ J/(kg} \cdot \text{K)}$ enables the local derived physical temperature T to be computed per particle from the internal energy according to Eq. (2.80).

Constitutive Strength and Hydrodynamic Target Treatment

- **Projectile Solid Strength:** The projectile is modeled as an elastic–viscoplastic solid using the hypoelastic Jaumann stress rate formulation. Elastic shear resistance is governed by shear modulus $G = 27.22$ GPa (derived from $E = 72.4$ GPa and $\nu = 0.33$). Plastic yield resistance is governed by the Johnson–Cook viscoplastic model (MAT002, §3.3.4), calibrated with Al 2024-T351 surrogate parameters: quasi-static yield stress $A = 0.265$ GPa (265 MPa), hardening modulus $B = 0.426$ GPa, exponent $n = 0.34$, strain-rate sensitivity $C = 0.015$, reference strain rate $\dot{\epsilon}_0 = 1.0 \times 10^{-3} \text{ ms}^{-1}$, and melting temperature $T_m = 775.0$ K.

- **Target Hydrodynamic Treatment:** The target disc is declared with `"strength": null`, removing deviatoric shear resistance ($G \equiv 0$). In the hypervelocity regime, the kinetic energy of the impactor so vastly exceeds plastic shear strength that the target material undergoes catastrophic hydrodynamic plastic flow and fluid-like cratering. Treating the target disc hydrodynamically accelerates execution by bypassing velocity gradient matrix tensor inversions and plastic radial return mapping across the 1.26 million target particles.

5.2.5 Numerical Regularisation and Time Integration

Hypervelocity impacts generate extreme deformation and shear banding that trigger meshless pairing instabilities unless robust numerical regularisers are active:

- **Particle Shifting Technology (PST):** Active in Collé velocity shifting mode (`mode: "colle"`), with a fixed Mach number $Ma = 0.05$, pair repulsion parameters $R = 0.2, n = 4$, and maximum shift displacement clamped to $0.005\Delta x_0$. Convective Arbitrary Lagrangian–Eulerian (ALE) transport corrections are enforced across both momentum (`aleMomentum: true`) and history fields (`aleHistory: true`).
- **Density Diffusion:** Molteni & Colagrossi δ -SPH diffusion is active at $\delta = 0.1$, restricted strictly to pairs sharing the same material index (`pairs: "sameMaterial"`) to eliminate artificial mass diffusion across the contact interface.
- **Shock Viscosities:** Quadratic von Neumann–Richtmyer bulk viscosity is configured with $C_Q = 1.5$ over length scale Δx_0 to smear shock discontinuities over 2–3 particle spacings. Monaghan pairwise artificial viscosity is active at low dissipation ($\alpha = 0.1, \beta = 0.0$).
- **Deformation Limiting:** The Erosion card operates in `"cap"` mode, clamping the volumetric deformation Jacobian $J = V/V_0 \in [0.3, 1.5]$ to prevent unphysical cell inversion in localized shear bands while retaining all mass within the simulation.
- **Explicit Adaptive Time Integration:** Equations of motion are advanced via the symplectic Euler scheme at Courant coefficient $CFL = 0.3$. Timestep sizing is dynamically clamped by dilatational wave speed, maximum particle acceleration, and the state-dependent Mie–Grüneisen acoustic speed $c(\rho, e)$.

5.2.6 Simulation Visualisation and Debris Cloud Morphometry

Figure 5.1 illustrates the three-dimensional debris cloud and temperature distribution produced by `hvi_3d_quarter_al2017.json` at physical time $t \approx 3.91 \mu s$.

The visual rendering clearly demonstrates key physical phenomena characteristic of hypervelocity impact:

1. **Forward Debris Bubble:** The fragmented remains of the spherical projectile and the punched target plug form an ellipsoidal forward debris cloud expanding at multi-kilometer-per-second velocity. Core fragments reach temperatures above 2500–4000 K, indicating shock melting and partial vaporisation.
2. **Perforated Target Lip and Petaling:** The target disc develops an inverted circular crater rim with significant plastic lip curling along the exit face.
3. **Backward Spall Ejecta:** Rarefaction waves off the projectile rear face eject an umbrella-shaped cone of fine backward-traveling fragments.

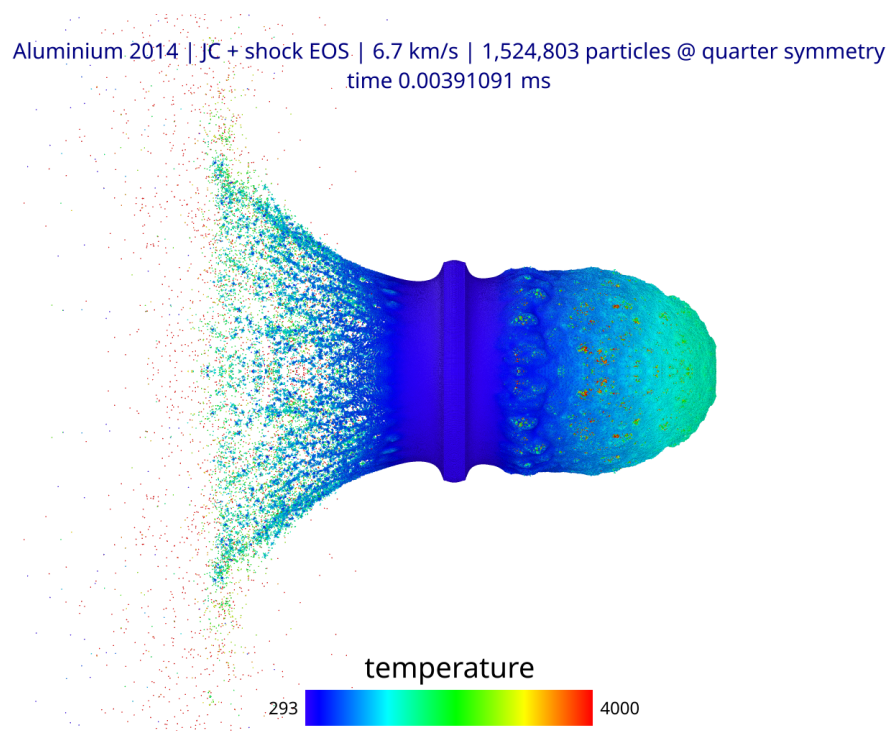


Figure 5.1: Visual rendering of the 3D quarter-symmetry hypervelocity impact simulation (`data/scenes/hvi_3d_quarter_al2017.json`) at $t = 3.91 \mu\text{s}$. The 1.52-million particle model captures the perforation of the 2 mm aluminium target plate, the formation of the forward-expanding hypervelocity debris bubble, backward spall ejecta, and the shock temperature field ranging from ambient (293 K, dark blue) to peak shock-heated states exceeding 4000 K (red).

5.2.7 Computational Execution Performance and Benchmarks

Because hypervelocity simulations with $> 1.5 \times 10^6$ particles represent computationally demanding workloads, `hvi_3d_quarter_al2017.json` has been subjected to rigorous performance profiling and hardware benchmarking in `CODE_DESCRIPTION.md` (§3.9, §3.12, §3.13, and §20.7).

Table 5.1 compiles the measured execution times, per-step throughput, memory footprints, and scaling behavior across modern GPU architectures.

Table 5.1: Measured computational execution performance and memory footprint for the 3D quarter Whipple bumper shield benchmark (`hvi_3d_quarter_al2017.json`, 1 524 803 particles).

Benchmark Metric	NVIDIA L40S (Server)	NVIDIA RTX 4050 (Mobile)
Active Particles (N)	1 524 803 (268k proj. + 1.26M target)	1 524 803 (268k proj. + 1.26M target)
Physical Duration	8.0 μ s (4 325 timesteps)	8.0 μ s (4 325 timesteps)
Total Wall-Clock Time	6 min 28 s (388.0 s)	—
Mean Step Execution Time	89.7 ms / step	648.6 ms / step
Throughput per Particle	58.8 ns / part / step	425.4 ns / part / step
Counted Field Memory	654.4 MB	654.4 MB
Peak VRAM Allocation	≈ 1.05 GB	≈ 1.05 GB
Axial Momentum Conservation (p_x)	$\Delta p_x / p_0 < 0.50\%$	$\Delta p_x / p_0 < 0.50\%$

Analysis of Hardware Scaling: On the data-center NVIDIA L40S GPU, the entire 1.52-million particle simulation completes 4 325 explicit timesteps in just 6 min 28 s (388 s), achieving an average throughput of 89.7 ms per global timestep (58.8 ns per particle per step).

Comparing this throughput against the 648.6 ms/step measured on the mobile RTX 4050 Laptop GPU reveals a $7.23\times$ speedup on the enterprise hardware. This performance scaling is directly attributable to the L40S’s massive memory bandwidth (864 GB/s vs. 192 GB/s) and larger L2 cache (48 MB vs. 12 MB), which accelerate the uncoalesced memory reads inherent in SPH neighbor cell traversals and particle sorting. Furthermore, thanks to the per-particle memory layout rules detailed in §1.3.2, the complete 1.52-million particle model requires only 654.4 MB of persistent GPU field memory, allowing large 3D simulations to execute comfortably even on entry-level mobile GPUs.

5.2.8 Symmetry Filtering and Physical Interpretation

When interpreting the results of quarter-symmetry simulations, two important theoretical considerations must be kept in mind:

- 1. Suppression of Non-Quadrant Azimuthal Modes:** By enforcing Cartesian symmetry across $y = 0$ and $z = 0$, the quarter model mathematically constrains all physical fields to azimuthal harmonic modes $m = 0, 4, 8, \dots$. Odd modes ($m = 1, 3, 5$) and the elliptical mode ($m = 2$) are suppressed by construction. While real physical debris clouds undergo chaotic, stochastic fragment break-up governed by low-order asymmetric modes, the quarter model generates a regularised debris cloud representing the exact solution to the symmetrised initial boundary-value problem.
- 2. Conservation of Axial Momentum:** As noted in §2.11, the transverse momentum components p_y and p_z carry physical boundary reaction forces against the symmetry cut planes; hence, their non-zero evolution represents physical outward expansion flux rather

than numerical drift. In contrast, the axial momentum p_x is strictly conserved, drifting by less than 0.5% of initial impactor momentum $|p_0|$ across the entire dynamic event.

5.2.9 Key Configuration Settings

The configuration deck `data/scenes/hvi_3d_quarter_al2017.json` establishes:

- **Domain & Symmetry:** `particleRadius:` 0.025 ($\Delta x_0 = 0.05$ mm), `symmetryPlanes:` "yz" (quarter-symmetry across $y = 0$ and $z = 0$), `boundary:` "open".
- **Time Integration:** Courant factor `CFL:` 0.3, duration 8.0 μ s, `pairSoundSpeed:` "off".
- **Deformation Limiting:** `Erosion.mode:` "cap", bounding volumetric Jacobian $J = V/V_0 \in [0.3, 1.5]$.
- **Viscosity & Shifting:** Bulk viscosity $C_Q = 1.5$, Monaghan viscosity ($\alpha = 0.1, \beta = 0.0$), Collé PST ($\text{Ma} = 0.05, R = 0.2, n = 4$) with ALE momentum and history transport.
- **Materials:** Projectile uses `al2017_t4` (Mie–Grüneisen EOS and Johnson–Cook strength); target disc uses hydrodynamic `al2017_t4` with `strength:` null.

5.3 Axisymmetric Tensile Dogbone Test (`tensile_axi_dogbone.json`)

The simulation deck `data/scenes/tensile_axi_dogbone.json` models the quasi-static uniaxial elongation, finite plastic necking, and ductile rupture of an axisymmetric metallic dogbone tensile specimen fabricated from austenitic stainless steel (AISI 304). Discretised as a two-dimensional axisymmetric continuum on the meridional (x, r) half-plane, this benchmark represents the canonical laboratory tensile test used throughout experimental mechanics and metallurgy to characterize work hardening, plastic flow localization, and ductile fracture.

Crucially, this deck serves as the primary verification benchmark demonstrating that the `sigmaSPH` solver can robustly sustain intense uniaxial and multi-axial tensile stress states without succumbing to the classical tensile pairing instability that has historically afflicted updated Lagrangian SPH codes.

5.3.1 Physical Problem and Tensile Stability in SPH

In continuum solid mechanics, tensile loading drives metals through elastic elongation into plastic yielding and, eventually, localized plastic necking governed by the Considère criterion. As the cross-sectional area contracts, the local stress state transitions from simple uniaxial tension into intense triaxial tension within the necking core, where mean hydrostatic pressure becomes strongly negative ($p = -\frac{1}{3}\text{tr}(\boldsymbol{\sigma}) < 0$). In physical metallurgy, this negative pressure drives void nucleation, micro-void coalescence, and eventual ductile cup-cone fracture.

The Classical Tensile Pairing Instability

In numerical meshless methods, however, simulating tensile stress states in solid bodies has historically proven to be one of the most formidable and stubborn failure modes. In their foundational mathematical analysis, Swegle, Hicks, and Attaway (1995) established that the discrete momentum equations of standard Smoothed Particle Hydrodynamics suffer from an intrinsic linear instability whenever the product of the stress tensor and the second derivative of the smoothing kernel satisfies the condition:

$$\sigma_{\alpha\beta} \frac{\partial^2 W}{\partial x_\alpha \partial x_\beta} > 0. \quad (5.12)$$

Because conventional bell-shaped smoothing kernels (such as the cubic spline or Wendland C^2 kernel) have a negative first derivative ($W' < 0$) and a positive second derivative ($W'' > 0$) across virtually the entirety of their compact interaction radius, any positive tensile stress ($\sigma > 0$) or negative hydrostatic pressure ($p < 0$) turns the effective spatial stiffness matrix negative.

Under this negative discrete stiffness, the physical restoring forces reverse sign: rather than repelling particles that approach one another, the discrete kernel forces cause neighboring particles along the tensile axis to attract each other. Particles spontaneously collapse into tight, unphysical pairs or linear chains, while unphysical numerical voids and artificial fracture surfaces tear open uncontrollably along coordinate directions. Within a few dozen timesteps, continuum stress coherence is completely annihilated, rendering standard SPH incapable of modeling ductile necking or tensile load paths.

Historical Workarounds and Their Limitations

To prevent this catastrophic pairing breakdown, past meshless literature has relied on several heuristic workarounds, each carrying severe physical or numerical compromises:

- **Negative Pressure Clamping (`allowNegativePressure: false`):** The simplest historical expedient clamps negative hydrostatic pressure to zero whenever $p < 0$. While this trivially eliminates tensile pairing forces, it completely strips the material of cohesive tensile resistance. Under uniaxial elongation, a solid with clamped pressure behaves like cohesionless dry sand or loose dust, disintegrating without transmitting tensile load or forming a physical neck.
- **Artificial Stress Formulations:** Monaghan (2000) and Gray et al. (2001) proposed adding an empirical repulsive stress term $R_{ij} \propto (\sigma/\rho)^n (W/W_0)^n$ directly to the momentum equation. Although artificial stress can stabilize regular lattices under mild tension, it introduces arbitrary tuning parameters, perturbs the physical plastic yield envelope, violates material frame indifference under large finite rotations, and can generate spurious isotropic swelling.
- **Total Lagrangian SPH (TL-SPH):** Formulating kernel gradients with respect to the initial undeformed reference configuration $\mathbf{X}_0$ avoids Eulerian pairing because neighbor connectivity is frozen. However, Total Lagrangian formulations cannot naturally accommodate topological separation, large plastic shear flow, or ductile fracture without invoking complex cohesive zone models or arbitrary particle deletion algorithms.

The σ SPH Solution: Updating Coordinates with PST and Hourglass Control

σ SPH resolves the tensile dilemma directly within the fully updated Eulerian hypoelastic-plastic formulation, running with true physical negative pressures (`allowNegativePressure: true`) through the coupled synergy of two advanced regularisers:

1. **Collé Velocity Particle Shifting Technology (PST):** Active in Collé mode (`mode: "colle"`), the solver continuously computes a regularising velocity perturbation $\delta \mathbf{v}_i^{\text{shift}}$ proportional to the gradient of the particle distribution index (see §2.4). This subtle geometric drift continuously nudges particles toward uniform, isotropic spatial packing, counteracting the pairing attractor before clumping can initiate. Crucially, the shifting is coupled with rigorous Arbitrary Lagrangian–Eulerian (ALE) transport corrections across both linear momentum (`aleMomentum: true`) and internal history variables (`aleHistory: true`, including deviatoric stress $\mathbf{S}$, equivalent plastic strain ε_p , and internal energy e , Eq. (2.31)). As established in §2.5.1, these convective transport fluxes prevent shifting from altering macroscopic momentum or introducing spurious stress relaxation.

2. **Eulerian Rate-Form Hourglass Damping:** The solver applies an explicit dissipative rate damper ($\alpha = 4.0$, §2.7) that penalises velocity fluctuations deviating from the locally affine velocity gradient:

$$\tilde{\mathbf{v}}_{ij} = \mathbf{v}_j - \mathbf{v}_i - \mathbf{L}_i(\mathbf{x}_j - \mathbf{x}_i). \quad (5.13)$$

Under homogeneous uniaxial extension, the physical deformation is strictly affine ($\mathbf{L} = \text{diag}(\dot{\epsilon}_{xx}, -\nu\dot{\epsilon}_{xx}, -\nu\dot{\epsilon}_{xx})$), meaning the hourglass damper exerts zero parasitic resistance against legitimate stretching or Poisson contraction. However, if any pair of particles attempts to rotate or clump into a high-frequency zero-energy pairing mode, a non-zero velocity residual $\tilde{\mathbf{v}}_{ij}$ immediately arises and is strongly damped (Eq. (2.47)).

Working together, Collé shifting prevents geometric particle clustering while Eulerian hourglass damping quenches kinematic zero-energy modes. As demonstrated in Figure 5.2, the specimen sustains extreme tensile stresses, necks down smoothly to a single-particle bridge, and cleanly separates into two halves without any unphysical void tearing or particle clumping.

5.3.2 Axisymmetric Kinematics and Dogbone Discretisation

The simulation exploits rotational symmetry by discretising the specimen on the two-dimensional meridional half-plane (x, r) , where x represents the axial coordinate along the specimen's longitudinal center-line and $r = y \geq 0$ represents the radial distance from the axis of symmetry.

Dogbone Profile and Geometric Proportions: The specimen geometry is instantiated using the specialized profile card `shape: "dogboneaxi"`. Its geometric parameters comprise:

- **Grip Tabs:** Tab length $L_{\text{tab}} = 10.0$ mm and tab radius (half-height) $r_{\text{tab}} = 3.6$ mm, providing broad grip surfaces for boundary displacement enforcement.
- **Gauge Section:** Gauge length $2L_{\text{gauge}} = 6.0$ mm (width 6.0 mm along x), centered symmetrically at $x = 0$.
- **Shoulder Transition:** Circular fillet transitions connecting the gauge section to the tabs with a blend radius $R = 20.36$ mm. The shoulder height is $y_j = 8.8$ mm with a total height of 24.8 mm, providing a smooth cross-sectional area reduction of 64.0% into the gauge region. The fillet joins the tab at an angle of 25.6° from the horizontal tab edge.

Particle Resolution and Radial Lattice Layout: The specimen is discretised with a nominal particle radius `Domain.particleRadius = 0.1` mm, establishing an initial uniform inter-particle spacing of:

$$\Delta x_0 = 2 \cdot \text{particleRadius} = 0.2 \text{ mm}. \quad (5.14)$$

The discretisation generates exactly:

$$N_{\text{specimen}} = 2\,508 \text{ particles}. \quad (5.15)$$

In accordance with the axisymmetric lattice placement convention established in §2.9.2 and Table 2.1, the innermost particle row is positioned at half a lattice spacing off the symmetry axis:

$$r_{\text{start}} = y_{\text{start}} = 0.1 \text{ mm} = 0.5\Delta x_0. \quad (5.16)$$

This placement prevents any particle from residing directly on the coordinate axis $r = 0$, eliminating potential numerical division-by-zero singularities in geometric source terms without requiring artificial radial offsetting. For all radial gradient evaluations, $1/r$ is floored at $0.1 \text{ mm} = 0.5\Delta x_0$.

Axisymmetric Ring Mechanics and Momentum Conservation: In the 2D axisymmetric formulation, each numerical particle represents a three-dimensional toroidal ring of radius r_i and cross-sectional area $A_i = \Delta x_0^2$. The volume per radian is:

$$V_i = r_i A_i = r_i \Delta x_0^2, \quad (5.17)$$

and the particle ring mass per radian is $m_i = \rho_0 V_i$.

Axial momentum $p_x = \sum m_i v_{x,i}$ represents the total axial momentum of the revolved 3D body divided by 2π . The solver enforces the conservative axial momentum formulation (**AXI_CONSERVATIVE_X**), evaluating pair interaction forces using the arithmetic mean pair radius $\bar{r}_{ij} = (r_i + r_j)/2$. This formulation ensures that the geometric hoop source term σ_{xr}/r cancels identically out of the pair sum, guaranteeing exact pairwise antisymmetry and strict conservation of total axial momentum p_x .

In contrast, radial momentum $p_r = \sum m_i v_{r,i}$ is a scalar measure of net radial flux rather than a conserved 3D vector component (in full 3D space, every radial momentum vector cancels identically against its diametric counterpart across the center-line). The scalar radial flux is directly driven by the hoop stress divergence term $(\sigma_{rr} - \sigma_{\theta\theta})/r$, which increases during radial contraction and reverses as hoop stresses adjust. Hoop strain rate $\dot{\epsilon}_{\theta\theta} = v_r/r$ enters the deviatoric stress rate tensor, correctly capturing the hoop stiffening that resists necking in cylindrical geometries.

5.3.3 Material Constitutive Model: AISI 304 Stainless Steel

The specimen is modeled as an elastic–viscoplastic solid using austenitic stainless steel (AISI 304 surrogate) from the material database (`library: "stainless_steel"`):

$$\rho_0 = 7.9 \times 10^{-6} \text{ kg/mm}^3 \quad (7900 \text{ kg/m}^3), \quad (5.18)$$

$$E = 193.0 \text{ GPa}, \quad (5.19)$$

$$\nu = 0.30. \quad (5.20)$$

The derived bulk modulus and shear modulus evaluate to:

$$K = \frac{E}{3(1-2\nu)} = 160.83 \text{ GPa}, \quad (5.21)$$

$$G = \frac{E}{2(1+\nu)} = 74.23 \text{ GPa}. \quad (5.22)$$

The corresponding acoustic wave speeds governing the Courant timestep are the bulk speed of sound $c_0 = \sqrt{K/\rho_0} = 4512 \text{ m/s}$ and the dilatational (longitudinal) elastic wave speed:

$$c_p = \sqrt{\frac{K + \frac{4}{3}G}{\rho_0}} = 5735 \text{ m/s}. \quad (5.23)$$

Johnson–Cook Viscoplasticity: Plastic yield resistance is governed by the Johnson–Cook constitutive relation (MAT002, §3.3.4), which expresses the dynamic flow stress as:

$$\sigma_y = \left(A + B \epsilon_p^n \right) \left[1 + C \ln \left(\frac{\dot{\epsilon}_p}{\dot{\epsilon}_0} \right) \right] \left[1 - \left(\frac{T - T_0}{T_m - T_0} \right)^m \right]. \quad (5.24)$$

The calibrated parameters for AISI 304 stainless steel are:

- Quasi-static yield stress: $A = 0.31 \text{ GPa}$ (310 MPa),
- Strain hardening modulus: $B = 1.00 \text{ GPa}$ (1000 MPa),

- Hardening exponent: $n = 0.65$,
- Strain rate sensitivity: $C = 0.02$,
- Reference strain rate: $\dot{\epsilon}_0 = 1.0 \times 10^{-3} \text{ ms}^{-1} = 1.0 \text{ s}^{-1}$,
- Melting temperature: $T_m = 1720 \text{ K}$.

The high work-hardening exponent $n = 0.65$ reflects the strong strain hardening capacity of austenitic stainless steel, which delays the onset of plastic instability and permits extensive uniform plastic elongation prior to localized neck formation. Plastic deformation is integrated via the standard radial return mapping on the deviatoric stress tensor $\mathbf{S}$, updating equivalent plastic strain ϵ_p and path-integrating the plastic dissipation work $W_{\text{plastic}} = \sum V_i \sigma_{\text{VM}} \Delta \epsilon_p$.

Johnson–Cook Fracture (DAM004): The deck adds the fracture card of §3.4.6 to the library material, with the AISI 316L constants of Umbrello et al. [30]: $D_1 = 0.05$, $D_2 = 3.44$, $D_3 = -2.12$, $D_4 = 0.002$, $D_5 = 0.61$. These give $\epsilon_f = 1.75$ at uniaxial triaxiality and 1.01 at $\eta = 0.6$. The failure displacement is $u_f = 0.1 \text{ mm}$, an assumption corresponding to $G_f \approx 50 \text{ N/mm}$. These are literature values, not a calibration of the library material, and the library entry itself carries no damage card, so the 2D and 3D dogbones are unaffected.

5.3.4 Loading Conditions and Controlled Displacement Ramping

Tensile elongation is applied through opposed displacement boundary conditions enforced on the rigid grip tabs:

- **Left Grip Tab:** All particles within the spatial window $x \in [-20.0, -10.0] \text{ mm}$ (325 particles) are constrained in both x and y velocity components, following a prescribed longitudinal displacement $\Delta x_{\text{left}} = -10.0 \text{ mm}$.
- **Right Grip Tab:** All particles within the spatial window $x \in [10.0, 20.0] \text{ mm}$ (300 particles) are constrained in both x and y velocity components, following a prescribed longitudinal displacement $\Delta x_{\text{right}} = +10.0 \text{ mm}$.
- **Deformable Gauge Region:** The central span $x \in (-10.0, 10.0) \text{ mm}$ contains exactly 1 883 freely deforming particles governed solely by internal continuum equations of motion.

Smoothstep Velocity Ramping: To prevent the generation of spurious acoustic shock waves and high-frequency inertial ringing upon grip motion, the boundary displacement is ramped smoothly using a cubic smoothstep polynomial over duration $T_{\text{ramp}} = 0.9 \text{ s}$ (`Load.rampTime: 0.9`):

$$s(\tau) = 3\tau^2 - 2\tau^3, \quad \text{where } \tau = \frac{t}{T_{\text{ramp}}} \in [0, 1]. \quad (5.25)$$

The prescribed grip displacement and velocity follow:

$$x_{\text{grip}}(t) = \Delta x \cdot s(\tau) = \Delta x (3\tau^2 - 2\tau^3), \quad (5.26)$$

$$v_{\text{grip}}(t) = \frac{\Delta x}{T_{\text{ramp}}} \frac{ds}{d\tau} = \frac{6\Delta x}{T_{\text{ramp}}} \tau(1 - \tau). \quad (5.27)$$

Because both initial acceleration $\ddot{x}(0)$ and initial velocity $v_{\text{grip}}(0)$ are identically zero, the tensile specimen transitions smoothly from rest into quasi-static stretching. Over the full simulation duration, total applied elongation is:

$$\Delta L = |\Delta x_{\text{left}}| + |\Delta x_{\text{right}}| = 10.0 \text{ mm} + 10.0 \text{ mm} = 20.0 \text{ mm}, \quad (5.28)$$

which represents a 100% nominal engineering elongation over the initial 20 mm span between the grip boundaries.

5.3.5 Hourglass Sub-Cycling and Super-Time-Stepping

To ensure complete suppression of the Swegle tensile pairing instability while sustaining large plastic necking, the deck configures a strong hourglass damping coefficient $\alpha = 4.0$ (`Hourglass.alpha: 4.0`).

The Damping Stability Bottleneck: As detailed in §2.7 and Eq. (2.50), explicit stability of the linear hourglass damping force is governed by the maximum relaxation rate across the particle ensemble:

$$\omega_{\text{HG}} = \frac{\alpha c_p S}{h}, \quad (5.29)$$

where S is the dimensionless neighbor kernel sum. For the present discretisation ($c_p = 5735$ m/s, $h = 0.6$ mm, $\alpha = 4.0$, and measured neighbor sum $S = 6.64$), the explicit relaxation rate evaluates to:

$$\omega_{\text{HG}} = 1.015 \times 10^6 \text{ s}^{-1}. \quad (5.30)$$

For a standard single-stage forward Euler integration, explicit numerical stability requires $\omega_{\text{HG}} \Delta t \leq 1.6$, imposing an hourglass timestep cap of:

$$\Delta t_{\text{HG}} = \frac{1.6}{\omega_{\text{HG}}} = 1.576 \times 10^{-6} \text{ s}. \quad (5.31)$$

However, the macroscopic acoustic CFL timestep governed by dilatational wave speed ($c_p = 5735$ m/s, $\Delta x_0 = 0.2$ mm, $\text{CFL} = 0.3$) is:

$$\Delta t_{\text{CFL}} = \text{CFL} \frac{\Delta x_0}{c_p} = 0.3 \times \frac{0.2 \times 10^{-3}}{5735} \approx 3.139 \times 10^{-5} \text{ s}. \quad (5.32)$$

Without algorithmic intervention, the stiff hourglass damper would restrict the global simulation timestep to Δt_{HG} , forcing the solver to take nearly 20× more timesteps to reach physical completion!

Super-Time-Stepping Sub-Cycling: To overcome this severe timestep penalty, `sigmaSPH` provides automated sub-cycling via the card setting:

```
"Hourglass": {
  "alpha": 4.0,
  "superTimeStepping": { "mode": "subcycle" }
}
```

Under sub-cycling mode, the global solver advances particle positions, stress divergence, and internal energy at the full acoustic CFL limit ($\Delta t = 3.139 \times 10^{-5}$ s, achieving $\Delta t / \Delta t_{\text{CFL}} = 1.0000$). Inside each global step, explicit hourglass damping forces are sub-cycled across $n_{\text{sub}} \approx 20$ high-frequency internal sub-steps. This completely decouples the hourglass damping stability constraint from the acoustic Courant limit, accelerating the simulation by an order of magnitude while maintaining complete stability against tensile pairing modes.

5.3.6 Simulation Visualisation and Ductile Rupture Progression

Figure 5.2 displays the visual rendering of the specimen during acute plastic necking at $t = 0.345$ ms, extracted directly from the simulation record in `ovito/dogbone_axi.mp4`. Figure 5.3 illustrates the subsequent post-rupture state at $t = 0.377$ ms.

The simulation captures the complete classical sequence of ductile metal fracture:

1. **Homogeneous Elastic Elongation** ($t < 0.05$ ms): The gauge section stretches elastically with uniform axial tensile stress $\sigma_{xx} > 0$ and modest lateral contraction dictated by Poisson's ratio $\nu = 0.30$.

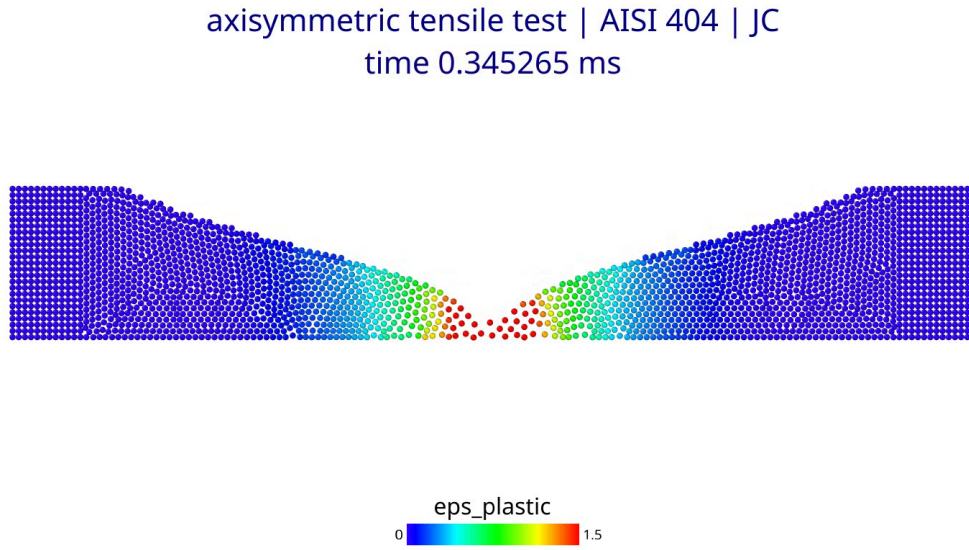


Figure 5.2: Visual rendering of the axisymmetric dogbone tensile test simulation (`data/scenes/tensile_axi_dogbone.json`) during acute localized necking at $t = 0.345$ ms ($\Delta x_{\text{grip}} \approx 7.67$ mm). Particles are coloured by equivalent plastic strain ε_p , illustrating intense plastic localization exceeding 1.50 (red) within the narrowing ligament while the bulk of the specimen retains regular interparticle spacing without tensile pairing instability.

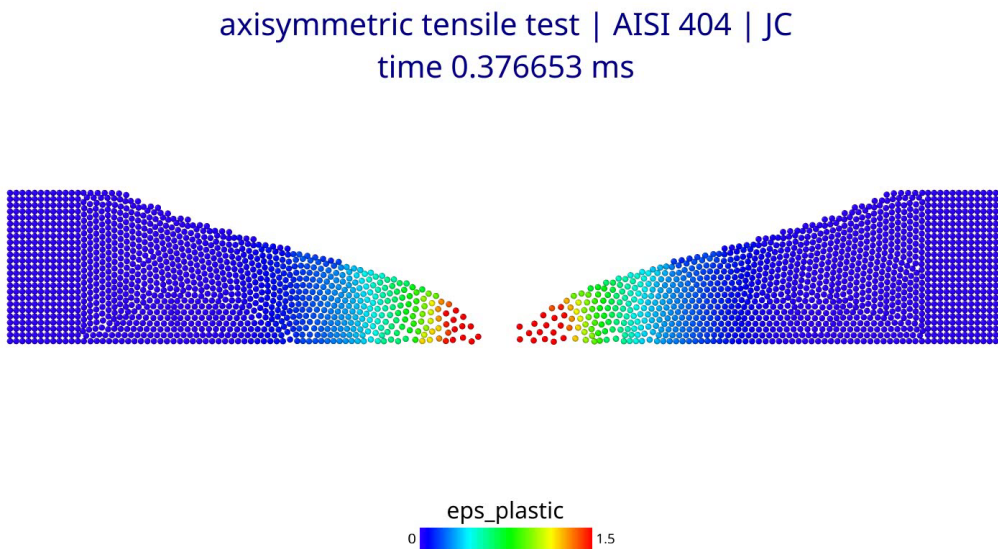


Figure 5.3: Visual rendering of the specimen immediately following ductile rupture at $t = 0.377$ ms ($\Delta x_{\text{grip}} \approx 8.37$ mm). When the narrowing neck thins beyond the compact support radius h , particles separate naturally into two symmetric halves without artificial fragmentation or numerical debris, leaving smooth conical cup-cone fracture surfaces.

2. **Uniform Plastic Yielding and Hardening** ($t \approx 0.05\text{--}0.20\text{ ms}$): Once axial stress reaches the yield threshold $A = 310\text{ MPa}$, plastic flow initiates throughout the reduced gauge section. Strong Johnson–Cook work hardening ($n = 0.65, B = 1000\text{ MPa}$) elevates flow stress past 600 MPa , spreading plastic strain uniformly along the gauge.
3. **Onset of Diffuse Necking** ($t \approx 0.20\text{--}0.30\text{ ms}$): As cross-sectional area contracts, the Considère instability criterion ($d\sigma/d\varepsilon = \sigma$) is attained. Plastic deformation concentrates into a diffuse neck centered at the symmetry plane $x = 0$.
4. **Acute Localized Necking and Triaxial Tension** ($t \approx 0.30\text{--}0.35\text{ ms}$): As seen in Figure 5.2 at $t = 0.345\text{ ms}$, deformation localizes into a narrow central ligament. Equivalent plastic strain peaks beyond $\varepsilon_p = 1.50$ (150% true plastic strain). Local hydrostatic pressure drops into severe tension ($p < -400\text{ MPa}$), yet interparticle spacing remains completely uniform, with zero pairing or diamond-pattern numerical tearing.
5. **Natural Ductile Rupture** ($t > 0.37\text{ ms}$): As seen in Figure 5.3 at $t = 0.377\text{ ms}$, the ligament thins until neighboring particles across the center-line naturally decouple beyond the compact support radius $h = 6.0 \cdot \text{particleRadius} = 0.6\text{ mm}$. The specimen cleanly separates into two halves with distinct conical tip profiles, followed by elastic strain energy recoil back into the grips.

5.3.7 Computational Performance and Energy Accounting

The complete simulation was executed on an NVIDIA CUDA GPU, advancing through 27 400 global solver steps over 548 render cycles to reach physical time $t = 0.860024\text{ s}$ (at 86% of the prescribed 1.0 s duration, at which point the interactive display window was closed).

Table 5.2 compiles measured execution throughput, solver stepping rates, momentum balance, and comprehensive energy partition recorded in the execution log `log.dogbone`.

Table 5.2 compiles key execution metrics, momentum conservation, and the primary plastic dissipation partition.

Analysis of the Energy Audit: The energy partition demonstrates three key physical and numerical insights:

1. **Dominance of Physical Plastic Heating:** Of the $+13.53\text{ J}$ of mechanical pair-force work delivered by moving grips, $+12.34\text{ J}$ (91.3%) is converted directly into plastic work (PW), path-integrated via the J_2 radial return map ($\sum V_i \sigma_{vM} d\varepsilon_p$). This plastic dissipation represents genuine, irreversible physical heat rather than numerical damping.
2. **Inactivity of Artificial Viscosity:** Artificial viscosity work accumulates only $4.3 \times 10^{-5}\text{ J}$ ($< 0.0003\%$ of internal energy), confirming that artificial viscosity is dormant during quasi-static elongation and does not artificially stiffen the response.
3. **Thermodynamic Cost of Regularisation:** The hourglass damper absorbs $+1.16\text{ J}$ (8.5% relative to internal energy) over 27 400 steps. This expenditure represents the exact thermodynamic cost of suppressing spurious pairing perturbations and maintaining regular interparticle spacing throughout extreme necking.
4. **Axial Momentum Conservation:** Total axial momentum drifts by only 2.0×10^{-5} across the entire 27 400-step simulation ($+0.425\%$ of gross momentum), confirming the exactness of the `AXI_CONSERVATIVE_X` formulation on meridional half-planes.

Table 5.2: Measured computational performance, momentum accounting, and energy balance for the axisymmetric tensile dogbone simulation (`tensile_axi_dogbone.json`, 2 508 particles).

Metric / Quantity	Physical Meaning / Role	Measured Value
Total Active Particles (N)	Revolved 2D axisymmetric dogbone	2 508 particles
Simulated Timesteps	Integration steps to $t = 0.8600$ s (86% pull)	27 400 steps
Wall-Clock Execution Time	Complete run (548 render updates)	2 min 44 s (164.3 s)
Mean Stepping Throughput	Global solver iterations per second	167 steps / s (5.99 ms/step)
Axial Momentum Drift (p_x)	Evaluated relative to gross momentum	+0.425%
Internal Energy Change (ΔIE)	Pair-force work of stress divergence	+13.53 J (100.0%)
Plastic Work Dissipation (PW)	Irreversible heat integrated via J_2 return map	+ 12.34 J (91.3% of IE)
Hourglass Damper Work (HE)	Energy removed to quench high-frequency pairing	+1.16 J (8.5% of IE)
Kinetic Energy (KE)	Residual post-rupture elastic recoil kinetic energy	+0.0067 J
Total System Energy (E)	KE + IE + HE + LW + WW	+14.35 J

5.3.8 Key Configuration Settings

The configuration deck `data/scenes/tensile_axi_dogbone.json` establishes:

- **Domain & Boundary:** `particleRadius: 0.1` ($\Delta x_0 = 0.2$ mm), `axisymmetric: true`, `boundary: "open"`. Innermost radial row placed at $r_{\text{start}} = 0.1$ mm = $0.5\Delta x_0$.
- **Time Integration:** Courant factor `CFL: 0.3`, duration 1.0 s, operating at full dilatational CFL timestep ($\Delta t = 3.139 \times 10^{-5}$ s).
- **Hourglass Sub-Cycling:** Parameter `Hourglass.alpha: 4.0` with `superTimeStepping.mode: "subcycle"`, sub-cycling explicit damping over ≈ 20 sub-steps to decouple damping stability from the acoustic step.
- **Material:** AISI 304 austenitic stainless steel (`stainless_steel`), with Johnson–Cook viscoplastic hardening ($A = 310$ MPa, $B = 1000$ MPa, $n = 0.65$).
- **Boundary Loading:** Opposed grip displacement $\Delta x = \pm 10.0$ mm ramped via cubic smoothstep over $T_{\text{ramp}} = 0.9$ s (`Load.rampTime: 0.9`) to avoid initial dynamic shock waves.

5.3.9 Resolution Study of the Force–Displacement Curve

The deck was run at $\Delta x_0 = 0.4, 0.2, 0.1$ and 0.05 mm (decks `tensile_axi_dogbone_dx0.4/_dx0.1/_dx0.05.json`; 640 to 40 114 particles, 42 s to 10 min). The runs were analysed with `tools/dogbone_convergence.py` (Figure 5.4); the full record is `reports/jc_failure_dogbone_convergence_2026-09-27.md`.

- The grip force rings at the bar’s transit period, $L/c_p \approx 0.007$ ms, and overshoots $\sigma_{y0}A_0 = 8.77$ kN to about 11 kN at first yield. Stiffness and yield forces are therefore read off the curve averaged over one transit time.

- $\Delta x_0 = 0.4$ mm lays a gauge radius of 3.2 mm, 13.8% more section than the 3.0 mm laid by the three finer levels, and it necks far later; it lies outside the asymptotic range.
- On the three finer levels the pre-peak curve rises monotonically and slowly with refinement. $F_{\max}$ is 16.26, 16.66 and 16.88 kN (GCI 2%); K and the departure yield force 8.50, 8.63, 8.74 kN rise by 1–4% per halving, at observed orders of 0.3–0.9.
- The fracture card leaves the pre-peak curve unchanged: a control without it agrees to the printed digit. It brings separation forward from 6.38 to 5.34 mm at 0.2 mm. The work to separation converges (75.8, 78.0, 78.9 J; $p = 1.4$), but the separation displacement still drifts, by +0.10 and then +0.16 mm per halving.

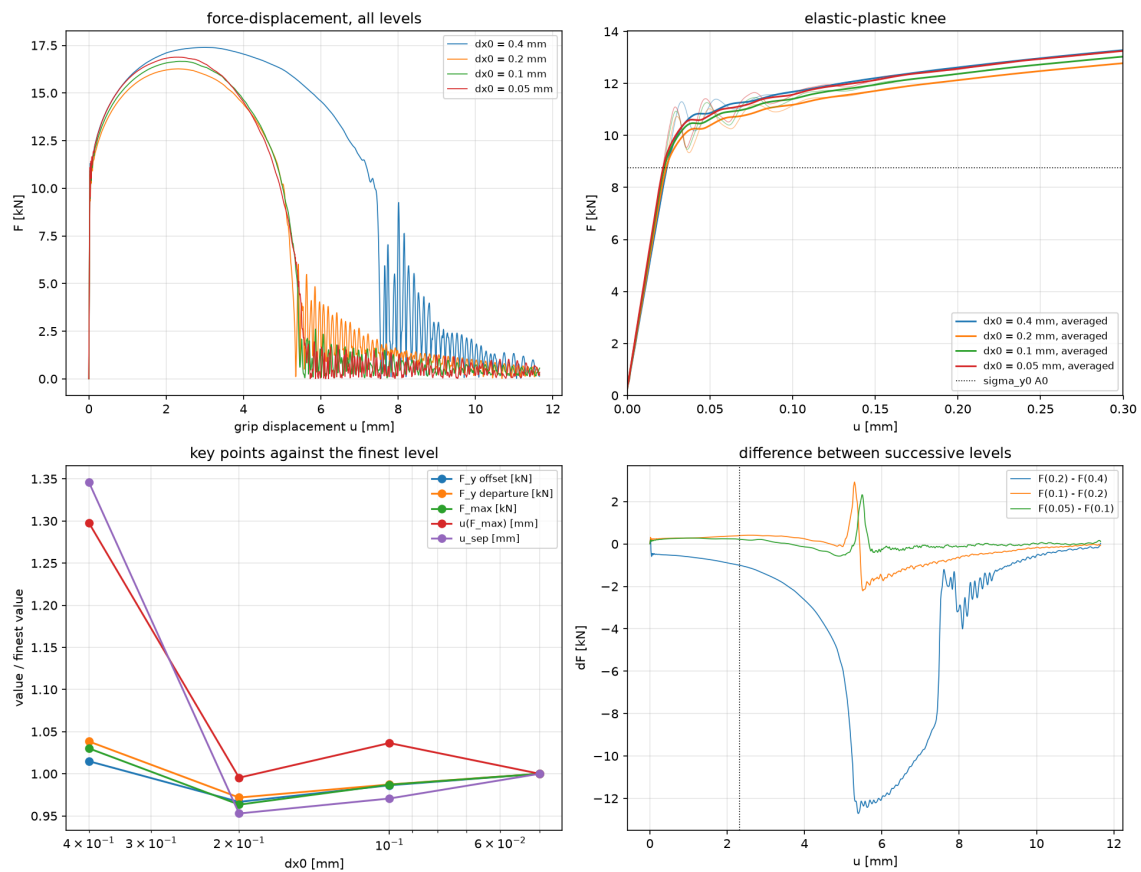


Figure 5.4: Resolution study of `tensile_axi_dogbone.json` with Johnson–Cook fracture. Top: raw force–displacement curves, and the elastic–plastic knee with the transit-averaged curves. Bottom: key points relative to the finest level, and differences between successive levels.

5.4 Hypervelocity Axisymmetric Spallation of Pure Tantalum (`hvi_axi_tantalum_qamar_r0.125.json`)

The benchmark deck `data/scenes/hvi_axi_tantalum_qamar_r0.125.json` recreates the hypervelocity impact and dynamic spallation experiment of Qamar et al. [27] (LLNL-PROC-865094 / AIAA SciTech Forum 2025). In this problem, a blunt cylindrical tantalum projectile strikes a circular pure tantalum plate at 5000 m/s (5.0 km/s). Discretised as a two-dimensional axisymmetric continuum on the meridional (x, r) half-plane, this benchmark serves as the primary

verification case for dynamic void nucleation, Cocks–Ashby void growth, unilateral pressure decomposition, and the sensitivity of dynamic fracture to the strength of rate-form hourglass control.

5.4.1 Physical Problem and Shock Rarefaction Mechanics

Under a 5000 m/s normal impact, strong planar shock waves propagate backward into the projectile and forward into the target plate. For pure tantalum ($c_0 = 3410$ m/s, $s = 1.20$, $\rho_0 = 16.69$ g/cm³), peak Hugoniot shock pressures reach tens of gigapascals.

Upon reaching the rear face of the projectile and the back free surface of the target plate, these compressive shock pulses reflect as unloading rarefaction fans traveling back toward each other. When the two tensile relief waves intersect inside the target plate, constructive interference produces intense transient hydrostatic tension ($\sigma_h = -p > 0$). In pure tantalum, pre-existing sub-micron defect sites nucleate voids when hydrostatic tension exceeds $\sigma_{hM} = 0.778$ GPa (778 MPa). Driven by continued tension and plastic deformation, voids expand rapidly according to the Cocks–Ashby law (§3.4.3) and coalesce into a macroscopic spall fracture. The detached spall scab flies off from the rear surface, leaving a characteristic free-surface velocity pullback signature Δv_{pb} measurable via VISAR interferometry.

5.4.2 Geometric Discretisation and Boundary Setup

The geometry is discretised on the meridional half-plane (x, r) revolved about the symmetry axis $r = 0$:

- **Projectile:** Tantalum cylinder of radius $r_i = 5.0$ mm and length $L_i = 10.0$ mm, spanning $x \in [-10.0, 0.0]$ mm and $r \in [0.125, 5.0]$ mm. It contains 800 particles initialized with uniform velocity $\mathbf{v}_0 = [5000.0, 0.0, 0.0]$ mm/ms.
- **Target Plate:** Circular tantalum plate of radius $r_t = 100.0$ mm and thickness $t_t = 50.0$ mm, spanning $x \in [0.0, 50.0]$ mm and $r \in [0.125, 100.0]$ mm. It contains 80 000 particles initialized at rest.
- **Total Discretisation:** Discretised with `Domain.particleRadius: 0.125` ($\Delta x_0 = 0.25$ mm), the total active particle count is $N = 80\,800$. In accordance with axisymmetric placement rules (§2.9.2), the innermost radial row is positioned at $r_{\text{start}} = 0.125$ mm = $0.5\Delta x_0$, eliminating division-by-zero singularities along the symmetry axis.
- **Boundary Constraints:** The outer radial boundary of the target plate ($r \in [98.0, 102.0]$ mm) is clamped in x and y to model a rigidly held target rim. All other boundaries specify `boundary: "open"`, allowing unconfined flight of the detached spall scab.

5.4.3 Material Model: Calibrated Pure Tantalum

Both projectile and target use the calibrated `tantalum_spall` library entry (§3.5, Table 3.5):

1. **Volumetric EOS:** Polynomial Mie–Grüneisen shock equation of state ($c_0 = 3410$ m/s, $s = 1.20$, $\Gamma_0 = 1.60$, $K_0 = 194.06$ GPa).
2. **Deviatoric Strength:** Hypoelastic Jaumann stress rate formulation with shear modulus $G_0 = 69.0$ GPa and Johnson–Cook viscoplasticity ($A = 0.733$ GPa, $B = 0.540$ GPa, $n = 0.28$, $C = 0.054$).
3. **Spall Damage:** Cocks–Ashby void growth with Chu–Needleman cumulative normal nucleation ($f_{n0} = 10^{-4}$, $\sigma_{hM} = 0.778$ GPa, $\sigma_{hS} = 0.0389$ GPa, $m = 4.0$).

4. **Unilateral Pressure Decomposition:** Effective pressure is computed via Eq. (3.59). For damaged particles ($D = 1$), tensile pressure capacity drops to zero ($p_{\text{eff}} \geq 0$), eliminating artificial tensile attraction across the spall plane while preserving 100% of compressive contact stiffness.

5.4.4 Numerical Regularisation and Physical Findings

Executing hypervelocity spallation in SPH revealed several crucial numerical and physical interactions:

1. **Hourglass Damping Must Stay Weak for Spall to Occur:** The deck sets `Hourglass.alpha: 0.0` (or weak ≤ 0.05) with `superTimeStepping.mode: "subcycle"`. Repeated runs of the deck (2026-09-25), with the Monaghan pairwise artificial viscosity at $\alpha = 1.0$, $\beta = 2.0$, show that values of `Hourglass.alpha` *above* 0.05 suppress spallation, while $\alpha = 0.0$ to 0.05 produces a well-formed spall. This reverses the finding recorded when the deck was first committed, which held that the damper was mandatory at $\alpha \approx 0.2$ and that spall vanished *below* 0.05. That earlier argument ran as follows: the SPH velocity gradient $\langle \nabla \mathbf{v} \rangle_i$ has a zero-energy null space, relative particle motion during the rarefaction could leak into non-affine modes that $\text{tr}(\langle \nabla \mathbf{v} \rangle)$ cannot see, and without damping the hydrostatic tension would therefore never reach the Chu–Needleman barrier $\sigma_{hM} = 0.778$ GPa. The runs do not support its conclusion at the present artificial-viscosity settings, and the argument should not be relied on.

The mechanism behind the upper bound has not been established. One plausible reading, which has not been tested, is that the pairwise velocity-difference damper resists exactly the non-affine relative separation of neighbouring particles that opens a spall plane. Its damage gating (§3.4) scales it by $1 - \max(D_i, D_j)$, which leaves it at full strength until damage has begun, so it acts undiminished against the incipient separation that precedes nucleation. What the runs do establish is narrower: the bound of 0.05 has been observed together with the Monaghan coefficients above and the legacy radial force form. Switching to the conservative radial form at the same α suppresses spall too (item 3 below), and the bound should be re-measured whenever the viscosity coefficients or the resolution change. Beyond that one observation, it is not known how far the bound moves with any of them.

2. **Dual Viscosity Partitioning:** The deck activates both von Neumann–Richtmyer bulk viscosity ($C_Q = 1.2$) and Monaghan pairwise artificial viscosity ($\alpha = 1.0$, $\beta = 2.0$). VNR bulk viscosity smears continuum shock compression across the bulk. Monaghan pairwise viscosity acts on approaching particle pairs ($\mathbf{v}_{ij} \cdot \mathbf{r}_{ij} < 0$), providing the localized quadratic contact penalty needed to prevent projectile particles from interpenetrating the target plate at 5 000 m/s. Crucially, both viscosity formulations evaluate to zero during rarefaction and crack opening ($\mathbf{v}_{ij} \cdot \mathbf{r}_{ij} \geq 0$ and $\nabla \cdot \mathbf{v} \geq 0$), ensuring that artificial viscosity never impedes tensile spallation.
3. **Radial Force Formulation: Conservative Radial Force Recovers Physical Spall upon Refinement:** The deck sets `Domain.conservativeRadialForce: true` (§2.9), which evaluates radial pair forces using symmetric midpoint volumes so that radial pair interactions telescope identically and conserve total energy (+2.76% drift). On the historical coarse grid ($\Delta x_0 = 0.50$ mm), the conservative form failed to spall because SPH kernel smoothing ($h = 1.5$ mm) and artificial viscosity smeared localized shear gradients during tensile reflection, keeping $\varepsilon_p \approx 0.12$ and kinematically freezing Cocks–Ashby void growth. At the resolved benchmark discretisation ($\Delta x_0 = 0.25$ mm, `Domain.particleRadius: 0.125`), non-planar shock fronts and lateral relief waves are sharply resolved: intense localized shear flow ($\varepsilon_p = 1.30$) develops in the tensile interference zone, driving void

growth to saturation ($f = 0.50, D = 1.0$) and cleanly detaching a dynamic spall scab with a classical pullback signature.

4. **Convective ALE Transport:** Convective ALE corrections for porosity f , deviatoric stress $\mathbf{S}$, and specific internal energy e are active under Collé particle shifting (`maFixed: 0.05`), preventing regularisation shifts from artificially blurring the sharp spall crack interface.

5.4.5 Computational Performance and Spall Verification

Figure 5.5 is a representative result of the resolved benchmark deck (`data/scenes/hvi_axi_tantalum_qamar_r0.125.json`) in its canonical configuration (`Domain.conservativeRadialForce: true`, baseline `rateMode: "effective"`, `Hourglass.alpha: 0.0`), rendered in OVITO from the NetCDF trajectory and coloured by the void volume fraction porosity f . The projectile has opened an extensive crater in the front face of the plate, whose rim is severely sheared and sheds a high-velocity spray of detached ejecta back along the approach path. Separately from the crater, and well inside the plate near its rear face, a cleanly detached curved lens of fully voided material ($f \rightarrow 0.50$) has formed: this is the dynamic spall scab, where constructive interference between the axial rarefaction wave and inward radial release waves drove intense multi-axial plastic shear flow ($\varepsilon_p = 1.30$) and void coalescence. The rear surface bulges outward behind it, separating the flying scab from the bulk target plate.

The simulation was executed to physical completion at duration $t = 0.075$ ms ($75 \mu\text{s}$) on an NVIDIA CUDA GPU. Table 5.3 compiles the core execution parameters, momentum accounting, and spallation characteristics for the resolved 80 800-particle configuration.

5.4.6 Key Configuration Settings

The configuration deck `data/scenes/hvi_axi_tantalum_qamar_r0.125.json` establishes:

- **Domain & Symmetry:** `particleRadius: 0.125` ($\Delta x_0 = 0.25$ mm), `axisymmetric: true`, `boundary: "open"`, and `conservativeRadialForce: true` (energy-conserving radial formulation).
- **Time Integration:** `CFL: 0.3`, duration 0.075 ms, adaptive timestep sizing based on dilational wave speed.
- **Hourglass Control:** `Hourglass.alpha: 0.0` (values above 0.05 have been observed to suppress spall at the present artificial-viscosity settings; ≤ 0.05 preserves dynamic fracture).
- **Viscosity Formulation:** Bulk viscosity $C_Q = 1.2$, Monaghan pairwise viscosity ($\alpha = 1.0, \beta = 2.0$).
- **Material Model:** `tantalum_spall` with explicit `rateMode: "effective"`.
- **Shifting & ALE:** Collé PST ($\text{Ma} = 0.05, R = 0.2, n = 4$) with `aleMomentum: true`, `aleHistory: true` and `aleEnergyConservative: true` (§2.5.2), which takes the total energy change from +6.0% to +0.6% of E_0 with the spall unchanged.
- **Output:** Binary AMBER NetCDF trajectory recording dynamic damage and porosity fields.

5.5 Aluminium Sphere into High-Strength Concrete (`hvi_3d_quarter_al_concrete150_chocron_gk_mm_GPa_ms.json`)

This deck reproduces the concrete validation case of Kim et al. [21], §3.2, after the experiment of Chocron et al. [8]: a 25.4 mm aluminium sphere strikes a 150 MPa high-strength concrete

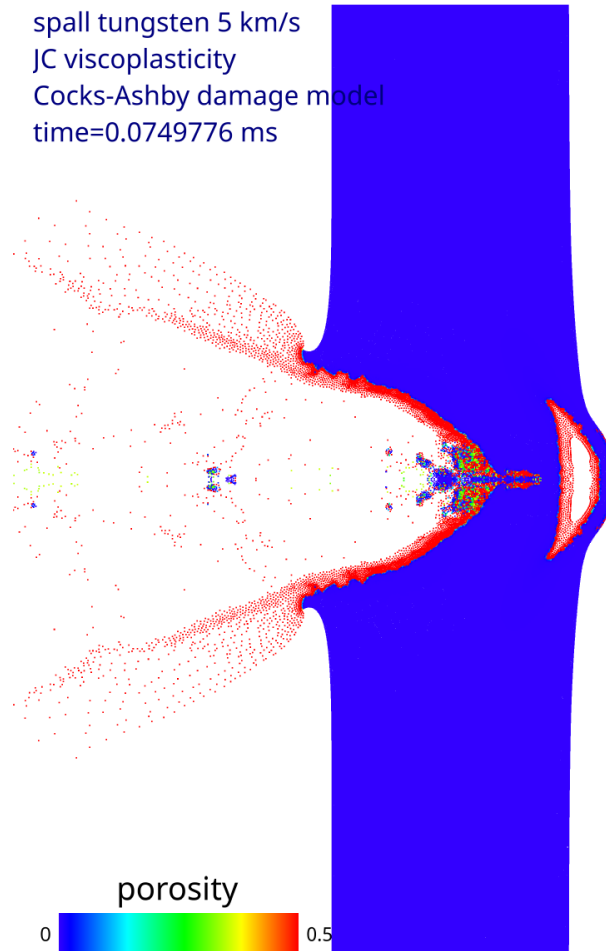


Figure 5.5: Representative result of the pure tantalum spallation benchmark (`data/scenes/hvi_axi_tantalum_qamar_r0.125.json`): a 5 km/s tantalum cylinder impacting a 50 mm tantalum plate, axisymmetric, with Johnson–Cook viscoplasticity and Cocks–Ashby dynamic void growth at resolved spacing $\Delta x_0 = 0.25$ mm (80 800 particles), `conservativeRadialForce: true`, and baseline `rateMode: "effective"`. The impact direction is left to right, and the meridional half-plane is shown mirrored about the symmetry axis at $t \approx 75 \mu\text{s}$. Particles are coloured by void fraction **porosity** f from 0 (blue) to 0.5 (red). The impact crater with backward ejecta is on the left; the cleanly separated lens of voided material near the rear face forms the detached spall scab, with the rear surface bulging outward.

Table 5.3: Key execution, momentum, and spallation metrics for the Qamar et al. pure tantalum spallation benchmark (hvi_axi_tantalum_qamar_r0.125.json, 80 800 particles).

Metric / Quantity	Description / Significance	Measured Value
Total Active Particles (N)	2D axisymmetric half-plane discretisation	80 800 particles
Impactor / Target Partition	800 projectile particles, 80,000 target plate	800 / 80 000
Nominal Lattice Spacing	Particle radius $r_p = 0.125$ mm, spacing $\Delta x_0 = 2r_p$	0.25 mm
Kernel Smoothing Radius (h)	Support radius factor 6.0 ($h = 3\Delta x_0$)	0.75 mm
Simulated Physical Duration	Physical duration to complete dynamic spallation	0.075 ms (75 μ s)
Recorded Trajectory Frames	AMBER NetCDF trajectory output cadence	223 frames
Axial Momentum Accounting	Reaction absorbed by outer radial rim clamp	Clamped boundary
Total Energy Conservation	Energy drift under conservative radial force	+2.76%
Peak Compressive Pressure	Hugoniot shock state under 5 000 m/s impact	> 150 GPa
Peak Hydrostatic Tension	Constructive interference of rarefaction fans	6.75 GPa
Peak Plastic Strain in Spall Zone	Multi-axial shear flow driving Cocks–Ashby rate	$\varepsilon_p = 1.301$
Maximum Porosity Attained	Saturated void volume fraction cutoff	$f_{\max} = 50.0\%$
Maximum Continuum Damage	Full material degradation at void coalescence	$D_{\max} = 1.000$
Severed Particle Count ($D \geq 0.99$)	Detached scab and crater ejecta particles	4 053 particles
Spall Pullback Velocity (Δv_{pb})	Free-surface velocity drop on target rear face	175.7 m/s ($\sigma_{\text{spall}} \approx 5.0$ GPa)
Asymptotic Scab Fly-Off Speed	Autonomous steady velocity of detached spall scab	$\approx 175\text{--}180$ m/s

plate, $\varnothing 300 \times 24$ mm, at 2 km/s. The plate is free, with no support. The experiment's crater, and the “completely damaged” diameter that the paper compares against it, is 80 mm. Where Kim et al. look at the crater at about 0.1 ms, this deck runs on to 1.1 ms after contact, because its purpose is the long-time fragmentation of the plate. It is therefore the verification scene of the Holmquist–Johnson–Cook model (§3.2.5–§3.4.4) together with its Grady–Kipp tension card (§3.4.5).

5.5.1 Geometry and Discretisation

The deck is a Cartesian quarter model with `symmetryPlanes`: “yz” (§2.9.4), the mirror planes being $y = 0$ and $z = 0$ and the impact running along $+x$. It uses Kim et al.'s own spacing, $\Delta x_0 = 1$ mm, with a kernel support of $3 \Delta x_0$. That gives 426 269 particles, of which 424 128 are plate and the rest projectile; the quarter holds a quarter of Kim et al.'s 1.7 million. The sphere starts $2h = 6$ mm clear of the impacted face, so contact is at about $3 \mu\text{s}$, and the duration of 1.103 ms ends the run 1.1 ms after it. The flaws that the tension card draws are mirrored across the planes together with the particles, so a fragment that meets a plane has its shape there imposed by the symmetry rather than drawn; the fragment tool flags those.

5.5.2 Materials and Settings

The projectile is `aluminium_kim2025` (Johnson–Cook strength on a Mie–Grüneisen EOS) and the plate `concrete_hjc_150` (§3.5.2, Table 3.6). HJC carries the compression: the compaction EOS, the pressure-dependent strength and the cumulative damage D_c . The tension is taken off HJC's uniform cutoff and given to the Grady–Kipp card (`damage.tension`, §3.4.5), set as follows:

- Weibull modulus $m = 8$. This is an assumption from the range of 5 to 12 quoted for concrete, not a sourced value.
- `referenceVolume` = $5.301 \times 10^6 \text{ mm}^3$, a 150×300 mm cylinder: the specimen on which the EOS tensile strength $T = 7.59 \text{ MPa}$ is taken to be the weakest-flaw strength. A particle of 1 mm^3 is far smaller, and its weakest flaw correspondingly stronger, by the size effect $(V_{\text{ref}}/V)^{1/m}$.
- `crackSpeedFactor` = 0.4, a crack growth speed of $0.4 c_l = 1.59 \text{ km/s}$.
- `seed`: 1, one sample of the flaw population, and `flawCap`: `false`, which switches off Benz and Asphaug's cap $D_t \leq n_{\text{act}}/n$ so that a crack band can separate what lies on either side of it. That is a deliberate departure from the original model.

The numerical settings are those of the Al 2017 hypervelocity deck:

- `pairSoundSpeed`: “off”, since the pair estimate reads nonsense across the aluminium–concrete interface, and `Erosion.mode`: “cap” with the volume ratio clamped into $[0.3, 5]$;
- bulk viscosity $C_Q = 1.5$ over Δx_0 , and Monaghan $\alpha = 1$, $\beta = 2$ on approaching pairs only, so that it does not resist a crack opening;
- the super-time-stepped hourglass damper at $\alpha = 2.0$ in `subcycle` mode;
- δ -SPH at 0.1 over same-material pairs, switched off by the solver across a cracked bond ($D_t \geq 0.99$);
- Collé PST at $\text{Ma} = 0.05$ with `aleMomentum`, `aleHistory` and `aleEnergyConservative`. With `aleHistory`, D_c , μ_{max} and D_t move with a shifted particle; the flaws do not, because they belong to the material.

The NetCDF trajectory carries `damage_t` and each particle's weakest-flaw strain `flaw_eps_min` alongside the usual fields, and the GUI colours the particles by `damageTension`.

5.5.3 Results

Figure 5.6 shows the plate from the impacted face and from the back, unfolded from the quarter into the full plate. Both faces show the same pattern: a crushed crater zone around the impact axis, surrounded by a ring of partially damaged concrete, and outside it radial cracks that branch as they run to the free edge of the plate. The damage maps of the run show this network fixed from about 0.1 ms after contact onwards; it does not change materially in the millisecond that follows.

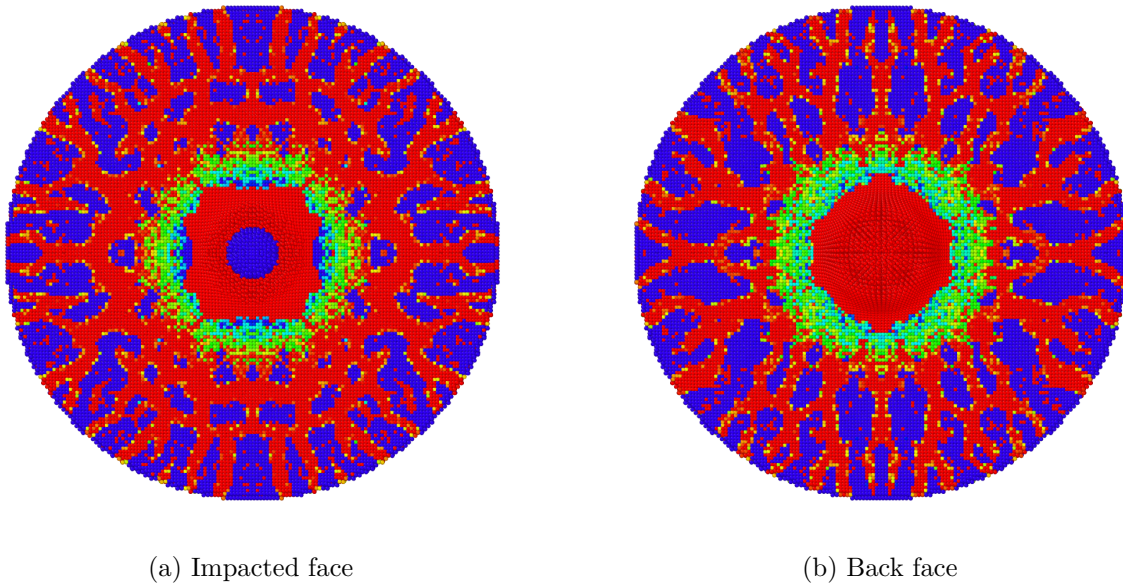


Figure 5.6: The Chocron plate of `hvi_3d_quarter_al_concrete150_chocron_gk_mm_GPa_ms.json`, mirrored from the quarter model into the full Ø300 mm plate and coloured by damage from blue (intact) to red (fully damaged). The aluminium projectile sits in the crater at the centre of the impacted face. Radial cracks branch from the crater zone to the free edge, but many branches end without meeting another crack.

Many of the crack branches end without meeting another, so the cells between them are not closed off and the plate does not break into sectors: it stays one connected body. The fragment measure of `tools/fragment_distribution.py` makes that quantitative. A fragment is a connected set of plate particles bonded within $1.5\Delta x_0$ whose combined damage $1 - (1 - D_c)(1 - D_t)$ is below a threshold, 0.99 by default; particles above it are counted as rubble, and the quarter is unfolded into the full plate. At 1.1 ms the largest fragment holds 65.5% of the plate, and still 57.5, 48.9 and 42.2% at the thresholds 0.9, 0.7 and 0.5, so the conclusion does not hang on the threshold. The rubble share is 36.8% at 0.1 ms and 34.4% at 1.1 ms, and no other fragment of ten particles or more exists.

The run drew 6 446 343 flaws, about fifteen per plate particle. Each particle's weakest flaw lies at 30, 50 and 64 MPa at the 1st, 50th and 99th percentile, while the weakest flaw of the whole plate is at 10.4 MPa. These particle strengths are what decides the outcome. At $\Delta x_0 = 2$ mm the same card, through the same reference volume, gives each particle a weakest flaw $8^{1/8} = 1.30$ times weaker (median 39 MPa), and that plate breaks into sectors; a 2 mm control run given the 1 mm strength distribution stays one body as this deck does. The card, calibrated through a reference pair, is therefore mesh-dependent through the Weibull size effect, by $2^{3/m}$ in particle

strength per halving of the spacing.

5.5.4 Computational Cost and Energy Conservation

The run to 1.1 ms took 57 minutes on a single GPU and wrote 7.8 GB of NetCDF in 176 frames. The total energy changed by +0.58% of E_0 . Of the budget terms, the advection of e_{int} by the ALE shift (`ie_ale_adv`, Eq. (2.34)) amounted to 17.5% of E_0 , which `aleEnergyConservative` fixed up (§2.5.2); without the fix-up, that term appears as an energy gain of several percent, made at the shock in the first few tens of microseconds after contact.

TRAP / OPERATIONAL NOTE: Not Yet a Fragment Distribution to Compare with Experiment

Every figure above is a single run of a single seed, with an assumed Weibull modulus. It is not established whether 50 MPa (one body) or 39 MPa (sectors) is the right particle-scale strength for this concrete: fragment data, or a measured dynamic tensile strength, is needed to decide it. A resolution-independent calibration would fix the flaw statistics per unit volume at a fixed length scale rather than per particle, for instance by scaling `referenceVolume` with the particle volume; the run that would test it is this deck at `referenceVolume/8`, which matches the 2 mm particle strength. The crater diameter is not measured on this deck (`tools/hjc_crater.py` does that for the axisymmetric deck `hvi_axi_al_concrete150_chocron_mm_GPa_ms.json`), and the deck is *not* tuned to the experiment's 80 mm.

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